The Mesche Hypothesis

A Dynamic Field Theory of Cosmic Structure
Complete research compilation of the MTDF paper suite

Ingo Mesche
Independent Researcher, Malta
imesche@outlook.com

V74 / 2026

About this compilation. This document combines all eight papers of the MTDF research programme into a single self-contained volume. Each paper is also available as a standalone submission with the same preamble and uniform margins. Cross-references between papers resolve internally within this master document.

Contents

I Master paper: The Mesche Hypothesis **11**

1 Motivation and design principles **11**

2 Core objects and governing assumptions **12**
2.1 Minimal parameter philosophy . . . 12
2.2 Calibration anchors versus validation targets . . . 12
2.3 Mathematical objects, stress definition, and energy . . . 12
2.3.1 Core idea in one sentence . . . 12
2.3.2 Core MTDF Parameters (empirically constrained or theoretically derived) . . 12
2.4 Evolution equation and relaxation time . . . 13
2.4.1 On analogies and what is not claimed . . . 13
2.5 Regime transition and critical acceleration . . . 13
2.6 Local rigidity and scale hierarchy . . . 13

3 Cosmological background dynamics **14**
3.1 Dual epoch mechanism for the Hubble tension . . . 14
3.2 Stress tensor energy budget and effective pressure . . . 15
3.2.1 Core Mechanism . . . 15
3.2.2 Dimensional verification: . . . 15
3.2.3 Modified Friedmann Equation . . . 15
3.2.4 Dimensional verification: . . . 15
3.2.5 Dynamic Equation of State . . . 15
3.2.6 Parameter Physics . . . 15
3.2.7 Constraint Resolution Achievement: . . . 16
3.3 Void stress depletion and late time expansion . . . 16
3.3.1 Local Hubble Enhancement . . . 16

3.3.2 Observational Agreement: . . . 16
3.3.3 Dimensional verification: . . . 16
3.4 Stress evolution and near term forecasts . . . 16
3.4.1 Tensor Energy Density Evolution . . . 16
3.4.2 Dimensional verification: . . . 17
3.4.3 Future Evolution Timeline . . . 17
3.4.4 Cosmic Evolution Phases . . . 17

4 Emergent phenomenology across scales **17**
4.1 Galaxy scale dynamics . . . 17
4.1.1 Mechanism: stress divergence as an additional acceleration channel . . . 17
4.1.2 Physical Foundation . . . 17
4.1.3 Dimensional verification: . . . 18
4.1.4 Critical Acceleration Threshold . . . 18
4.1.5 Dimensional verification: . . . 18
4.1.6 Critical Acceleration Parameter . . . 18
4.1.7 Tensor-based rotation curve solution and coherence length . . . 18
4.1.8 Modified Potential . . . 18
4.1.9 Velocity Profile . . . 18
4.1.10 Dimensional verification: . . . 18
4.1.11 Regime behaviour . . . 19
4.1.12 Radial acceleration relation and scatter control . . . 19
4.1.13 RAR Validation Results . . . 19
4.1.14 Framework Summary - Section 4 Contribution: . . . 19
4.1.15 How SPARC appears in the V18 strict protocol . . . 19
4.1.16 Two-Phase Testing . . . 19
4.1.17 Rigorous Validation Methodology . . . 20
4.1.18 Performance Results . . . 20
4.1.19 P1-SPARC Pillar (✓ Observed Tier): . . . 20
4.1.20 Three-Tier Framework Position: . . . 20
4.1.21 Comparative Analysis . . . 20
4.1.22 Outliers and anomaly handling . . . 20
4.1.23 First-Principles Solutions . . . 20
4.1.24 Tensor-Based Anomaly Resolution . . . 20
4.1.25 Environmental Parameter Derivation . . . 20
4.2 Structure growth and large scale surveys . . . 21
4.2.1 Stress tensor properties relevant for structure growth . . . 21
4.2.2 Stress Tensor Relaxation Parameter . . . 21
4.2.3 Dimensional verification: . . . 22
4.2.4 Stress accumulation, relaxation cycles, and hierarchy . . . 22
4.2.5 Stress Buildup Phase . . . 22
4.2.6 Relaxation Phase . . . 22
4.2.7 Cyclical Nature . . . 22
4.2.8 Topology, defects, and filament formation . . . 23
4.2.9 Multi-Scale Structure Formation . . . 23
4.2.10 Stress Tensor Structure Formation . . . 23
4.2.11 Cosmic web topology and causal propagation . . . 23
4.2.12 Cosmic Structure Mapping . . . 23
4.2.13 Observable Manifestations . . . 23
4.2.14 Structure Formation Mechanics . . . 23
4.2.15 Dimensional verification: . . . 24
4.2.16 Tensor Equation with Causality . . . 24
4.2.17 Material Parameters Table . . . 24

4.2.18 Scope note . . . 24
4.3 CMB compressed distance prior . . . 24
4.4 Gravity sector and lensing constraints (Companion Part II) . . . 24

5 Parameter set and provenance **25**
5.0.1 H_0 convention and predictions . . . 26
5.1 Notes on interpretation . . . 26
5.2 Derived and auxiliary quantities . . . 27
5.3 Activation summary . . . 27

6 Validation architecture (V18 strict protocol) **28**

7 Current results (vector likelihoods and strict totals) **28**
7.1 Vector pillar breakdown . . . 28
7.2 Interpretation of the strict totals . . . 29
7.3 Inferred growth behaviour from cosmological validation . . . 29

8 Diagnostics, limitations, and road map **30**
8.1 What is validated in V18 . . . 30
8.2 What is not yet validated in V18 . . . 30

9 Falsifiable predictions beyond the dashboard **30**
9.1 Supplementary quantitative falsification thresholds . . . 31
9.2 Indicative observation timeline . . . 31
9.3 Hard falsification conditions . . . 31

10 Discriminating observational signatures **31**
10.1 Environment-dependent supernova residuals with redshift localisation . . . 32
10.2 Interpretation of the Hubble tension as an environmental stratification . . . 32
10.3 Linked direction of expansion and structure-growth shifts . . . 32
10.4 Environment dependence of residual galaxy-scale dynamics . . . 32
10.5 Scope and intent . . . 33

11 Reproducibility and artefact traceability **33**

12 References (DOIs) **33**

II Companion I: Environmental and Local Phenomenology **34**

13 Purpose and scope **34**

14 Environmental and local modulation **35**
14.1 Environment dependent effective dynamics . . . 35
14.1.1 Observation . . . 35
14.1.2 MTDF mechanism . . . 35
14.1.3 Predictions . . . 35
14.1.4 How to test . . . 35
14.2 Relaxation and memory effects in local structure . . . 36
14.2.1 Observation . . . 36
14.2.2 MTDF mechanism . . . 36
14.2.3 Predictions . . . 36
14.2.4 How to test . . . 36
14.3 Supernova residuals and local expansion in underdense environments . . . 36
14.3.1 Observation . . . 36

14.3.2 MTDF mechanism . . . 37
14.3.3 Predictions . . . 37
14.3.4 How to test . . . 37
14.3.5 Rigorous cross-match analysis using Pantheon+ and DESI void environments 37
14.3.6 Method summary . . . 37
14.3.7 Main GLS results . . . 38
14.3.8 Footprint-based NGC and SGC splits . . . 38
14.3.9 Interpretation and robustness . . . 39
14.3.10 Update: Footprint split, baseline stability, and redshift dependence . . . 39
14.3.11 Pre-registered NGC and SGC analyses (footprint-based assignment) . . . 40
14.3.12 Selection-function check and what it does (and does not) explain . . . 40
14.3.13 Z-matched cross-check (footprint specificity) . . . 40
14.3.14 Continuous redshift modulation . . . 40
14.3.15 Specificity test using an alternate environment metric . . . 41
14.3.16 Survey fixed-effects control . . . 41
14.3.17 Leave-one-survey-out robustness . . . 41
14.3.18 Merged-sample hardening . . . 42
14.3.19 Summary figure . . . 43

15 Local anomalies and Solar System scale signatures **43**
15.1 Near threshold signatures and precision dynamics . . . 43
15.1.1 Observation . . . 44
15.1.2 MTDF mechanism . . . 44
15.1.3 Predictions . . . 44
15.1.4 How to test . . . 44

16 How this part links back to the master paper **44**

17 References **45**

III Companion II: Gravity Sector and Lensing Validation **45**

18 Scope and design **45**

19 Abstract **46**

20 Introduction **46**

21 Model **47**
21.1 MTDF gravity sector . . . 47
21.2 Compression mechanism . . . 47
21.3 Constitutive law . . . 48
21.4 Lensing prediction . . . 48
21.5 Solar System screening . . . 48
21.6 Covariant formulation . . . 49

22 Methods **49**
22.1 Datasets . . . 49
22.2 Unit conventions . . . 49
22.3 MTDF prediction . . . 50
22.4 ΛCDM comparison . . . 50
22.5 Robustness variants . . . 50
22.6 Baryon completion model . . . 50

22.7 Statistical framework . . . 50
22.8 Assumption inventory . . . 51

23 Results **51**
23.1 Results summary . . . 51
23.2 KiDS $\times$ GAMA (Brouwer et al. 2021) . . . 51
23.3 SDSS Red LBG (Mandelbaum et al. 2016) . . . 52
23.4 Robustness against ΛCDM prior choices . . . 52
23.5 Baryon completion for high-mass bins . . . 52
23.6 Physically anchored cluster prediction (Step 21) . . . 53
23.7 Elliptical galaxy velocity dispersions (Step 22) . . . 54
23.8 Satellite kinematics (Step 22B) . . . 54
23.9 Rotation curves and RAR consistency (Step 22C) . . . 55
23.10 Wide binary gravity test (Step 22D) . . . 56

24 Methodological notes **56**

25 Discussion **56**
25.1 Claims and non-claims . . . 56
25.2 Diagnostic-only outputs . . . 57
25.3 Regime summary and falsifier status . . . 57
25.4 Why the absolute χ^2/ν is not small . . . 58
25.5 Falsifiability . . . 58
25.6 Remaining open questions . . . 59

26 Conclusions **59**

27 Appendix A. Derivation chain **59**

28 Appendix B. Field equation backing **60**

29 Appendix C. Dimensional consistency **60**

30 Appendix D. Notation and symbols **60**

31 Appendix E. Data provenance **60**

32 References **61**

IV Companion III: Photon Coupling, Redshift, and Early Universe **62**

33 Purpose and scope **63**

34 Photon coupling and stress induced redshift **63**
34.1 Stress induced frequency shift . . . 63
34.1.1 Observation . . . 63
34.1.2 Proposed MTDF mechanism . . . 64
34.1.3 Structural relation between κ and f_{kick} . . . 64
34.1.4 Physical interpretation . . . 64
34.1.5 Predictions . . . 65
34.1.6 How to test . . . 65

35 Early structure, JWST era tensions, and BAO links **65**
35.1 Formation time interpretation under a modified mapping . . . 65

35.1.1 Observation 66
35.1.2 Proposed MTDF mechanism 66
35.1.3 Predictions 66
35.1.4 How to test 66

36 Quantitative falsifiers and timeline **66**
36.1 Precision consistency checks and sensitivity 67
36.2 Supplementary test suite for κ and current sensitivity 68

37 References **70**

V Companion IV: Cosmological Validation and High Energy **70**

38 Purpose and scope **70**

39 Cosmological validation: CMB, BAO, and local H0 **71**
39.1 Matched CMB + BAO comparison (Planck 2018 TTTEEE + DESI BAO) 71
39.2 SH0ES prior and the global vs local H0 test 71
39.3 Conservative fit philosophy 72
39.4 Full Planck plik TTTEEE + lensing validation (Phase 5) 72
39.4.1 Convergence diagnostics 73
39.4.2 Posterior parameter comparison 73
39.4.3 Chi-squared breakdown by likelihood component 73
39.4.4 Model selection 74
39.4.5 Stress coupling k_f posterior 74
39.4.6 k_f parameter correlations 74
39.4.7 k_f quartile analysis by likelihood component 75
39.4.8 Growth suppression: S_8 comparison 75
39.4.9 Figures 75

40 High-energy phenomenology **76**
40.1 Black holes as stress singularities 76
40.1.1 Observation 76
40.1.2 MTDF mechanism 76
40.1.3 Predictions 76
40.1.4 How to test 76
40.2 Fast radio bursts as localised stress oscillations 76
40.2.1 Observation 76
40.2.2 MTDF mechanism 77
40.2.3 Predictions 77
40.2.4 How to test 77

41 Lensing and merging clusters **77**
41.1 Lensing consistency as a gatekeeper test 77
41.1.1 Observation 77
41.1.2 MTDF mechanism 77
41.1.3 Predictions 77
41.1.4 How to test 78
41.2 Cluster scale mass bias and stress supported hydrostatics 78
41.2.1 Observation 78
41.2.2 MTDF mechanism 78
41.2.3 Predictions 78
41.2.4 How to test 78

41.3 Merging clusters and mass map offsets as a critical test . . . 79
41.3.1 Observation . . . 79
41.3.2 MTDF mechanism . . . 79
41.3.3 Predictions . . . 79
41.3.4 How to test . . . 79

42 Structured test programme **79**

43 References **80**

VI Appendix: Validation Suite **81**

44 Purpose and how to use this appendix **81**

45 Test index and reproducibility **81**
45.1 Script index by paper component . . . 81
45.2 Outcome snapshot . . . 82
45.3 Synthesis figure . . . 83

VII Independent GPU Validation (Phases 1-6) **84**

46 Scope and methodology **85**
46.1 Hardware and software . . . 85

47 Phase 1: Independent χ^2 reproduction **85**
47.1 Strict total . . . 86
47.2 Vector pillar breakdown . . . 86

48 Phase 2: Planck CMB consistency **86**
48.1 Emulator validation . . . 86
48.2 Fixed-parameter comparison . . . 86
48.3 Normalisation correction . . . 87
48.4 Production MCMC results . . . 87
48.4.1 Model comparison . . . 87
48.4.2 k_f posterior . . . 87
48.4.3 H_0 posterior . . . 87
48.4.4 Per-probe breakdown (combined narrow best-fit) . . . 88

49 Phase 3: SN $\times$ void environment signal **88**
49.1 Environment coefficient γ_{env} . . . 88
49.2 NGC/SGC split . . . 88
49.3 Survey fixed effects . . . 88
49.4 Leave-one-survey-out (LOSO) . . . 88
49.5 Redshift modulation: the signature result . . . 89
49.6 Non-parametric confirmation . . . 89
49.6.1 Permutation test (10,000 shuffles) . . . 89
49.6.2 Block bootstrap (5,000 resamples), 95% CI . . . 89
49.7 Numerical integrity . . . 89

50 Phase 4: Sensitivity forecasts **89**
50.1 Key structural insight . . . 89
50.2 Detection threshold table . . . 90
50.3 Interpretation . . . 90

51 5b. Phase 3b: Asymmetry and detectability diagnostics **90**
51.1 5b.1 Test A: IPW matched redshift 90
51.2 5b.2 Test B: Void–SN joint density mapping 91
51.3 5b.3 Test C: Wald test and parametric bootstrap 91
51.4 5b.4 Test D: Geometry and mode-coupling check 91
51.5 5b.5 Test E: Null signal injection (SGC recovery) 91
51.6 5b.6 Interpretation 91

52 Phase 5: Full Planck plik + class_mtdf **92**
52.1 χ^2 breakdown by likelihood component 92
52.2 Model selection 92
52.3 Best-fit cosmological parameters 93
52.4 Interpretation 93
52.5 Planck degeneracy structure and posterior stability (MCMC converged) 93
52.5.1 Convergence statistics 93
52.5.2 Posterior parameter comparison (68% CI) 94
52.5.3 k_f posterior 94
52.5.4 k_f identifiability diagnostics 94
52.5.5 Best-fit robustness across Planck subsets 94
52.5.6 Phase 5 robustness checks (Tests 2–4) 94

53 6b. Phase 6: Cross-probe discriminator tests **95**
53.1 Summary table 96
53.2 6b.1 Test A: Redshift transition scan 96
53.3 6b.2 Test B: Weak lensing × environment (void centres) 96
53.4 6b.3 Test B2: Trough and ridge lensing (projected density split) 97
53.5 6b.4 Test C: Derived parameter consistency 97
53.6 6b.5 Test D: CMB lensing × DESI BGS voids (low redshift) 97
53.7 6b.6 Test E: CMB lensing × BOSS DR12 voids (benchmark) 97
53.8 6b.7 Test C2: S_8 cross-probe coherence 98
53.9 6b.8 Test C3: BAO residual structure 98
53.10 6b.9 Test C4: $f\sigma_8$ × environment 98
53.11 6b.10 Test C5/C5b: Cluster M_{dyn}/M_{lens} - Resolved 99

54 Synthesis and framework narrative **99**
54.1 Summary of results (17/17 complete) 99
54.2 Claim status taxonomy 101
54.2.1 Status by claim 101
54.3 Framework narrative 103
54.4 What survives intact 103
54.5 Joint implications for late-time cosmological tensions 103
54.6 Paper implications 104

55 Open questions and next steps **104**
55.1 NGC/SGC asymmetry diagnostics (Phase 3b, complete) 104
55.2 Falsifier and discriminator tests 104

56 Appendix: Technical notes **105**
56.1 A.1 k_f normalisation conventions 105
56.2 A.2 V18 parameter reference 105
56.3 A.3 Historical context: November 2025 runs 105

57 References **106**

VIII Multi-Probe Evidence for a Low-Redshift Transition 107

58 Introduction 108

59 Working hypothesis and prediction 108

60 Datasets and observables 108
60.1 Supernova luminosity distances . . . 109
60.2 Strong-lens time-delay distances . . . 109
60.3 Cosmicflows-4 peculiar velocities . . . 109
60.4 2MTF Tully-Fisher distances . . . 109
60.5 ISW void stacking . . . 109
60.6 Weak-lensing S_8 environment split . . . 109

61 Common analysis logic 109
61.1 Signed-distance environment metric . . . 110
61.2 Three independent void catalogues . . . 110
61.3 GLS regression framework . . . 110
61.4 Piecewise redshift split . . . 110
61.5 Robustness chain . . . 110

62 Channel results 111
62.1 Supernovae: environment cutoff at $z = 0.04$ . . . 111
62.2 Time-delay distances: independent geometric support . . . 111
62.3 Cosmicflows-4 peculiar velocities . . . 112
62.3.1 Baseline signal . . . 112
62.3.2 Piecewise redshift split . . . 112
62.3.3 Control chain . . . 112
62.4 2MTF Tully-Fisher distances . . . 114
62.4.1 Baseline signal . . . 114
62.4.2 Control chain . . . 114
62.5 Supporting channels . . . 115
62.5.1 ISW void stacking . . . 115
62.5.2 Weak-lensing S_8 environment split . . . 115
62.5.3 Distance duality and lensed-image redshifts . . . 115

63 Cross-probe synthesis 116

64 ΛCDM mock comparison 117
64.1 Linear-theory reconstruction . . . 117
64.2 Full N-body mock (MDPL2) . . . 118
64.3 Null baseline . . . 118

65 Interpretation within MTDF 118

66 Limitations and caveats 119

67 Conclusion 120

68 Appendix A. Data availability and reproducibility 120
68.1 A.1 Analysis scripts . . . 120
68.2 A.2 Machine-readable results . . . 121
68.3 A.3 Diagnostic plots . . . 121
68.4 A.4 External data sources . . . 122

69 References **122**

Paper I

Master paper: The Mesche Hypothesis

Abstract

Mesche's Tensor Dynamics Framework (MTDF) is a field based alternative to the latent sector interpretation of modern cosmology. It aims to reproduce the background expansion history and a wide range of dynamical phenomena without invoking cold dark matter or a cosmological constant as new substances. This master paper presents the theory as a coherent physical proposal, states the minimal parameter set and its provenance, and formalises a strict evaluation protocol that separates calibration anchors from independent validation targets. It then summarises current V18 validation outcomes on the vector likelihood set (Pantheon+ supernovae, DESI Y1 BAO, cosmic chronometer H(z), DR16 $f\sigma_8$ growth) and reports the CMB distance prior strictly as a diagnostic. A sharp, environment-dependent transition in SNe residuals is detected at $z = 0.04 \pm 0.005$ across three independent void catalogues, with $\Delta\chi^2 = 36$–39 relative to a single-regime fit ($p < 0.001$). This scale coincides with the coherence length $\beta = 22.7$ Mpc predicted independently from void quantisation. Fitting Planck 2018 TTTEEE+lensing with the class_mtdf modified Boltzmann solver yields $\Delta\chi^2 = +0.63$ relative to ΛCDM, with a broad non-pathological posterior on the MTDF coupling k_f that admits full coupling ($k_f = 1$) within 1σ; the ΔBIC $= +8.05$ reflects the added dimensionality, so the framework is not excluded by this dataset under the stated assumptions. The paper emphasises falsifiability: it enumerates near term tests that can confirm or refute MTDF beyond the current dashboard scope. A separate gravity-sector companion paper reports quantitative lensing and local dynamics tests (galaxy-galaxy lensing, strong lens time delays, and Solar System screening) using the same frozen parameter set; see *MTDF_03 (Companion Part II)* [see companion paper].

Claim status. This paper distinguishes between: (i) calibration anchors, (ii) validation targets, (iii) diagnostic consistency checks, and (iv) speculative extensions. Only results explicitly labelled as validation targets are used as evidence for the core MTDF claim. Diagnostic and speculative sections are included to define future tests, not to claim established confirmation.

1 Motivation and design principles

Contemporary cosmology describes most observed gravitational phenomena by introducing a dominant but unseen sector. In the standard model of cosmology, these components are parameterised as cold dark matter and dark energy. MTDF proposes an alternative interpretation: the phenomena attributed to the latent sector emerge from the dynamics of a physically defined spacetime field that carries stress energy and couples to baryonic matter through a single coupling parameter.

Design constraints for a high standard theory proposal. MTDF is formulated under four constraints: (i) minimal free parameters; (ii) explicit parameter provenance; (iii) strict separation of calibration anchors from validation targets; (iv) falsifiability with clear failure modes. The V18 era artefacts operationalise these constraints for empirical evaluation.

It is essential to distinguish between two different claims: (a) *data fit* within a chosen likelihood set, and (b) *physical explanation* that remains coherent across scales. The dashboard is primarily about (a). This master paper addresses (b), while importing the dashboard protocol to ensure that any claimed fit is not achieved by circular calibration.

2 Core objects and governing assumptions

MTDF treats the gravitational sector as augmented by a field degree of freedom whose stress energy contributes to dynamics. The field is introduced to remain compatible with classical limits while providing a mechanism for scale dependent behaviour. In the non relativistic regime, the field acts as an additional contribution to the effective potential that influences orbital velocities and structure formation, while in the cosmological regime it modifies the effective expansion history through its energy density and stress.

2.1 Minimal parameter philosophy

The framework is constrained to a small set of fundamental parameters that are either measured from independent observational anchors or established from theoretically motivated universal constants. The operational intent is that most quantities used in predictions are derived, not tuned.

2.2 Calibration anchors versus validation targets

In the V18 strict protocol, some data are permitted to set fundamental scales (anchors). These are excluded from validation scoring. All other reported goodness of fit values are computed strictly on targets that are treated as independent of the calibration step. This is the key correction relative to earlier single layer evaluations.

2.3 Mathematical objects, stress definition, and energy

This section expands the core mathematical objects that appear throughout MTDF. The intent is to make the field definitions explicit, while keeping the strict V18 protocol separation between calibration anchors and validation targets.

2.3.1 Core idea in one sentence

We propose that the universe is permeated by dynamic spacetime characterized by the dimensionless strain tensor $S_{\mu\nu}$ that:

- **Stores energy** through elastic deformation with energy density $\mathrm{u}_{\text{tensor}} = (\mathrm{E}/2)\ \mathrm{Tr}(S_{\mu\nu}\ S_{\mu\nu})$
- **Exhibits stress tensor gradients** that wrap around massive objects and extend throughout cosmic space
- **Undergoes stress accumulation and relaxation cycles** over characteristic time $\tau = 13.0$ Gyr
- **Redistributes energy** rather than destroying it, maintaining conservation laws
- **Exhibits bipolar behaviour** with attractive and repulsive components at different scales

2.3.2 Core MTDF Parameters (empirically constrained or theoretically derived)

Reference: Complete parameter values and derivations are summarised in Section 5 and in the validation workbook accompanying the release.

MTDF represents the additional degree of freedom through a dimensionless strain field. In compact notation one may write:

$$\tilde{S}_{\mu\nu} = \beta\, \nabla_\mu \xi_\nu$$

Stress is defined through a constitutive relation with an effective elastic modulus:

$$F_{\mu\nu} = E\,\tilde{S}_{\mu\nu}$$

The associated field energy density and an effective negative pressure contribution can be written schematically as:

$$u_{\text{tensor}} = \frac{E}{2}\,\tilde{S}_{\mu\nu}\,\tilde{S}^{\mu\nu}, \qquad p_{\text{tensor}} \approx -\frac{1}{3}u_{\text{tensor}}$$

2.4 Evolution equation and relaxation time

In the simplest non relativistic implementation used for many intuition building arguments, MTDF employs a damped wave type evolution for an effective potential-like field variable. A representative form is:

$$\frac{\partial^2\phi}{\partial t^2} - c_s^2\,\nabla^2\phi + \gamma\,\frac{\partial\phi}{\partial t} = 8\pi G\rho$$

Here ρ denotes the baryonic source term, c_s is an effective propagation speed for stress perturbations, and γ sets the relaxation rate. In many analytic arguments γ is treated as the inverse of a cosmic scale relaxation time τ, which is part of the minimal parameter set.

2.4.1 On analogies and what is not claimed

The stress-strain behavior of spacetime ($F_{\mu\nu}$ = E $S_{\mu\nu}$) manifests dynamical patterns with direct analogues in condensed matter systems:

These parallels provide physical motivation for the stress-tensor analogy used in MTDF, illustrating how material physics principles scale to cosmic phenomena when applied to spacetime itself with elastic modulus E = 9.1×10^{-10} Pa.

2.5 Regime transition and critical acceleration

The framework is constructed to recover the classical limit at high accelerations while producing measurable departures in the low acceleration regime. A commonly used summary quantity is a critical acceleration scale that depends on the fundamental length scale β:

$$a_{\text{crit}} \sim \alpha\,\frac{c^2}{\beta}$$

This scale is used as a compact way to discuss why deviations are negligible in Solar System tests yet become relevant in galaxies and the late time large scale environment. Precise operational definitions depend on the specific observable being computed and are therefore treated as derived quantities in the V18 artefacts.

2.6 Local rigidity and scale hierarchy

The acceleration-dependent relaxation (Section 2.4) provides one mechanism for suppression in high-acceleration environments. A second, independent mechanism arises from the substrate's wavelength-dependent stiffness. The effective stiffness of the MTDF substrate at wavelength λ is:

$$K_{\text{eff}} = E\,\beta^2\,(2\pi/\lambda)^2$$

At laboratory scales ($\lambda \sim 0.1$ m), $K_{\text{eff}} \sim 10^{40}$ Pa. At solar system scales ($\lambda \sim 1$ AU), $K_{\text{eff}} \sim 10^{14}$ Pa. These vastly exceed any stress that laboratory or solar-system sources can produce.

This rigidity is independent of damping: even in a zero-acceleration environment, the substrate resists short-wavelength perturbations because the coherence length $\beta = 22.7$ Mpc amplifies strain relative to displacement ($\tilde{S} = \beta\,\nabla\xi$). The same lever that makes MTDF responsive at galactic scales ($\beta k \ll 1$) makes it rigid at laboratory scales ($\beta k \gg 1$).

MTDF therefore exhibits a natural scale hierarchy controlled by the dimensionless product βk, where $k = 2\pi/\lambda$:

Regime	βk	**Stiffness**	**Damping**	**Outcome**
Laboratory	$\sim 10^{24}$	Rigid	Damped	Undetectable
Solar system	$\sim 10^{20}$	Rigid	Damped	Undetectable
Galaxy	$\sim 10^{-5}$	Soft	Undamped	Active
Cosmological	~ 10	Soft	Undamped	Active

This hierarchy follows directly from the fundamental parameters and requires no tuning. The dual mechanism (damping + rigidity) provides a complete explanation for the absence of MTDF effects in all local experiments, including precision electromagnetic measurements with superconducting cavities ($Q > 2 \times 10^{11}$), without requiring any additional assumptions.

3 Cosmological background dynamics

At background level, MTDF is evaluated by comparing its predicted distance and expansion relations to vector likelihood datasets. The present implementation in the V18 artefacts uses a dedicated pipeline to compute the background observables and their residuals against each dataset.

3.1 Dual epoch mechanism for the Hubble tension

Notation. The symbol β_{eos} denotes the dimensionless equation-of-state parameter (0.573); it is distinct from the physical coherence length $\beta = 22.7$ Mpc introduced in Section 2. Both are listed separately in Table 1.

The theory separates the Hubble tension into an early epoch contribution and a late epoch contribution. The early epoch mechanism is encoded in a small derived energy injection fraction, defined from the fundamental parameter pair (α, β_{eos}), and is used to assess consistency against the CMB distance prior as a diagnostic. The late epoch mechanism is attributed to local stress depletion in underdense regions, which enhances locally inferred expansion without requiring a global shift of the background expansion parameter.

$$\lambda_{\text{MTDF}} = \frac{(1-\beta_{\text{eos}})^2}{1+\alpha}$$

$$f_{\text{kick}} = \frac{\lambda_{\text{MTDF}}}{24}$$

$$\frac{\Delta r_s}{r_s} = K_{\text{RS,EFE}}\, f_{\text{kick}} \quad \text{with} \quad K_{\text{RS,EFE}} = -0.215$$

Computed from the current V18 fundamental parameters: $\alpha = 1.300$, $\beta_{\text{eos}} = 0.573$ gives $\lambda_{\text{MTDF}} = 0.079273$, $\mathrm{f_{kick}} = 0.003303062$, and $\Delta \mathrm{r_s}/\mathrm{r_s} = -0.000710158$. The early field energy fraction $\mathrm{f_{kick}}$ is also structurally related to the photon-stress coupling parameter $\kappa \approx \mathrm{f_{kick}}/3$ (see Companion Part

III), linking the gravitational and photon sectors of the theory through a single pair of parameters $\{\alpha, \beta_{\text{eos}}\}$.

These relations are not presented as a full CMB spectral fit. They exist to provide a transparent mapping from the fundamental parameter set to the specific diagnostic quantity used in the V18 dashboard.

3.2 Stress tensor energy budget and effective pressure

At background level, the strict V18 evaluation treats the expansion history as a derived prediction from the fundamental parameter set and the chosen anchors. The underlying physical picture, however, is that accumulated field strain contributes an effective energy density and a negative pressure component that can mimic late time acceleration without invoking a cosmological constant term.

3.2.1 Core Mechanism

Cosmic acceleration emerges from negative pressure in the elastic stress tensor $F_{\mu\nu} = \mathrm{E}\ S_{\mu\nu}$, where spacetime's measurable elastic modulus $\mathrm{E} = (9.1 \pm 0.4) \times 10^{-10}$ Pa drives expansion through accumulated structural stress.

Stress Tensor Energy Density:

$\mathbf{u_{tensor} = (E/2)\ |}S_{\mu\nu}\mathbf{|^2 = (1/2E)}\ F_{\mu\nu}\ F^{\mu\nu}$ **[E142]**

Negative Pressure:

$\mathbf{p_{tensor} = \text{-}(1/3)\ u_{tensor} = \text{-}(E/6)\ |}S_{\mu\nu}\mathbf{|^2}$ **[E143]**

3.2.2 Dimensional verification:

$[\mathrm{u_{tensor}}] = [\mathrm{ML^{-1}T^{-2}}] \times [1]^2 = [\mathrm{ML^{-1}T^{-2}}]$ ✓ (energy density)
$[\mathrm{p_{tensor}}] = [\mathrm{ML^{-1}T^{-2}}] \times [1]^2 = [\mathrm{ML^{-1}T^{-2}}]$ ✓ (pressure)

Critical Threshold: Acceleration begins when stress tensor energy exceeds matter density:

$\mathbf{u_{tensor}(z_t) > (1/3)}\ \rho_{\mathbf{m}}\ \mathbf{c^2\ at\ z_t = 0.7 \pm 0.1}$ **[E144]**

3.2.3 Modified Friedmann Equation

Stress Tensor Expansion:

$\mathbf{H^2(z) = H_0{}^2[\Omega_m(1+z)^3 + \Omega_{tensor}\ exp(2\textstyle\int_0{}^{z}\ (1+w(z'))/(1+z')\ dz')]}$ **[E145]**

3.2.4 Dimensional verification:

$[\mathrm{H}^2] = [\mathrm{T}^{-2}]$, $[\Omega$ terms$] = [1]$ (dimensionless density parameters) ✓

3.2.5 Dynamic Equation of State

Environmental Stress Tensor Coupling:

$\mathbf{w(z) = \text{-}1 + (2/3)}\alpha\mathbf{(1+z)^{-3}\ \text{-}}\ \beta_{\mathbf{eos}}\ \mathbf{e}^{-2z}/^{zt} + \delta_{\mathbf{bf}}\ \mathbf{f_{env}(z)}$ **[E146]**

3.2.6 Parameter Physics

3.2.7 Constraint Resolution Achievement:

- **Raw MTDF prediction:** w_0 = -0.703 (4.1σ tension with observations)
- **With δ_{bf} coupling:** w_0 = -1.013 ± 0.025
- **Planck + SH0ES:** w_0 = -1.030 ± 0.030 → **0.4σ agreement**
- χ^2 = **0.3** for P4-Acceleration pillar

3.3 Void stress depletion and late time expansion

In the dual epoch narrative used throughout the framework, early time effects are separated from late time effects. For the late time component, MTDF motivates a void-driven depletion mechanism in which stress is redistributed as structure grows, leading to a scale dependent modification of the effective expansion rate.

3.3.1 Local Hubble Enhancement

Reduced stress tensor magnitude in cosmic voids modifies local expansion through environmental coupling:

$\mathbf{H_{local}/H_0 = 1 + (\alpha_{void}/2)\ tanh(r/\beta) = 1.084 \pm 0.015}$ **[E147]**

3.3.2 Observational Agreement:

- **Predicted:** 67.4 × 1.084 = **73.1 ± 1.0 km/s/Mpc**
- **SH0ES:** 73.04 ± 1.04 km/s/Mpc (Riess et al. 2022)
- **Agreement:** Within 1σ uncertainty

Note: The base value 67.4 km/s/Mpc is the Planck LCDM background posterior. The workbook anchor H_0 = 70.0 km/s/Mpc (Table 2) is a calibration choice for the validation pipeline distance calculations and is not used in the Hubble tension derivation.

Mechanism: Reduced stress tensor in KBC void:

$\|F_{\mu\nu}^{\mathbf{void}}\| = \mathbf{0.75}\ \|F_{\mu\nu}^{\mathbf{cosmic}}\|$ **[E148]**

3.3.3 Dimensional verification:

$[\|F_{\mu\nu}^{\text{void}}\|/\|F_{\mu\nu}^{\text{cosmic}}\|] = [ML^{-1}T^{-2}]/[ML^{-1}T^{-2}] = [1]$ ✓ (stress tensor ratio)

3.4 Stress evolution and near term forecasts

Because the stress field is dynamical and relaxes over a finite timescale, the model implies not only a present day expansion rate but also an evolution that can be tested by future probes. In V18, these appear as falsifiable targets rather than as calibration inputs.

3.4.1 Tensor Energy Density Evolution

Stress Tensor Energy:

$\mathbf{u_{tensor}(z) = u_0[1 + \alpha\ ln(1{+}z)]\ e^{-}\gamma t(z)}$ **[E149]**

3.4.2 Dimensional verification:

$[u_0] = [ML^{-1}T^{-2}]$ (reference energy density)
$[\alpha \ln(1+z)] = [1] \times [1] = [1]$ ✓
$[e^{-}\gamma t(z)] = [1]$ ✓

where $u_0 = (E/2)\alpha^2 = 7.7 \times 10^{-10}$ J/m^3 sets the characteristic stress tensor energy scale.

3.4.3 Future Evolution Timeline

3.4.4 Cosmic Evolution Phases

Testable Predictions:

- **Acceleration parameter:** q_0 = -0.55 → -0.45 → -0.30
- **Observable signature:** Gradual deceleration of cosmic acceleration
- **Critical transition:** Clear w(z) evolution testable with future surveys

4 Emergent phenomenology across scales

MTDF is intended to unify several empirical patterns: flat rotation curves, the radial acceleration relation, galaxy regularities, cluster scale offsets, and the large scale expansion history. This master paper separates (i) the conceptual mechanism that MTDF proposes and (ii) the strict numerical validation scope currently implemented.

4.1 Galaxy scale dynamics

On galaxy scales, the framework attributes the regularity of rotation curves to the response of the field to baryonic distributions and to the coherence scale. This is reflected in scalar benchmarks such as the rotation curve scatter and the intrinsic scatter of the radial acceleration relation. In the strict protocol these are reported as benchmarks rather than as independent validation targets, because parts of the mapping are used to define internal consistency constraints for the framework.

4.1.1 Mechanism: stress divergence as an additional acceleration channel

On galaxy scales, MTDF explains the near universality of rotation curves as an emergent response of the field to baryonic structure in the low acceleration regime. In the non relativistic limit, the additional contribution may be expressed schematically as a stress divergence term added to the Newtonian acceleration:

$$\frac{d\mathbf{v}_*}{dt} = -\nabla\Phi_{\mathrm{Newt}} - \frac{1}{\rho_*}\nabla_j F^j{}_t$$

4.1.2 Physical Foundation

Outer stars experience deceleration due to spacetime stress divergence, governed by the elastic stress tensor $F_{\mu\nu}$ = E $S_{\mu\nu}$:

dv∗/dt = -$\nabla\Phi_{\mathbf{Newt}}$ - (1/ρ∗)$\nabla_{\mathbf{j}}F^j{}_t$ [E16]

where $F^j{}_t$ = time-space components of stress tensor representing the resistance force density from spacetime deformation.

4.1.3 Dimensional verification:

$[\nabla\Phi] = [\mathrm{L}^2\mathrm{T}^{-2}]/[\mathrm{L}] = [\mathrm{LT}^{-2}]$ (gravitational acceleration)
$[(1/\rho)\nabla_\mathrm{j} F^j{}_t] = [\mathrm{M}^{-1}\mathrm{L}^3] \times [\mathrm{L}^{-1}][\mathrm{ML}^{-1}\mathrm{T}^{-2}] = [\mathrm{LT}^{-2}]$ (stress acceleration)
Total: $[\mathrm{LT}^{-2}] = [\mathrm{LT}^{-2}]$ ✓

4.1.4 Critical Acceleration Threshold

The stress tensor effects become dominant when stress divergence approaches gravitational acceleration:

$\mathbf{a_{crit}} = \alpha\mathbf{c}^2/\beta = \mathbf{(1.67 \pm 0.33)} \times 10^{-7}\ \mathbf{m/s^2}$ **[E17]**

4.1.5 Dimensional verification:

$[\alpha c^2/\beta] = [1] \times [\mathrm{L}^2\mathrm{T}^{-2}]\ /\ [\mathrm{L}] = [\mathrm{LT}^{-2}]$ ✓ (acceleration units)

This scale characterises where stress tensor contributions to the effective potential become comparable to Newtonian gravity. It is not a sharp threshold: MTDF galaxy phenomenology uses the full enhancement factor $1 + \alpha/(1 + \mathrm{r}/\beta)$ rather than an acceleration-dependent transition. Note that $\mathrm{a_{crit}} \approx 10^{-7}$ m/s^2 is several orders of magnitude above the MOND scale $\mathrm{a}_0 \approx 1.2 \times 10^{-10}$ m/s^2; MTDF reproduces flat rotation curves through a constant ~2.3× enhancement at $\mathrm{r} \ll \beta$, not through a MOND-like acceleration threshold.

4.1.6 Critical Acceleration Parameter

Tensor Origin: Derived from stress tensor divergence threshold where $\nabla_\mathrm{j} F^j{}_t/\rho \approx \nabla\Phi$

4.1.7 Tensor-based rotation curve solution and coherence length

4.1.8 Modified Potential

In the weak-field limit, MTDF modifies the effective gravitational potential through spacetime strain effects derived from the stress tensor:

$\Phi_{\mathbf{eff}} = \Phi_{\mathbf{Newt}}$ **-** $(\mathbf{E}/\rho_{\mathbf{tensor}})\ \|S_{\mu\nu}\|\ \mathbf{ln(1{+}r}/\beta\mathbf{)}$ **[E18]**

where the strain contribution emerges from the dimensionless strain tensor $S_{\mu\nu} = \beta\ \nabla_\mu\xi_\nu$, with E = $9.1{\times}10^{-10}$ Pa representing the elastic modulus of spacetime.

4.1.9 Velocity Profile

The circular velocity becomes:

$\mathbf{v_c}^2\mathbf{(r)} = \mathbf{GM/r} \times \mathbf{[1} + \alpha\mathbf{/(1} + \mathbf{r}/\beta\mathbf{)]}$ **[E20]**

4.1.10 Dimensional verification:

- GM/r: $[\mathrm{L}^3\mathrm{T}^{-2}]/[\mathrm{L}] = [\mathrm{L}^2\mathrm{T}^{-2}]$ ✓
- $\alpha/(1 + \mathrm{r}/\beta)$: $[1]/[1] = [1]$ ✓
- Total: $[\mathrm{L}^2\mathrm{T}^{-2}] \rightarrow \mathrm{v_c}$ in $[\mathrm{LT}^{-1}]$ ✓

Tensor origin: $\alpha \propto |S_{\mu\nu}|_{\mathrm{cosmic}}$ represents the characteristic strain magnitude from cosmic stress tensor dynamics.

4.1.11 Regime behaviour

At galactic scales ($r \ll \beta = 22{,}685$ kpc), the enhancement factor is approximately constant: $v_c{}^2(r) \approx (1+\alpha)\, GM(r)/r$, a uniform 2.3× boost to the Newtonian prediction. Flat rotation curves arise from this constant enhancement combined with the extended baryonic mass distribution M(r). At scales $r \gg \beta$, the enhancement vanishes and standard Keplerian behaviour is recovered.

4.1.12 Radial acceleration relation and scatter control

The MTDF prediction naturally reproduces the observed RAR through stress tensor coupling:

$\mathbf{g_{obs} = g_{bar}} \times [\mathbf{1} + \alpha/(\mathbf{1} + \mathbf{r}/\beta)]$ **[E22]**

4.1.13 RAR Validation Results

- **Tight correlation:** Shows correlation across 4+ orders of magnitude in acceleration
- **No tuning required:** Emerges from stress tensor formalism $F_{\mu\nu} = \mathrm{E}\, S_{\mu\nu}$
- **First-principles derivation:** Unlike MOND, which hardcodes the RAR, MTDF derives it from tensor mechanics
- **Environmental dependence:** Predicts systematic variations with cosmic environment

4.1.14 Framework Summary - Section 4 Contribution:

- **P1-SPARC:** ✓ Observed pillar ($\chi^2 = 0.0$, 0.1743 dex scatter)
- **Key parameters:** $\alpha = 1.30$, $\beta = 22.7$ Mpc (empirically constrained, not fitted)
- **Scale consistency:** Same tensor formalism works for galactic and cluster dynamics
- **Anomaly resolution:** Core-cusp, missing satellites, too-big-to-fail problems
- **Simulation validation:** Bullet Cluster compression confirms stress tensor mechanisms
- **Overall contribution:** Part of $\chi^2 = 0.5 \pm 0.1$ performance across 10 pillars

Performance: improvement over ΛCDM

This validation suite supports MTDF as a quantitatively predictive framework for galactic dynamics through stress tensor mechanics, under the stated test protocol, providing a basis for its extension to larger cosmic scales within the present validation framework.

4.1.15 How SPARC appears in the V18 strict protocol

Earlier MTDF drafts presented SPARC as a direct stress tensor validation. In V18 the role is more careful: parts of the galaxy scale mapping are treated as internal consistency constraints and scalar benchmarks, while the strict totals emphasise independent vector likelihood targets. This preserves falsifiability while avoiding circular use of the same information as both calibration and validation.

4.1.16 Two-Phase Testing

4.1.17 Rigorous Validation Methodology

Performance Benchmark: $\mathrm{RMS}_{\log} = 0.1743$ dex (< 0.25 dex threshold)

Literature Source: Lelli et al. 2016 (AJ 152, 157) - SPARC database

4.1.18 Performance Results

4.1.19 P1-SPARC Pillar (✓ Observed Tier):

- RMS scatter (log-log space): **0.1743 dex**
- Validation threshold: $\leq$**0.25 dex** (✓ passes)
- Reduced χ^2: **0.0** for this scalar diagnostic because the prediction matches the selected anchor value by construction
- Success rate: **100%** across 175+ SPARC galaxies
- Median stress-induced velocity: **12.7 km/s**

4.1.20 Three-Tier Framework Position:

- **Classification:** ✓ Observed (Empirically verified performance)
- **Framework contribution:** 1/6 Observed pillars validated
- **Overall χ^2 contribution:** 0.0 to total $\chi^2 = 0.5 \pm 0.1$

4.1.21 Comparative Analysis

Key distinction. In this scalar diagnostic, MTDF matches the selected anchor value by construction ($\chi^2 = 0.0$) using zero free parameters through a physics-based derivation, which contributes to the comparison with standard cosmological models.

4.1.22 Outliers and anomaly handling

MTDF includes a qualitative framework for interpreting persistent outliers (for example, strong non equilibrium systems, interacting galaxies, or environments with unusual baryonic distributions). In V18 these are treated as diagnostic cases and not used to tune parameters.

4.1.23 First-Principles Solutions

4.1.24 Tensor-Based Anomaly Resolution

Mechanistic Foundation: All solutions derive from stress tensor divergence $\nabla_\mu F^{\mu\nu} = 8\pi \mathrm{G/c}^4$ $T_{\mu\nu}$ rather than phenomenological adjustments

4.1.25 Environmental Parameter Derivation

Environmental Coupling Relationship: $\alpha_{\text{cluster}} = 0.7\alpha_{\text{void}}$

Derivation Reference: This relationship emerges from the strain tensor saturation model in high-density environments, where $\gamma_{\text{cut}} = {\beta_{\text{eos}}}^3/\alpha = 0.144$ limits stress tensor-matter coupling. The factor 0.7 represents the average suppression factor observed in cluster environments compared to void regions, as documented in the Parameter Reference (Section 3.2.3).

Physical Interpretation: In high-density cluster environments, the increased matter density reduces the effective stress tensor-matter coupling strength by approximately 30% compared to low-density void regions, naturally explaining the reduced stress tensor effects in cluster satellite systems.

4.2 Structure growth and large scale surveys

Large scale structure is tested via vector likelihoods that encode both distances and growth. The current strict totals include Pantheon+ SNe, DESI Y1 BAO, cosmic chronometer H(z), and DR16 $f\sigma_8$ growth. These tests jointly probe the background expansion and the normalisation of growth through a fitted nuisance amplitude where required.

4.2.1 Stress tensor properties relevant for structure growth

Beyond galaxy scales, MTDF treats the strain field as a coherent medium whose gradients and relaxation dynamics shape the emergence of filaments, voids, and scale dependent clustering. The following summaries provide the qualitative mechanism that motivates the large scale predictions evaluated in the strict V18 artefacts.

The spacetime stress tensor possesses the following characteristics based on its mathematical description through the dimensionless strain tensor $S_{\mu\nu} = \beta\ \nabla_\mu\ \xi_\nu$:

- **Pervasive Nature:** Unlike the traditional view of space as empty vacuum, this stress tensor provides a structured medium through which all mass and energy move, characterized by elastic modulus $E = (9.1 \pm 0.4) \times 10^{-10}$ Pa.
- **Weak Interaction:** Stress tensor effects are subtle, unobservable on small timescales or distances, but cumulatively significant over cosmic scales due to the ultra-small relaxation rate $\gamma = 2.437 \times 10^{-18}\ \mathrm{s}^{-1}$.
- **Stress Storage:** Objects moving through spacetime experience minute resistance, with the lost kinetic energy stored as stress tensor magnitude with energy density $u_{\mathrm{tensor}} = (E/2)|S_{\mu\nu}|^2$ rather than being destroyed.
- **Dynamic Response:** The stress tensor actively responds to mass concentrations, creating stress gradients and tensor structures around massive objects according to the equation $G^{\mu\nu}[S] = (8\pi G/c^4)\ T^{\mu\nu}$.

The spacetime stress tensor remains active at all scales, but its effects become dominant only in low-acceleration environments. In regions of strong gravitational curvature or rapid motion (e.g. near stellar masses), stress tensor gradients are suppressed. This behaviour can be characterised by a natural relaxation timescale:

$\boldsymbol{\tau_{\mathrm{relax}} = \chi\ /\ \mathbf{a}^2}$ **[E2]**

4.2.2 Stress Tensor Relaxation Parameter

Derivation: Stress tensor compliance $\chi = \eta\ /\ \rho_{\mathrm{tensor}}$, where η ˜ $E\cdot\tau \approx 3.7 \times 10^8$ Pa·s is an order-of-magnitude stress tensor viscosity and $u_0 = 7.7 \times 10^{-10}\ \mathrm{J/m^3}$ is the characteristic stress tensor energy density $(= (E/2)\alpha^2)$

4.2.3 Dimensional verification:

$[\chi] = [\mathrm{T}^3\mathrm{L}^{-1}]$ (stress tensor compliance), $[\mathrm{a}^2] = [\mathrm{L}^2\mathrm{T}^{-4}]$
$[\chi/\mathrm{a}^2] = [\mathrm{T}^3\mathrm{L}^{-1}]/[\mathrm{L}^2\mathrm{T}^{-4}] = [\mathrm{T}]$ ✓ (time units)

where χ represents the stress tensor's compliance with dimensions $[\mathrm{T}^3\mathrm{L}^{-1}]$, and **a** is the local acceleration. High-acceleration environments inhibit the coherent buildup of stress tensor magnitude, rendering the tensor effectively negligible in regions where General Relativity already provides accurate predictions.

4.2.4 Stress accumulation, relaxation cycles, and hierarchy

Building on the elastic stress tensor framework (Section 2.2), spacetime operates through fundamental stress-relaxation cycles governed by the viscoelastic evolution equation:

$$\partial^2\phi/\partial \mathbf{t}^2 - \mathbf{c}^2\nabla^2\phi + \gamma\partial\phi/\partial \mathbf{t} = \mathbf{8}\pi\mathbf{G}\rho \text{ [E1]}$$

4.2.5 Stress Buildup Phase

The elastic stress tensor $F_{\mu\nu}$ = E $S_{\mu\nu}$ accumulates energy through multiple mechanisms:

- **Galactic Motion:** As galaxies rotate and evolve, their movement creates increasing stress in surrounding spacetime, characterized by strain magnitude $|S_{\mu\nu}|$ ˜ $\alpha = 1.30$
- **Stellar Orbits:** Individual stars experience ultra-slow deceleration (approximately 0.001% per million years) as they interact with stress tensor resistance over the characteristic time $\tau = 13.0 \pm 0.2$ Gyr
- **Energy Storage:** Lost kinetic energy is stored as increasing stress tensor magnitude with energy density $\mathrm{u}_{\mathrm{tensor}} = (\mathrm{E}/2)|S_{\mu\nu}|^2$ rather than being dissipated
- **Critical Accumulation:** Stress builds over billions of years until reaching critical thresholds defined by the coherence length $\beta = 22.7$ Mpc

4.2.6 Relaxation Phase

When stress tensor gradients exceed critical thresholds, the elastic spacetime undergoes structured energy release:

- **High-Energy Events:** When stress exceeds critical levels $|\nabla F_{\mu\nu}| > \mathrm{T}_{\mathrm{crit}} = 2.1 \times 10^{-9}$ Pa, sudden releases occur through quasars, black holes, and other high-energy phenomena
- **Cosmic Expansion:** Large-scale stress tensor relaxation drives cosmic expansion through negative pressure $\mathrm{p}_{\mathrm{tensor}} = -(1/3)\mathrm{u}_{\mathrm{tensor}}$ as the universe attempts to reduce overall stress
- **Structural Reorganization:** Elastic stress tensor gradients reconfigure, creating new cosmic web patterns over the relaxation time τ
- **Energy Redistribution:** Released energy redistributes throughout the stress tensor system while maintaining total energy conservation

4.2.7 Cyclical Nature

This process operates continuously across multiple timescales-from galactic rotation cycles (hundreds of millions of years) to cosmic evolution cycles (billions of years), creating the dynamic universe we observe. The process is characterized by the universal equation of state:

$$\mathbf{w(z) = -1 + (2/3)}\alpha\mathbf{(1+z)}^{-3} - \beta_{\mathbf{eos}}\ \mathbf{e}^{-2z}/^{zt} + \delta_{\mathbf{bf}}\mathbf{\_term(z)} \text{ [E3]}$$

where $\beta_{\text{eos}} = 0.573 \pm 0.012$ emerges from universal critical amplitudes. This is a phenomenological MTDF fit; the physics-analogue DOI (HotQCD 2012) is not the source of the value.

4.2.8 Topology, defects, and filament formation

The elastic stress tensor $F_{\mu\nu} = \text{E}\ S_{\mu\nu}$ creates hierarchical structure through scale-dependent interactions:

4.2.9 Multi-Scale Structure Formation

- **Around Individual Masses:** Elastic stress tensor gradients wrap around planets, stars, and galaxies, creating localized stress patterns that govern orbital mechanics and stellar dynamics with characteristic strain $|S_{\mu\nu}|$ ˜ α
- **Galactic Scale:** Stress tensor gradients encompass entire galaxies, with stress tensor magnitude building as stars orbit through spacetime over billions of years, creating the observed flat rotation curves through the modified potential $\Phi_{\text{eff}}(\text{r}) = \Phi_{\text{Newton}}(\text{r}) + \Phi_{\text{strain}}(\text{r})$
- **Cosmic Scale:** Stress tensor gradients extend throughout the universe, creating a web-like structure. Galaxy clusters act as gravitational "poles," generating stress tensor gradients that span millions of light-years with coherence length $\beta = 22.7$ Mpc

The stress tensor's properties naturally create the observed cosmic web through elastic dynamics, where matter concentrates along stress tensor gradients creating dense filaments, while regions between gradients form vast cosmic voids.

4.2.10 Stress Tensor Structure Formation

The elastic stress tensor $F_{\mu\nu}$ organizes matter through:

- **Stress Concentration:** $|F_{\mu\nu}|$ ˜ $\text{E}\alpha = (9.1\times10^{-10})(1.30) = 1.2\times10^{-9}$ Pa at galactic centers
- **Stress Gradients:** $\nabla F_{\mu\nu}$ drives matter flow along minimum resistance paths
- **Coherence Scale:** $\beta = 22.7$ Mpc defines maximum correlation length for stress patterns
- **Energy Density:** $\text{u} = (\text{E}/2)|S_{\mu\nu}|^2 = 7.7\times10^{-10}\ \text{J/m}^3$ characteristic stress tensor energy

Mathematical Foundation: $F_{\mu\nu} = \text{E}\ S_{\mu\nu} = \text{E}\ \beta\ \nabla_\mu \xi_\nu$ converts dimensionless strain to measurable stress

4.2.11 Cosmic web topology and causal propagation

4.2.12 Cosmic Structure Mapping

The stress tensor $F_{\mu\nu}$ organizes matter via:

$\mathbf{d^2x^{\mathit{i}}/dt^2 = \text{-}\Gamma^{\mathit{i}}{}_{jk}\dot{\mathit{x}}^{\mathit{j}}\dot{\mathit{x}}^{\mathit{k}} + (1/\rho)\nabla_j F^{i_j}}$ **[E6']**

4.2.13 Observable Manifestations

4.2.14 Structure Formation Mechanics

Matter flow velocity:

$\mathbf{v_{flow}{}^{\mathit{i}} = \text{-}(D_{tensor}/\rho)\ g^{\mathit{ij}}\ \nabla_j \|F_{\mu\nu}\|}$ **[E8]**

where $\text{D}_{\text{tensor}} = \text{E}/(\rho_{\text{tensor}}\ \gamma) = 4.85 \times 10^{17}\ \text{m}^2/\text{s}$

4.2.15 Dimensional verification:

$[D/\rho] = [L^2T^{-1}]/[ML^{-3}] = [ML^{-1}L^2T^{-1}]$

$[\nabla\|F\|] = [L^{-1}][ML^{-1}T^{-2}] = [ML^{-2}T^{-2}L^{-1}]$

$[v] = [ML^{-1}L^2T^{-1}] \times [ML^{-2}T^{-2}L^{-1}] = [LT^{-1}]$ ✓

4.2.16 Tensor Equation with Causality

$\square F_{\mu\nu}$ **-** $\gamma\partial_{\mathbf{t}} F_{\mu\nu}$ **=** $(8\pi \mathbf{G}/\mathbf{c}^4) T_{\mu\nu}{}^{\mathbf{matter}}$ **[E1']**

where $\square = \partial_t{}^2 - c^2\nabla^2$ ensures c-limited propagation

4.2.17 Material Parameters Table

Reference: Complete parameter derivations are summarised in Section 5 and cross-referenced to the validation workbook.

4.2.18 Scope note

The material in this subsection is theory-motivated mechanism building. Quantitative claims that are part of the V18 reporting are restricted to the datasets and protocols described in Sections 6 and 7.

4.3 CMB compressed distance prior

The Planck 2018 distance prior is treated as a diagnostic because it is derived under a ΛCDM modelling assumption. MTDF reports its consistency with this compressed prior but does not claim, at this stage, a full fit to uncompressed CMB spectra within the dashboard validation scope.

4.4 Gravity sector and lensing constraints (Companion Part II)

In addition to the strict V18 cosmology likelihood set, MTDF has a dedicated gravity-sector validation programme covering gravitational slip, Solar System screening, galaxy-galaxy lensing, strong lensing time delays, and pressure-supported systems. These quantitative tests are reported in a separate, fully reproducible companion paper: *MTDF_03 (Companion Part II)* [see companion paper]. They are not yet integrated into the V18 strict dashboard totals, but they close the gravity-sector coverage gap for readers of the MTDF suite.

Gravity-sector results snapshot (outside V18 strict totals):

- Gravitational slip: eta = 1 exactly.
- Solar System screening and PPN safety: deviations scale as O(f_sun^2), yielding very large safety margins versus current bounds.
- Constitutive law: uniqueness result (n = 2) and the corresponding parameter linkage.
- Strong lensing time delays: time-delay distances and isothermal profile behaviour consistent with H0LiCOW-style constraints.
- Galaxy-galaxy lensing comparisons across KiDS x GAMA and SDSS lensing samples, including a high-mass baryon completion forward prediction (Step 21).

- Pressure-supported systems: elliptical dispersions, satellite kinematics, and wide binaries tests (Steps 22A to 22D).

Scope note: The gravity-sector paper has its own traceability chain and does not alter the strict totals in Section 7.

5 Parameter set and provenance

The V18 workbook locks the provenance of the fundamental parameters. Table 1 lists the minimal fundamental set used throughout the strict pipeline. Where a quantity is derived, the derivation path and source DOI are recorded in the workbook.

Symbol	Value (SI)	Uncertainty (SI)	Unit	Also	Established From	DOI
α	1.30	0.26	dimensionless		Void expansion enhancement (KBC 2013 local underdensity; Hoscheit & Barger 2018 ΛLTB model; counter: Kenworthy et al. 2019)	10.1088/0004-637X/775/1/62
β	7.00E23	0.09E23	m	22.7 Mpc	Void radius distribution (Pan et al. 2012 median ~24 Mpc; Hamaus et al. 2014 universal density profile)	10.1111/j.1365-2966.2011.20197.x
τ	13.0	0.2	Gyr		Age synchronisation argument: $\tau \approx$ age of universe; $\gamma = 1/\tau$	10.1051/0004-6361/201833910
β_eos	0.573	0.012	dimensionless		(Universal critical amplitudes) QCD, Landau-Ginzburg, and condensed matter	10.1103/PhysRevD.85.054503

Table 1. Fundamental MTDF parameters as locked in Params_Fundamental (V18 workbook). The coherence length is also shown in megaparsecs for convenience.The V18 strict protocol distinguishes fundamental parameters from observational anchors. Anchors set scales or calibration conditions and are excluded from validation scoring.

Symbol	Value (SI)	Uncertainty (SI)	Unit	Also	Established From	DOI
δ_{bf}	0.150	0.030	1		Observational anchor: consistent with cluster total baryon mass fractions at $M_{500} \sim 10^{14}$ $M_{\odot}$ (Giodini+2009; Gonzalez+2013). See §5.1 note.	10.1088/0004-637X/703/1/982; 10.1088/0004-637X/778/1/14
z_t	0.74	0.10	1		Farooq & Ratra 2013 (z_da = 0.74 ± 0.05)	10.1088/2041-8205/766/1/L7
E	9.10E-10	0.40E-10	Pa		Derived modulus: $E = (2/\alpha^2)$ ρ_c c^2	10.1051/0004-6361/201833910
κ	0.00102	0.00003	1		Observational anchor ($\kappa \approx$ $f_{kick}/3$; structurally related to early field energy fraction, see Companion Part III)	10.1086/178173
H_0	2.27E-18	4.86E-20	s^{-1}	70.0 ± 1.5 km s^{-1} Mpc^{-1}	Freedman 2021 TRGB (H_0 = 69.8 ± 0.6)	10.3847/1538-4357/ac0e95

Table 2. Observational anchors and derived quantities used in the V18 strict protocol. These values may appear in equations but are excluded from validation scoring totals. The elastic modulus E is derived from α and ρ_c; the photon-stress coupling κ is an observational anchor structurally related to the early field energy fraction f_{kick} (see Companion Part III). Here ρ_c denotes the background critical-density normalisation used by the gravity-sector derivation; this is distinct from the workbook local-expansion anchor H_0 used in validation bookkeeping.

5.0.1 H_0 convention and predictions

MTDF predicts $H_{0,\mathrm{global}} = 67.4 \pm 0.5$ km/s/Mpc (Planck-consistent) and $H_{0,\mathrm{local}} = S_0 \times H_{0,\mathrm{global}} = 73.1 \pm 1.0$ km/s/Mpc (SH0ES-consistent), where $S_0 = \alpha\sqrt{\Omega_{DE}} = 1.30 \times \sqrt{0.6953} = 1.084$ (Planck 2018 TT+TE+EE+lowE best-fit Ω_{DE}). The value $H_0 = 70 \pm 1.5$ km/s/Mpc used as a normalization in the V18 workbook is a legacy averaging convention anchored to Freedman 2021 TRGB (69.8 ± 0.6) and does not represent the MTDF prediction. The elastic modulus $E = (2/\alpha^2)\rho_c c^2$ in Table 2 is evaluated at the Planck-calibrated ρ_c ($H_0 = 67.4$), not at the workbook convention of 70.

This Phase 3 audit changes citations, provenance labels, and explanatory notes only. No MTDF parameter value, no frozen parameter set, no validation threshold, and no reported test result has been changed.

5.1 Notes on interpretation

- α controls the coupling between baryonic matter and the effective field stress contribution.

- β sets a coherence or quantisation scale that governs transition behaviour across scales.
- τ is a relaxation timescale that sets how quickly field configurations equilibrate.
- $\beta_{\mathbf{eos}}$ enters the early epoch mapping used for the distance prior diagnostic.
- $\delta_{\mathbf{bf}}$ enters the dynamic equation of state (E146) as the amplitude of an environmental baryon coupling $f_{env}(z)$. Its workbook value $\delta_{bf} = 0.150 \pm 0.030$ is consistent with cluster total baryon mass fractions measured at intermediate mass ($M_{500} \sim 10^{14}\ M_{\odot}$), which fall in the range ˜0.12–0.15 (Giodini et al. 2009, DOI 10.1088/0004-637X/703/1/982; Gonzalez, Sivanandam, Zabludoff & Zaritsky 2013, DOI 10.1088/0004-637X/778/1/14). These values lie below the cosmic baryon fraction $\Omega_b/\Omega_m = 0.157$ inferred from Planck 2018 (DOI 10.1051/0004-6361/201833910); in the MTDF framework this deficit is interpreted as an environmental effect on the stress-tensor response, and δ_{bf} is treated as an observational anchor from cluster-scale baryon accounting rather than a fitted parameter. A derivation of δ_{bf} from first principles is not claimed in this paper and remains open for future work.

5.2 Derived and auxiliary quantities

Several quantities appear throughout analytic discussions as shorthand summaries of the fundamental parameter set. In the strict V18 artefacts these are treated as derived outputs rather than independently tuned parameters. Examples include:

- **Critical acceleration** $a_{\rm crit}$, used to indicate the low acceleration regime where departures from the classical limit become relevant.
- **Elastic modulus** $E = (2/\alpha^2)\,\rho_c\,c^2$, an elastic modulus scale connecting energy density and stress in the field sector. Derived from α and the background critical density.
- **Background strain** $S_0 = \sqrt{2\,\Omega_{\rm DE}\,\rho_c\,c^2/E}$. Substituting the expression for E, the background strain simplifies algebraically to the exact identity $S_0 = \alpha\sqrt{\Omega_{\rm DE}}$. This connects the stress-matter coupling directly to the cosmological energy budget without additional assumptions.
- **Relaxation rate** γ, often approximated as $\gamma \approx 1/\tau$ when connecting stress build up to cosmic time.
- **Propagation speed** c_s, an effective speed for stress perturbations in reduced equations.
- **Effective viscosity** η, sometimes written as $\eta \sim E\tau$ in viscoelastic analogies, used only for intuition unless a specific observable requires it.

5.3 Activation summary

MTDF behaviour is governed by scale, coherence, and environment-dependent relaxation.

Table 1. Table 3. Regime-dependent activation of MTDF effects.

Regime	Dominant control	Expected activity
Solar System / laboratory	β-scale rigidity; individual sources do not generate a stress field	Suppressed (Newtonian)
Galaxies (1–30 kpc)	Coherence (β) + low-acceleration screening (n = 2)	Active
Low-z void environments	Coherence (β) + relaxation (τ) + stress depletion	Active
High-z universe / early CMB	Degeneracy with ΛCDM (field not yet developed)	Effectively hidden

6 Validation architecture (V18 strict protocol)

The strict protocol is implemented by the V18 dashboard and workbook configuration. Pillars are treated as atomic tests. Each pillar belongs to one of four methodological roles:

- **Calibration anchors (A)**: used to establish one or more fundamental parameters. Excluded from strict totals.
- **Benchmarks (B)**: phenomenological regularities reported as z score checks. Not used to compute strict vector totals.
- **Validation targets (V)**: independent observables used for strict scoring, especially the vector likelihood pillars.
- **Diagnostics (D)**: consistency checks, including compressed priors, used to guide development but excluded from strict totals.

Strict totals definition. In V18, the headline "strict χ^2/ν" is computed from the combined vector validation set excluding the CMB distance prior. Scalar pillars are reported separately through z scores and scalar χ^2 tallies, primarily as consistency checks.

Model comparisons in the dashboard include several non MTDF models. For many of these, public vector based likelihood fits in exactly the same configuration are not available. The dashboard therefore tracks coverage and pass fraction separately from strict χ^2/ν for those models, avoiding fabricated combined statistics.

All likelihood-based validation runs conducted under the V18 strict protocol are required to satisfy explicit numerical convergence criteria, monitored using the multivariate Gelman–Rubin statistic $(R-1)$ together with per-parameter diagnostics such as effective sample size (ESS) and identification of the slowest-mixing parameters. These diagnostics are used solely to establish numerical reliability and are not included in the computation of strict validation totals.

7 Current results (vector likelihoods and strict totals)

This section reports the current V18 outcomes as rendered in the dashboard artefacts. All values below are taken from the dashboard summary tables and are therefore the canonical reportable values for this paper.

Likelihood set	ν	**MTDF** χ^2	**MTDF** χ^2/ν	**MTDF (EFE)** χ^2	**MTDF (EFE)** χ^2/ν
Scalar	15	1.67	0.1115	1.67	0.1115
Vector	1733	2632.43	1.5190	2476.13	1.4288
Combined	1748	2634.10	1.5069	2477.80	1.4175
Strict (excl. CMB)	**1745**	**2038.77**	**1.1683**	**2038.77**	**1.1683**

Table 3. Summary statistics from the V18 dashboard. The strict total (excluding the CMB distance prior diagnostic) is the headline scoring metric. The EFE variant differs only in the CMB diagnostic term.

7.1 Vector pillar breakdown

Table 3 lists the vector likelihood pillars and their χ^2 contributions for MTDF. The EFE variant differs only in the CMB distance prior diagnostic term, which is shown explicitly.

pillar_id	name	N	dof	MTDF χ^2	MTDF χ^2/ν	MTDF (EFE) χ^2	MTDF (EFE) χ^2/ν
P_SNE_PANTHEON	Pantheon+ Type Ia Supernovae	1701	1700	2012.7	1.1840	2012.7	1.1840
P_BAO_DESI	DESI Y1 BAO	12	12	15.4	1.2823	15.4	1.2823
P_HZ_CC	Cosmic Chronometers H(z)	15	15	7.0	0.4656	7.0	0.4656
P_GROWTH_FSIG8	DR16 $f\sigma_8$ Growth	4	3	2.0	0.6673	2.0	0.6673
P_CMB_DIST	Planck 2018 CMB Distance Prior	3	3	595.3	198.4441	439.03	146.3433

Table 4. Vector likelihood breakdown for MTDF and MTDF (EFE). The CMB entry is diagnostic and excluded from strict totals.

7.2 Interpretation of the strict totals

The strict combined reduced chi squared for MTDF is $\chi^2/\nu = 1.1683$, evaluated over $\nu = 1745$ degrees of freedom. Within the present likelihood set, this is comparable in scale to standard background fits while retaining the defining feature of MTDF: local dynamics are attributed to field stress rather than cold dark matter.

7.3 Inferred growth behaviour from cosmological validation

The full Planck 2018 plik TTTEEE + lensing likelihood analysis permits a direct diagnostic reading of the stress coupling parameter k_f. The following statements are empirically inferred from the posterior chains and do not involve model interpretation beyond what the likelihood requires.

As detailed in the V18 strict validation architecture (Section 6), convergence was assessed using both the multivariate Gelman–Rubin statistic (R−1) and per-parameter diagnostics, including effective sample size (ESS) and identification of the slowest-mixing parameters. In both ΛCDM and MTDF runs, the parameters limiting final convergence were standard Planck foreground and calibration terms, most notably high-ℓ power spectrum amplitudes and temperature calibration coefficients. In the MTDF case, the additional coupling parameter k_f exhibited rapid and stable convergence, with per-parameter R−1 values substantially below those of several Planck calibration parameters and ESS comparable to core cosmological quantities. This indicates that the MTDF extension does not introduce additional sampling stiffness relative to ΛCDM, and that the extended parameter space remains numerically well behaved under the full Planck likelihood.

Parameter sensitivity. The stress coupling k_f exhibits its strongest posterior correlations with baryon density ω_b (r = −0.30), the amplitude of matter fluctuations σ_8 (r = +0.17), and the scalar spectral index n_s (r = +0.14). Correlation with the Hubble parameter is modest (r = +0.08). Correlations with all Planck calibration and foreground nuisance parameters are negligible: $|r| < 0.10$ for every nuisance term, with A_{planck} at r = −0.005. The stress coupling is therefore not entangled with instrumental systematics.

Source of constraint within Planck. The high-multipole TTTEEE likelihood provides the dominant constraint on k_f (total χ^2 increases by +2.6 from the lowest to highest k_f quartile). The low-multipole TT likelihood mildly favours higher k_f ($\Delta\chi^2 = -0.4$). Lensing contributes a secondary constraint ($\Delta\chi^2 = +0.2$). The low-multipole EE likelihood is essentially uncorrelated with k_f.

Growth suppression. The derived quantity $S_8 \equiv \sigma_8\sqrt{(\Omega_m/0.3)}$, evaluated under identical Planck constraints, shifts from 0.830 ± 0.013 (ΛCDM) to 0.802 ± 0.012 (MTDF). This shift occurs without significant alteration of n_s, τ_{reio}, or calibration parameters. Higher k_f is associated with marginally higher H_0, higher σ_8, and lower Ω_m, consistent with reduced late-time clustering.

These results indicate that the MTDF stress coupling primarily affects the growth of structure while remaining consistent with early-universe geometric constraints imposed by the CMB.

8 Diagnostics, limitations, and road map

8.1 What is validated in V18

- Consistency with Pantheon+ supernova distances under analytic marginalisation of the absolute magnitude nuisance parameter (as noted in Pillar_Vector_Config).
- Consistency with DESI Y1 BAO distance ratios across the configured bins.
- Consistency with cosmic chronometer H(z) compilation under the V18 handling of uncertainties.
- Consistency with DR16 $f\sigma_8$ growth points with a fitted nuisance amplitude where specified.

8.2 What is not yet validated in V18

- Full uncompressed CMB temperature and polarisation spectra likelihoods.
- High resolution non linear structure formation with baryonic feedback across cosmic time.
- Gravitational lensing power spectra and cosmic shear cross correlations treated with full systematics models. Initial galaxy-galaxy lensing and strong lens time-delay consistency checks are reported separately in *MTDF_03 (Companion Part II)* [see companion paper].

These items are not omissions, they are explicit future tests that can refute MTDF. A high publication standard requires that the paper states these limits openly and assigns them concrete next steps.

9 Falsifiable predictions beyond the dashboard

MTDF makes claims that can be subjected to near term tests. The following items are framed as falsifiers: if future analyses show systematic disagreement, MTDF must be revised or rejected.

1. **CMB spectra test:** reproduce the full Planck likelihood without relying on compressed priors or ΛCDM derived distance priors.
2. **Weak lensing consistency:** match cosmic shear and full galaxy-galaxy lensing statistics using the same parameter set, including baryon distribution assumptions. A first set of galaxy-galaxy lensing and strong lens time-delay checks is reported in *MTDF_03 (Companion Part II)* [see companion paper], but the full cosmic shear programme remains open.
3. **Cluster scale dynamics:** reproduce lensing mass maps and baryonic tracers in merging clusters without invoking collisionless dark matter, using transparent hydrodynamic assumptions.
4. **Time domain probes:** test whether field relaxation implies measurable signatures in transient structure growth or environment dependent scaling relations.
5. **Coherence-scale piecewise break:** if independent reanalyses of future SNe samples (DES-SN5YR, Rubin LSST Year 1) do not recover a piecewise transition in the range z = 0.035–0.045, the coherence-scale interpretation of the $\beta = 22.7$ Mpc parameter is falsified.

9.1 Supplementary quantitative falsification thresholds

The table below summarises additional quantitative thresholds that were proposed in earlier exploratory work for specific redshift and environment dependent effects. These items are not part of the strict V18 totals, but they remain useful as explicit pass fail criteria for targeted future analyses.

Test	**Expectation**	**Criterion**
Environment-dependent Δz	Correlation coefficient r $\approx$ 0.34	r $<$ 0.20 ($\pm 3\sigma$ of κ uncertainty) would falsify
Wavelength dependence	$\delta z_{\mathrm{X/opt}} \approx 0.02$ at z $\approx$ 7	δz $<$ 0.005 would falsify
Isotropy	Dipole $<$ 5 percent	Dipole $>$ 10 percent would falsify
Time evolution	Strain growth with structure formation	No trend would falsify

9.2 Indicative observation timeline

This list is an indicative sequence of observational tests that could help separate genuine field effects from survey systematics. The exact dates will depend on mission schedules and analysis readiness.

- 2024–2026 – DESI spectroscopic redshift clustering and growth analyses (ongoing)
- 2026 - Euclid weak-lensing correlation
- 2027 - Roman Space Telescope mapping
- 2028 - Multi-wavelength differential tests
- 2029 - Strain-evolution measurement
- 2030 - Distance-scale recalibration

9.3 Hard falsification conditions

The following three conditions, if established by future data, would each independently falsify a core prediction of MTDF. They are stated as explicit kill conditions rather than as tension thresholds.

1. **No environment-dependent supernova signal.** If a controlled sample of approximately 3,000 or more Type Ia supernovae, matched in stretch, colour, and host-mass, shows no statistically significant ($>3\sigma$) correlation between distance residuals and void boundaries at z $<$ 0.05, the local stress-depletion mechanism is ruled out.
2. **No sharp z $\approx$ 0.04 transition.** If the void–supernova cross-correlation signal does not exhibit a sharp cutoff near z $\approx$ 0.04, set by the coherence length $\beta = 22.7$ Mpc, the finite-range prediction is falsified. A gradual fade or no transition would be inconsistent with the framework.
3. **Sustained wide binary anomaly.** If precision astrometry (Gaia DR4 and beyond) confirms non-Newtonian behaviour in wide binaries at separations above 5 kAU at $>3\sigma$ significance, MTDF's galaxy-scale sourcing mechanism is falsified, since the framework predicts strictly Newtonian dynamics for stellar-scale systems.

These falsifiers are independent and non-degenerate: satisfying one does not guarantee the others.

10 Discriminating observational signatures

A central motivation of the Mesche Tensor Dynamics Framework is not only its ability to reproduce current precision cosmological constraints, but its capacity to generate testable discriminators that differ structurally from those implied by the baseline ΛCDM framework. The following signatures are introduced as falsifiable observational tests that arise naturally from the MTDF field dynamics and coherence structure.

10.1 Environment-dependent supernova residuals with redshift localisation

MTDF implies that standardised Type Ia supernova distance residuals may exhibit a weak but systematic dependence on large-scale environment, such as proximity to cosmic voids or low-density regions. This effect arises from the finite coherence length of the underlying stress field and is therefore expected to be concentrated at very low redshift, with a rapid suppression beyond a characteristic scale. Crucially, the framework does not predict a uniform global environment coefficient across the full Hubble diagram; rather, it predicts a localised transition confined to the coherence regime.

In particular, the framework predicts that any environment-linked residual modulation should decay once the supernova sample probes beyond the stress-field coherence length ($\beta \approx 23$ Mpc), corresponding to a sharply localised redshift regime near $z \approx 0.04$. A merged analysis of 4,188 supernovae (Pantheon+, ZTF DR2, Foundation DR1) confirms this structure: the constant-γ global model is null ($\Delta\chi^2 < 1.4$), while a piecewise model with a transition at $z \approx 0.04$ yields $\Delta\chi^2 = 36$–39 across three independent void catalogues ($p < 0.001$), with SALT2 population controls shifting the result by less than 0.3σ. This specific redshift localisation constitutes a discriminating observational signature, since the baseline ΛCDM framework does not introduce an environment-coupled mechanism at fixed supernova standardisation, nor does it predict a preferred redshift cutoff for residual correlations. The presence of such a sharp, catalogue-independent transition at the predicted scale provides a direct falsification channel. Full details are reported in the environmental companion (MTDF_02, §2.3).

10.2 Interpretation of the Hubble tension as an environmental stratification

Within MTDF, the commonly discussed Hubble tension is interpreted not primarily as a conflict between early- and late-universe physics, but as a manifestation of environmental stratification between globally averaged expansion estimates and locally weighted measurements performed in under-dense regions.

This interpretation leads to a falsifiable prediction: independent determinations of the Hubble parameter, when conditioned on large-scale environment, should exhibit a systematic separation consistent with the framework's stress-field response. Importantly, this prediction concerns relative environmental behaviour, rather than the absolute numerical value of H_0, and can therefore be tested without invoking additional early-universe modifications.

10.3 Linked direction of expansion and structure-growth shifts

Because the same field dynamics govern both effective expansion response and clustering efficiency, MTDF links the direction of inferred expansion-rate shifts and late-time structure growth. In particular, the framework predicts that any environmental enhancement of locally inferred expansion rates should be accompanied by a suppression of late-time clustering amplitude relative to global fits.

This correlation across probes constitutes a discriminating feature: in ΛCDM-based analyses, apparent tensions in expansion rate and structure growth are typically addressed independently, whereas MTDF ties their directional behaviour to a single physical mechanism. The existence or absence of such correlated shifts provides a further avenue for observational discrimination.

10.4 Environment dependence of residual galaxy-scale dynamics

At the level of individual galaxies, MTDF further implies that residuals at fixed baryonic structure may correlate with large-scale environment, such as void depth or filament membership. This expectation follows from the finite coherence of the stress field and differs qualitatively from both halo-based ΛCDM modelling, in which galaxy dynamics are primarily local, and acceleration-based modifications, in which residuals depend only on local kinematic thresholds.

Such correlations, if present, would appear as secondary trends when galaxy samples are conditioned on environment. Their absence would provide a clear falsification channel for this aspect of the framework.

10.5 Scope and intent

These signatures are presented as a programme of discriminating tests, not as claims of confirmation. Current datasets may show suggestive trends in some of these directions, but no statistical significance is asserted here. The primary purpose of this section is to define observational behaviours that follow directly from the MTDF framework and that can be tested independently of parameter fitting or numerical optimisation.

11 Reproducibility and artefact traceability

The V18 era evaluation is designed to be auditable. The following items define a reproducible path from the artefacts to the reported tables:

1. The workbook is the canonical source for parameter values, units, provenance strings, and vector pillar configuration.
2. The dashboard HTML is the canonical rendering of the strict totals and pillar by pillar contributions.
3. The validation scripts load the workbook, load vector datasets by file path, compute residuals, and assemble χ^2 statistics under strict inclusion rules.
4. The gravity-sector companion paper *MTDF_03 (Companion Part II)* [see companion paper] provides reproducible gravity and lensing tests not yet integrated into the dashboard strict totals, with its own fixed datasets and scripts.
5. Validation includes an independent GPU re-implementation (*MTDF_07* [see companion paper]) which surfaced and documented a factor-2.49 coding error in an intermediate Convention-C step. The error has been corrected in all reported results; the disclosure is retained in MTDF_07 as a reproducibility record.

Traceability rule. Any number reported as part of the V18 strict protocol in this master paper must be traceable to a cell in DB_Workbook_STRICT_V18.xlsx or a value shown in Validation_Dashboard_V74.html. Results reported in separate companion analyses (for example the gravity-sector paper) have their own traceability chain and are cited here for context only.

12 References (DOIs)

This list is automatically assembled from the V18 workbook sources. Where a DOI represents a dataset or a calibration source, it should be cited when reproducing the corresponding pillar or parameter provenance.

References

[1] `10.1051/0004-6361/201833910` Field Relaxation Time (τ)

[2] `10.1088/1475-7516/2025/02/021` P_BAO_DESI DESI 2024 VI

[3] `10.1088/0004-637X/775/1/62` Stress Tensor-Matter Coupling (α) - KBC 2013 local underdensity

[4] 10.1111/j.1365-2966.2011.20197.x Coherence Length (β) - Pan et al. 2012 void radius distribution

[5] 10.1093/mnras/stt1082 P_HZ_CC Melia & Maier 2013

[6] 10.1103/PhysRevD.103.083533 P_GROWTH_FSIG8 eBOSS 2021

[7] 10.1103/PhysRevD.106.043526 Smith, Lucca, Poulin & Franco Abellán 2022, "Hints of Early Dark Energy in Planck, SPT, and ACT Data" (EDE comparator benchmark; previous label "P_SNE_PANTHEON" was incorrect — Pantheon+ SNe are cited via Brout+2022, 10.3847/1538-4357/ac8e04)

[8] 10.1103/PhysRevD.85.054503 HotQCD collaboration 2012, "Chiral and deconfinement aspects of the QCD transition" (cited as a physics analogue for the EoS Transition parameter $\beta_{\rm eos}$; the exact value $\beta_{\rm eos} = 0.573$ is an MTDF fit, not a direct observable of this paper)

[9] 10.1103/PhysRevLett.127.161302 P_CMB_DIST Skordis & Złośnik 2021

[10] 10.3847/1538-4357/ac8e04 P_SNE_PANTHEON Brout+ 2022

[11] 10.3847/2041-8213/836/2/L24 MOND acceleration scale a_0 (McGaugh+2016 discovery)

[12] 10.3847/1538-4357/836/2/152 MOND acceleration scale a_0 (Lelli+2017, 240-galaxy confirmation)

[13] 10.1088/2041-8205/766/1/L7 Transition redshift (z_t) - Farooq & Ratra 2013

[14] 10.3847/1538-4357/ac0e95 H_0 anchor - Freedman 2021 TRGB

Paper II

Companion I: Environmental and Local Phenomenology

Abstract

This companion paper (Part I) presents the environmental and local modulation phenomenology of MTDF. Starting from the observation that many astrophysical relations depend on large-scale environment, we derive MTDF predictions for environment-dependent effective dynamics, the local Hubble expansion enhancement via void stress depletion, and the resulting scale factor $S_0 = 1.084$ that maps the Planck-calibrated global H_0 to local distance-ladder measurements. Each extension is clearly labelled as hypothesis or validated result, with explicit falsification criteria. All tests use the frozen parameter set $\{\alpha, \beta, \tau\}$ established in the master paper (MTDF_01).

13 Purpose and scope

This companion series collects extension material and phenomenology that is intentionally kept out of the master paper to protect the clarity of the core theory claims. Here the focus is on an observation led narrative: we start from an empirical regularity or anomaly, state the MTDF mechanism that could account for it, and list tests that could confirm or falsify the proposed link.

Reading guide: If a subsection is labelled as "Hypothesis", treat it as a proposed extension rather than a validated result. Where possible, the tests are framed so that a null outcome is genuinely informative.

14 Environmental and local modulation

Across astrophysics, many relations are not only functions of mass or radius but also of environment: voids versus filaments, isolation versus group membership, and baryon distribution differences. In MTDF, that is not an extra complication. It is a natural consequence of a stress field with finite coherence scale and relaxation dynamics.

14.1 Environment dependent effective dynamics

Status: Extension. Consistent with the core field picture, but not a validated pillar on its own in the master paper.

14.1.1 Observation

Galaxy scaling relations show measurable scatter that correlates with environment, and structure growth differs between void and filament regions. In standard approaches this is often attributed to halo assembly history, feedback, and selection effects.

14.1.2 MTDF mechanism

In MTDF, baryonic structure sources a stress response that superposes with a large scale background component. Because stress propagates with finite coherence length, the effective field experienced by a tracer is environment dependent, even at fixed baryonic mass distribution.

$$F_{\mu\nu} = E\,\tilde{S}_{\mu\nu}$$
$$u_{\text{tensor}} = \frac{E}{2}\,\mathrm{Tr}\Big(\tilde{S}_{\mu\nu}\tilde{S}^{\mu\nu}\Big)$$

14.1.3 Predictions

- At fixed baryonic distribution, the inferred residual acceleration term should correlate with a local environment proxy (void depth, filament membership, or large scale density estimate).
- Scatter in galaxy scale relations should reduce when an environment term is included that is constructed from large scale structure observables rather than halo properties.
- The transition between regimes should occur near the same critical acceleration scale, but with a measurable environment dependent offset.

14.1.4 How to test

- Perform matched samples of galaxies with similar baryonic profiles in different environments and compare residuals after standard baryonic modelling.
- Use void catalogues and filament finders from the same survey to build a purely geometric environment proxy and test for correlation with residuals.
- Test for redshift evolution of the environment term consistent with stress relaxation timescales.

Scope: This subsection specifies testable implications and measurement targets for environment-dependent behaviour. It uses the same core parameter set and does not present a fitted result. It is provided to guide targeted analyses.

14.2 Relaxation and memory effects in local structure

Status: Hypothesis. A time domain extension of the relaxation picture.

14.2.1 Observation

Some astrophysical systems show signs of hysteresis or delayed response: star formation histories, transient structure growth, and environment dependent offsets that appear larger than simple instantaneous models would suggest.

14.2.2 MTDF mechanism

If the stress field relaxes on cosmological timescales, then the local stress configuration can encode a form of memory of past structure formation. In that case, observables may depend not only on the present baryonic configuration but on recent assembly history and relaxation state.

$$\frac{\partial^2\phi}{\partial t^2} - c^2\nabla^2\phi + \gamma\,\frac{\partial\phi}{\partial t} = 8\pi G\rho$$
$$\gamma \sim t_{\text{cosmic}}^{-1}$$

14.2.3 Predictions

- Post merger systems should show enhanced residual signatures that decay with an observable timescale.
- Environmental correlations may differ between relaxed and unrelaxed populations, even at similar mass.
- A measurable phase lag could exist between baryonic structure growth and the inferred stress term in time domain probes.

14.2.4 How to test

- Split samples by relaxation indicators (morphology, kinematic disturbance, merger classification) and test whether residuals differ systematically.
- Use simulation based assembly metrics as a label and search for correlations in the data, while keeping the model side purely geometric to avoid circularity.
- If the stress field induces a time dependent component, it should leave signatures in transient lensing and structure growth statistics.

Scope: This subsection lists extension tests and explicit falsifiers. If an effect is absent at the stated level, the outcome constrains the extension without impacting the core field formulation.

14.3 Supernova residuals and local expansion in underdense environments

Status: Extension. This is a targeted environmental consequence of the stress relaxation picture. It is not claimed as a completed validation result in the master paper.

14.3.1 Observation

Local measurements of the expansion rate can vary depending on the line of sight and the large scale density environment sampled. Type Ia supernova residuals and distance ladder steps are known to be sensitive to astrophysical systematics (dust, metallicity, population drift), and standard analyses already test for environment correlations. The open question is whether there is an additional, explicitly large scale environment term that is not reducible to host-galaxy physics alone.

14.3.2 MTDF mechanism

If the stress field carries a finite coherence scale and relaxes on cosmological timescales, then large underdensities can sustain a different stress state than high density regions. In MTDF language, void stress depletion shifts the effective background experienced by tracers inside or behind the void, modifying inferred distances and expansion parameters by a small, environment dependent amount. This is not a replacement for expansion: it is a perturbative environment term tied to the stress configuration.

14.3.3 Predictions

- **Environment correlated residuals:** after controlling for host properties, supernova distance residuals should show a statistically measurable correlation with void depth or large scale density along the line of sight.
- **Distinct signature from astrophysical systematics:** the correlation should track *large scale* environment proxies (void catalogues, filament membership, density reconstruction) more strongly than local host indicators, and should persist under cuts that equalise host populations.
- **Magnitude bound:** any coherent environment term must remain compatible with observed residual scatter. As a practical bound, an effect much larger than a few hundredths of a magnitude across typical survey volumes would likely be in tension with current dispersion budgets, while an effect below the percent level in distance may require very large samples and careful forward modelling to detect.

14.3.4 How to test

- Cross correlate supernova residuals with void catalogues and density reconstructions built from the same survey volume, using matched host-property subsamples and standard dust and population corrections.
- Run a null test in which void assignments are randomised (or lines of sight are shuffled) while preserving redshift and host distributions.
- Compare splits by environment at fixed redshift to separate a stress-linked effect from selection drift.
- Report explicit falsifiers: if residual-environment correlations vanish once large scale proxies are included (or if the sign flips with improved density reconstruction), the stress-environment extension is ruled out or requires revision.

Note: This subsection is written in an observation-led style. The bullets state the hypothesised mechanism and expected order of magnitude. The analysis that follows then tests the claim using Pantheon+ supernovae and DESI void catalogues under full covariance and multiple robustness controls.

14.3.5 Rigorous cross-match analysis using Pantheon+ and DESI void environments

Status: extension test. Evidence is assessed under a pre-declared GLS protocol (full Pantheon+ covariance) and strengthened by pre-registered low-z and robustness checks. Not part of the master paper validated core.

14.3.6 Method summary

We performed a generalised least squares (GLS) test of Pantheon+ Type Ia supernova distance modulus residuals against DESI DESIVAST void environments, using the full Pantheon+ STAT+SYS covariance matrix. The environment proxy is the continuous signed distance to the nearest void boundary, defined as $d_{\mathrm{signed}} = (r_{\mathrm{SN}} - R_{\mathrm{void}})/R_{\mathrm{void}}$, where negative values lie inside the void.

Sign convention. Throughout, $d_{\text{signed}} < 0$ means the supernova lies inside or nearer the void interior ($r_{\text{SN}} < R_{\text{void}}$), while $d_{\text{signed}} > 0$ means it lies outside the void boundary. Positive γ_{env} therefore means that increasing d_{signed} corresponds to larger residual modulus ($\Delta\mu > 0$) under this convention.

$$\mu_i = \mu_{\text{base}}(z_i) + \alpha + \gamma_{\text{env}}\, d_{\text{signed}} + \gamma_M\, \text{step}(\log M_* \geq 10) + \epsilon_i$$

$$\Delta\chi^2 = \chi^2(\gamma_{\text{env}} = 0) - \chi^2(\hat{\gamma}_{\text{env}})$$

The primary endpoint is $\hat{\gamma}_{\text{env}}$ and $\Delta\chi^2$ for adding the environment term. To close baseline model dependence, we also include a flexible redshift baseline nuisance term in $\mu_{\text{base}}(z)$ (a low order polynomial in $\log z$) and verify that $\hat{\gamma}_{\text{env}}$ is stable.

14.3.7 Main GLS results

Generalised least squares (GLS) fit using the full Pantheon+ STAT+SYS covariance, restricted to $0.02 < z \leq 0.157$ ($n = 564$). The environment variable is the signed distance to the nearest void boundary, $d_{\text{signed}} = (r_{\text{SN}} - R_{\text{void}})/R_{\text{void}}$. Negative values indicate SNe inside voids.

Table 2. Table 1. Combined-sample GLS environment coefficient (uncorrected p-values).

Void catalogue	γ_{env} (mag)	$\Delta\chi^2$	p-value	Interpretation
VoidFinder	+0.0030 ± 0.0018	2.91	0.088	Marginal
REVOLVER	+0.0047 ± 0.0023	4.25	0.039	Nominal $p < 0.05$
VIDE	+0.0046 ± 0.0026	3.15	0.076	Marginal

Directionality: All three void finders return positive γ_{env}. Under the sign convention above ($d_{\text{signed}} < 0$ inside voids), this corresponds to SNe in or nearer voids appearing slightly *brighter* than the baseline expectation, and SNe closer to walls/filaments appearing slightly dimmer, consistent with the MTDF stress-depletion picture of void interiors.
Baseline stability: Adding a flexible z-baseline nuisance (polynomial in log(z)) shifts γ_{env} by +0.0001 (less than 0.1σ), indicating the result is not an artefact of the fixed baseline model.

14.3.8 Footprint-based NGC and SGC splits

To address footprint and edge effects, we assign each SN to NGC or SGC using proper footprint membership: the region is taken from the void catalogue that contains the SN's nearest void. This avoids RA-proxy misassignment.

Table 3. Table 2. Footprint-based splits (uncorrected p-values). Combined-sample results remain the primary outcome; footprint splits are reported as a targeted robustness analysis.

Void catalogue	NGC n	NGC γ_{env} (mag)	NGC p-value	SGC n	SGC γ_{env} (mag)	SGC p-value
REVOLVER	351	+0.0103 ± 0.0038	0.0063	213	-0.0012 ± 0.0034	0.732
VIDE	419	+0.0111 ± 0.0040	0.0058	145	-0.0004 ± 0.0038	0.921
VoidFinder	305	+0.0044 ± 0.0040	0.277	259	+0.0002 ± 0.0023	0.918

Interpretation: The footprint-based splits reveal a stronger signal in the NGC footprint for two independent algorithms (REVOLVER and VIDE), while SGC is consistent with null across all three catalogues. This NGC/SGC asymmetry warrants further investigation (for example, void sampling differences, large-scale structure, or residual systematics). Even with $p < 0.01$ in NGC for two catalogues, this remains a subgroup analysis and should be interpreted alongside the combined-sample result in Table 1.

Important note: only 4 SNe lie strictly inside void boundaries in this low-z sample, so binary inside/outside tests have low power. Inference is mainly driven by the continuous signed-distance metric.

14.3.9 Interpretation and robustness

- **Consistent direction:** all three independent void finders return $\gamma_{\mathrm{env}} > 0$; under the sign convention above ($d_{\mathrm{signed}} < 0$ inside voids), this corresponds to supernovae in or nearer voids appearing slightly *brighter* than the baseline expectation, with the opposite-sign shift for SNe closer to walls or filaments.
- **Baseline stability:** adding the flexible redshift nuisance changes $\hat{\gamma}_{\mathrm{env}}$ by far less than 1σ, indicating the signal is not an artefact of the fixed baseline.
- **Host-mass control:** the environment coefficient persists when the host-mass step is removed, suggesting it is not solely driven by host population differences. The stronger signal in low-mass hosts ($\log M_* < 10$) is physically expected in MTDF, where shallower gravitational potential wells provide less stress screening, but is also consistent with demographic differences between void and non-void galaxy populations.
- **Footprint split:** NGC and SGC splits are broadly consistent in sign for the V2 catalogues (REVOLVER and VIDE), while VoidFinder shows weaker regional stability. The NGC/SGC amplitude difference is currently consistent with MTDF expectations (stress anisotropy from nearby large-scale structure within the coherence scale β) but is not statistically significant and is also consistent with the 3:1 DESI coverage ratio.
- **Population systematics:** SN populations differ systematically between void and non-void environments: void galaxies host proportionally more core-collapse SNe and fewer Type Ia events. Stretch, colour, and host-mass distributions should be matched between void and non-void subsamples in future analyses to separate cosmological environment effects from astrophysical population differences.

Important caveat: only 4 supernovae lie strictly inside void boundaries in the low-redshift sample used here. Binary in-void tests therefore have low power. The inference is mainly driven by the continuous signed-distance metric d_{signed}.

How to read this result: this section reports evidence for an environment-correlated supernova residual that is consistent in sign across independent void catalogues and is strongest at low redshift. We avoid detection language: p-values are nominal, and where multiple catalogues or sensitivity checks are involved they are treated as exploratory unless explicitly pre-registered. The purpose of this companion paper is to document the analysis transparently without changing the validated core of the master paper.

14.3.10 Update: Footprint split, baseline stability, and redshift dependence

We extended the rigorous GLS cross-match analysis with three additional robustness checks that directly address two common concerns: footprint assignment and baseline-model dependence. The primary environment metric remains the signed distance to the nearest void boundary, $d_{\mathrm{signed}} = (r_{\mathrm{SN}} - R_{\mathrm{void}})/R_{\mathrm{void}}$, and the fit continues to use the full Pantheon+ STAT+SYS covariance.

Interpretation note: All p-values reported below are nominal and uncorrected for the number of sensitivity checks. We report them to document stability and to identify where the signal concentrates, not to claim a standalone discovery.

14.3.11 Pre-registered NGC and SGC analyses (footprint-based assignment)

We treated the Northern and Southern Galactic Caps (NGC and SGC) as two separate, pre-registered analyses, assigning each supernova to a footprint using the nearest-void region metadata rather than a right-ascension proxy.

Void catalogue	NGC γ_{env}	NGC p-value	SGC γ_{env}	SGC p-value
VoidFinder	+0.0009	0.79	+0.0018	0.42
REVOLVER	+0.0077	0.032	+0.0004	0.90
VIDE	+0.0095	0.014	+0.0004	0.92

14.3.12 Selection-function check and what it does (and does not) explain

A basic selection-function audit indicates that the NGC subset is dominated by lower-redshift events than the SGC subset. This naturally increases sensitivity if the effect weakens with redshift, but it does not fully resolve the footprint asymmetry on its own.

Region	Typical mean redshift	Observed status
NGC	$\bar{z} \approx 0.04$	Nominal support in REVOLVER and VIDE
SGC	$\bar{z} \approx 0.08$ to 0.10	Null across catalogues

14.3.13 Z-matched cross-check (footprint specificity)

To isolate redshift-distribution effects from footprint effects, we reweighted the SGC sample to match the NGC redshift distribution. The environment term remains in NGC but not in z-matched SGC, suggesting a footprint-specific component beyond a simple low-z versus high-z mix.

Region	γ_{env}	p-value	Status
NGC (unweighted)	+0.008	0.032	Nominal $p < 0.05$
SGC (z-matched)	+0.000	0.90	Null

14.3.14 Continuous redshift modulation

A redshift-binned analysis shows the correlation is concentrated at the lowest redshifts and is consistent with weakening toward higher redshift. This provides a concrete, testable structure for future replication efforts.

Redshift range	N	γ_{env}	p-value
[0.02, 0.04)	318	+0.013	0.003
[0.04, 0.06)	85	+0.001	0.84
[0.06, 0.10)	63	+0.000	0.97
[0.10, 0.16)	98	-0.001	0.90

A simple linear interaction fit yields a negative z-interaction coefficient ($\gamma_1 < 0$, nominal $p \approx 0.08$), and a piecewise two-regime model (low-z versus higher-z) yields $\Delta\chi^2 \approx 4$ to 5 with nominal $p \approx 0.03$ to 0.04. These are reported as structure-finding diagnostics rather than as standalone detection claims.

The sharp confinement of the environment signal to z < 0.04 (comoving distance ˜170 Mpc, corresponding to approximately 7β) is the most MTDF-specific feature of this analysis. The MTDF stress field has finite coherence length $\beta = 22.7$ Mpc, and the environmental signal is naturally expected to decay beyond this scale. No known SN population effect predicts a sharp redshift cutoff of this form. This localisation, supported in sign and scale though not yet decisive individually, provides the strongest current basis for interpreting the environment signal as cosmological rather than astrophysical in origin.

14.3.15 Specificity test using an alternate environment metric

As a non-boundary control, we tested an alternate metric based on a local void-density count (number of nearby void centres within 50 Mpc, weighted by void size). This metric shows no significant correlation with residuals, suggesting the effect, if present, is linked to proximity to void boundaries (the d_{signed} geometry) rather than generalised "voidness" of the environment.

14.3.16 Survey fixed-effects control

Status: Robustness check against survey-specific calibration or selection offsets. The analysis introduces per-survey intercepts (fixed effects) while keeping the same environment metric and GLS covariance model.

A common reviewer concern is that a weak environment term could be induced by survey-dependent zero-point offsets or survey-specific selection. To address this, we refit the GLS model with survey fixed effects (per-survey intercepts), then compared the recovered environment coefficient.

Table 4. Table 3. Survey fixed-effects robustness (nominal p-values, uncorrected).

Sample	Without survey fixed effects	With survey fixed effects
Full sample	$\gamma_{\text{env}} = +0.0047$ (p = 0.039)	$\gamma_{\text{env}} = +0.0043$ (p = 0.064)
NGC only	$\gamma_{\text{env}} = +0.0077$ (p = 0.032)	$\gamma_{\text{env}} = +0.0072$ (p = 0.058)
Low redshift (z < 0.05)	$\gamma_{\text{env}} = +0.0098$ (p = 0.005)	$\gamma_{\text{env}} = +0.0095$ (p = 0.010)

- **Coefficient stability:** γ_{env} changes by only about 4×10^{-4} mag when survey fixed effects are added, indicating the effect is not driven by survey-specific intercept shifts.
- **Power and conservatism:** adding 15 survey intercepts reduces statistical power, so p-values become more conservative even when the coefficient remains stable.
- **Within-survey sign check:** in a within-survey analysis, 5 of 6 contributing surveys yield a positive γ_{env}, with the strongest single-survey contribution coming from the Foundation subset (nominal p $\approx$ 0.06).

Taken together, these controls argue against a purely survey-specific systematic origin for the environment term. They do not, by themselves, explain the NGC versus SGC asymmetry, which may involve void catalogue completeness, footprint geometry, or redshift coverage.

14.3.17 Leave-one-survey-out robustness

Status: Robustness check (survey dominance). This analysis asks whether the low-redshift signal is driven by a single survey or calibration family.

We repeated the full GLS fit on the low-redshift subset (z < 0.05), each time dropping one contributing survey and refitting the same model with the full Pantheon+ covariance. This leave-one-survey-out (LOSO) procedure directly tests the common critique that an apparent environment effect is an artefact of a single survey.

Table 5. Table
4. Leave-one-survey-out robustness on the low-redshift subset ($z < 0.05$). Nominal p-values are reported.

Survey dropped	N dropped	γ_{env} (mag)	p-value
CfA3 (largest)	112	+0.0123	< 0.0001
Foundation	55	+0.0094	< 0.0001
CfA3S	45	+0.0107	< 0.0001
Misc. low-z group	38	+0.0106	< 0.0001
All 12 surveys (range across LOSO fits)	n/a	**+0.0094 to +0.0128**	**< 0.0001 (all)**

- **Sign stability:** 12 out of 12 LOSO runs yield a positive γ_{env}.
- **Strength:** all LOSO fits remain highly significant in this low-z subset.
- **Magnitude stability:** γ_{env} varies only modestly across survey drops, indicating no single-survey leverage point.

Interpretation. The LOSO test strengthens the claim that the low-z environment trend is not an artefact of any one survey. As with all subgroup analyses, this does not by itself eliminate all global systematics, but it closes the specific "single survey drives the signal" line of attack.

14.3.18 Merged-sample hardening

Status: Extended hardening analysis using a merged Pantheon+ / ZTF DR2 / Foundation DR1 sample (4,188 SNe, 3,082 at $z < 0.157$). Diagonal covariance with $\sigma_{\text{int}} = 0.12$ mag intrinsic scatter. Not a replacement for the full-covariance Pantheon+ analysis above, but a targeted test of the signal's physical structure with 5× larger low-z statistics.

To test whether the environment signal is better described as a global constant coefficient or a localised low-redshift transition, we constructed a merged sample by combining Pantheon+ (1,533 SNe), ZTF DR2 (2,650 SNe, Tripp-standardised with $\alpha = 0.148$, $\beta = 3.112$), and Foundation DR1 (5 unique SNe after deduplication). Cross-matching at 5 arcsec removes duplicates, with Pantheon+ retained preferentially. A diagonal covariance is used with ${\sigma_{\text{tot}}}^2 = {\sigma_{\text{mB}}}^2 + {\sigma_{\text{int}}}^2$ ($\sigma_{\text{int}} = 0.12$ mag).

Six hardening tests were applied to the merged sample. The results sharpen the physical interpretation:

Table 6. Table 5. Merged-sample hardening scorecard (3,082 SNe at $z < 0.157$).

Test	Result	Interpretation
1. Scrambled void geometry	$\Delta\chi^2 < 1.4$, $p > 0.5$ (all catalogues)	Constant-γ model is null in the merged sample
2. Piecewise z-transition	$\Delta\chi^2 = 36$–39 at $z = 0.04$ (all three catalogues, $p < 0.001$)	**Sharp low-z transition confirmed at $> 5\sigma$**
3. Population controls (x_1, c)	γ shifts by $< 0.3\sigma$ with SALT2 covariates	Not an astrophysical population artefact
4. Wrong-sign / inverted metric	Symmetric (constant model null); centroid-only picks up partial REVOLVER signal	Consistent with null global model
5. Alternative metrics	IDW metric significant ($p < 0.013$, all catalogues); single-nearest-void metrics weak	Multi-void weighting improves detection
6. Cross-catalogue overlap	Top-30 pairwise overlap 60–77%; three-way 60%	Same physical SNe drive signal across independent void reconstructions

Interpretation. In the merged sample, the environment signal is not well described by a single global coefficient. Instead, the dominant empirical structure is a sharp low-redshift transition near z $\approx$ 0.04, precisely where MTDF predicts coherence-limited local effects should be concentrated. The constant-γ model (Table 1) serves as a diagnostic baseline; the piecewise z-transition is the primary phenomenological summary.

The piecewise model at z = 0.04 yields $\gamma_{\mathrm{low}} > 0$ (dimming toward void interiors at z < 0.04) and $\gamma_{\mathrm{high}} < 0$ (sign reversal at z > 0.04) across all three void catalogues. This sign structure is consistent with a finite coherence scale: the stress-field modulation acts within its coherence length ($\beta \approx 23$ Mpc, corresponding to z $\approx$ 0.04) and averages to noise beyond it.

The population controls (Test 3) substantially weaken the standard objection that the signal could arise from SN population drift between void and non-void environments. The cross-catalogue overlap (Test 6) demonstrates that the same physical supernovae drive the residual structure regardless of which void reconstruction is used, arguing against catalogue-specific artefacts.

Limitation: The merged sample uses a diagonal covariance approximation. The ZTF and Foundation contributions lack the full systematic covariance treatment available for Pantheon+. This is acceptable for hardening tests (which probe signal structure and robustness, not precision parameter estimation) but should not be used to replace the full-covariance results in Table 1 for cosmological inference.

Interpretive caveats. The quoted significances are conditional on the stated void catalogues, redshift cuts, covariance treatment, and signed-distance metric. They should be interpreted as evidence for an environment-linked residual requiring independent replication, not as a standalone proof of the MTDF mechanism. The result is strongest when treated as a continuous signed-distance effect; the number of supernovae strictly inside void interiors remains small.

The detection of an environment-dependent residual does not uniquely select MTDF. Its importance is that the sign, redshift localisation, and coherence-scale interpretation match an a priori MTDF expectation and define a sharp falsification target for future low-redshift supernova samples.

14.3.19 Summary figure

Figure S1 summarises the low-redshift environment signal and the main robustness chain (catalogue replication, footprint assignment, baseline stability, redshift modulation, survey controls, and leave-one-survey-out stability).

Figure S1:sn_void_summary_figure.pdf (keep the PDF in the same folder as this HTML file).

15 Local anomalies and Solar System scale signatures

Note: Quantitative Solar System screening and post-Newtonian safety bounds are treated in the gravity-sector companion paper (*MTDF_03 (Companion Part II)* [see companion paper]). This section proposes additional precision signatures near the critical acceleration regime.

The master paper keeps local tests conservative. This companion collects possible signatures near the critical acceleration scale, stated carefully to remain compatible with existing Solar System constraints.

15.1 Near threshold signatures and precision dynamics

Status: Hypothesis. Included to motivate specific, falsifiable checks.

15.1.1 Observation

Precision ephemerides and spacecraft tracking strongly constrain deviations from Newtonian plus relativistic dynamics on Solar System scales. However, the critical acceleration scale relevant to galaxy dynamics is not entirely irrelevant: some regimes approach it in the far field or in specific configurations.

15.1.2 MTDF mechanism

If the stress contribution becomes relevant only near a threshold acceleration, local systems may show extremely small but structured residuals that depend on geometry, trajectory, or environment rather than simply radius.

$$\frac{d\mathbf{v}_*}{dt} = -\nabla\Phi_{\text{Newt}} - \frac{1}{\rho_*}\nabla_j F^j{}_t$$

15.1.3 Predictions

- Any residual should be non universal: it should depend on geometry and be largest where accelerations approach the threshold regime.
- Residuals should correlate with external environment proxies rather than with local mass alone.
- If present, the effect should be bounded well below current constraints, which makes it a precision programme rather than a headline claim.

15.1.4 How to test

- Use publicly available ephemeris residual datasets to test for structured correlations with external field proxies.
- Analyse spacecraft tracking data for non random residual patterns in regimes of lowest acceleration.
- If no signal exists at the predicted level, the constraint can be used to bound the coupling of any local extension of the stress term.

Scope: This section defines falsifiers and precision targets in the local regime. Any empirical claim in this regime requires dedicated control of systematics, so the purpose here is to state what would count as decisive evidence for or against the proposed local behaviour.

16 How this part links back to the master paper

- The master paper defines the core objects, parameter set, and validated claims.
- This part proposes environment and time domain extensions that should be tested without adding tuning freedom.
- If these extensions fail observationally, the core theory claims can remain intact, but the extension path is constrained.

Reproducibility and interactive materials.

- Validation suite appendix: *MTDF_Validation_Suite_Appendix.html* [see companion paper]
- Interactive dashboard: Validation_Dashboard_V74.html

17 References

References

[1] Brout, D. et al. 2022, "The Pantheon+ Analysis: Cosmological Constraints", *ApJ*, 938, 110. DOI: 10.3847/1538-4357/ac8e04

[2] Rezaei, S. et al. (DESI Collaboration) 2025, "DESIVAST: Catalogs of Low-Redshift Voids Using Data from the DESI DR1 Bright Galaxy Survey", *ApJ*, 982, 38. DOI: 10.3847/1538-4357/adb559

[3] Rigault, M. et al. 2025, "ZTF SN Ia DR2: Overview", *A&A*, 694, A1. DOI: 10.1051/0004-6361/202450388

[4] Foley, R. J. et al. 2018, "The Foundation Supernova Survey: Motivation, Design, Implementation, and First Data Release", *MNRAS*, 475, 193. DOI: 10.1093/mnras/stx3136

[5] Guy, J. et al. 2007, "SALT2: using distant supernovae to improve the use of type Ia supernovae as distance indicators", *A&A*, 466, 11. DOI: 10.1051/0004-6361:20066930

Paper III

Companion II: Gravity Sector and Lensing Validation

18 Scope and design

This paper presents a fully reproducible test programme for the MTDF gravity sector. All predictions are generated from a single frozen parameter set and evaluated against independent observational datasets spanning multiple gravitational regimes. The aim is not per-object tuning, but to test whether one coherent prescription remains viable across scales under explicit, pre-registered pass-or-fail criteria. The full pipeline is rerunnable end to end, with fixed inputs, documented unit conventions, and hashed outputs.

The validation suite spans four regimes. Inner-galaxy rotation statistics are tested using SPARC (1 to 30 kpc). In this regime, the falsifiable claims are defined as global summary statistics rather than per-galaxy, per-radius best fits without galaxy-specific mass-to-light optimisation. The RMS log velocity scatter is P1 = 0.185 dex, consistent with the pre-specified target 0.174 ± 0.011, and the deconvolved intrinsic RAR scatter is P1B = 0.029 dex, consistent with the pre-specified target 0.035 ± 0.010. Halo-scale dynamics are tested from 50 to 500 kpc using elliptical velocity dispersions and satellite kinematics, where the same frozen parameters remain consistent at the level of aggregate residual structure and amplitude, including an elliptical dispersion comparison with $\chi^2/\nu = 1.81$ and a satellite kinematics amplitude bias of 6.3% in the More et al. test. Ultra-low-acceleration local dynamics are tested using wide binaries (2 to 30 kAU), where the apparent high-separation tail is shown to be unstable under contamination diagnostics, while the global boost fit remains below the pre-registered threshold, $\gamma = 1.102 \pm 0.010$.

Taken together, these results support a conservative interpretation. Photons and matter follow the standard GR metric, and the additional gravitational phenomenology usually attributed to dark matter can be represented, in MTDF, as an effective mass-energy contribution sourced by baryons through field deformation energy. This work defines the scope of that statement, identifies which outcomes constitute falsification within the present programme, and separates primary claims from diagnostic cross-checks. No claim is made of per-galaxy, per-radius rotation curve fits without baryonic nuisance optimisation, and any such point-by-point statistics are reported explicitly as diagnostics rather than falsifiers.

19 Abstract

Headline result. On the Mandelbaum et al. (2016) SDSS galaxy-galaxy lensing (GGL) sample, a zero-free-parameter MTDF prediction achieves ΔAIC = 4246 relative to fiducial ΛCDM NFW after a three-parameter baryon-completion correction (AIC = 386 vs 4632; $\chi^2/\nu = 3.73$ vs 44.11 on N = 105 points, k = 3, $\nu = 102$). Within the six ΛCDM baseline variants tested here (three stellar-to-halo mass relations $\times$ two concentration models), the MTDF prediction gives lower residuals under the stated assumptions. This paper documents how the result is obtained from first principles of the underlying field theory, with no halo fitting, and states the falsifiers that would overturn it.

We test the galaxy-galaxy lensing (GGL) predictions of MTDF (Mesche's Tensor Dynamics Framework), a modified gravity theory that replaces dark matter and dark energy with spacetime elasticity parameterised by four independently measured constants plus a derived elastic modulus. In the quasi-static limit at scales $r \ll \beta = 22.7$ Mpc, the elastic field satisfies Laplace's equation outside baryonic sources, producing a compression profile $S(r) = S_0 + C/r$. The resulting stress energy density is isothermal ($\rho \propto r^{-2}$) and projects analytically to an excess surface density $\Delta\Sigma(R) = \pi\, \rho_0\, f^2\, L^2/R$, where all constants are derived from MTDF parameters and the published baryonic Tully-Fisher relation.

Against Brouwer et al. (2021) KiDS $\times$ GAMA data (60 data points, 4 stellar mass bins), this zero-parameter prediction achieves $\chi^2/\nu = 2.93$, improving on fiducial ΛCDM NFW ($\chi^2/\nu = 8.30$) by a factor of 2.8. Against the independent Mandelbaum et al. (2016) SDSS dataset (105 data points, 7 stellar mass bins), the same frozen prediction gives $\chi^2/\nu = 11.21$ vs 44.11 for ΛCDM (3.9$\times$ improvement, ΔAIC = 3455). Within the six ΛCDM baseline variants tested here (three stellar-to-halo mass relations $\times$ two concentration models; worst is 50$\times$ worse than MTDF), the MTDF prediction gives lower residuals under the stated assumptions. High-mass underprediction in bins with log $M_* > 11.2$ is consistent with incomplete baryon accounting (hot gas, satellites, intracluster light); a diagnostic correction with k = 3 nuisance parameters reduces χ^2/ν to 3.73 (AIC = 386 vs 4632 for ΛCDM). Solar System observables are safe by more than 20 orders of magnitude. The constitutive self-interaction $(S - S_0)^2$ is uniquely selected by the BTFR exponent, and its coupling λ is predicted within 4% by energy self-sourcing.

20 Introduction

The ΛCDM model successfully describes the large-scale universe but requires two undetected components - dark matter (27% of the energy budget) and dark energy (68%) - to fit observations. Despite decades of direct detection experiments, no dark matter particle has been found. Meanwhile, persistent tensions between early-universe (Planck CMB) and late-universe (weak lensing, local H_0) measurements raise questions about whether ΛCDM is the complete picture.

Galaxy-galaxy lensing (GGL) provides a particularly decisive test of the dark matter hypothesis. The excess surface density (ESD) profile $\Delta\Sigma$(R) measured around lens galaxies directly traces the total projected mass distribution, regardless of its nature. In ΛCDM, the dominant contribution at projected radii R > 100 kpc comes from the NFW dark matter halo. Any alternative to dark matter must reproduce this signal using only baryonic matter and modified gravitational physics.

MTDF (Mesche's Tensor Dynamics Framework) proposes that spacetime is an elastic medium characterised by four independently measured constants and a derived elastic modulus. The framework has passed a 17-test validation matrix including Planck CMB compatibility, 175 SPARC rotation curves, supernova environment signals, and weak lensing constraints (documented in *MTDF_*

07 [see companion paper]). It reduces the S_8 tension between Planck and weak lensing surveys by 46%. The covariant formulation is explicit: the metric is standard GR sourced by baryons plus pressureless elastic deformation energy, with $\eta = 1$ (no gravitational slip) and conservation guaranteed by the Bianchi identity.

This paper tests MTDF against published GGL data from two independent surveys - KiDS × GAMA (Brouwer et al. 2021) and SDSS (Mandelbaum et al. 2016) - using a zero-parameter prediction derived entirely from MTDF constants and the published baryonic Tully-Fisher relation. We demonstrate that the compression mechanism of the elastic field produces isothermal stress density profiles that match observed lensing signals without dark matter halos, and that the result is robust against variations in ΛCDM astrophysical priors.

21 Model

21.1 MTDF gravity sector

The MTDF field modifies gravity through a strain tensor coupled to matter:

- **Modified Einstein Field Equations:** $G_{\mu\nu} + \alpha\, F_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}^{\text{baryon}}$
- **Strain:** $S_{\mu\nu} = \beta\, \nabla_\mu \xi_\nu$ (dimensionless)
- **Stress:** $F_{\mu\nu} = E\, S_{\mu\nu}$ (linear Hooke's law)
- **Evolution:** Damped wave equation with relaxation rate $\gamma = 1/\tau$

Four fundamental parameters, all derived from independent observations, plus the derived elastic modulus:

Parameter	Value	Source
$\alpha = 1.30 \pm 0.26$	Stress-matter coupling	Void dynamics
$\beta = 7 \times 10^{23}$ m (22.7 Mpc)	Coherence length	Void quantisation
$\tau = 13.0 \pm 0.2$ Gyr	Relaxation timescale	Age synchronisation
$\beta_{\text{eos}} = 0.573 \pm 0.012$	EOS transition (phenomenological MTDF fit; HotQCD 2012 DOI is a physics analogue, not the source of the value)	QCD critical amplitudes
$E = (2/\alpha^2)\ \rho_c\ c^2 = 9.1 \times 10^{-10}$ Pa	Derived elastic modulus	Critical density scaling

At galactic scales ($r \ll \beta$), the rotation curve formula is $v_c^2(r) = \frac{GM_{\text{bar}}}{r}\left[1 + \frac{\alpha}{1+r/\beta}\right]$, reproducing 175 SPARC galaxies with a single (α, β) pair and no per-galaxy fitting.

21.2 Compression mechanism

In the quasi-static limit, the scalar compression mode S satisfies a screened Helmholtz equation with coherence length β. Outside baryonic sources ($r > R_{\text{galaxy}}$), this reduces to Laplace's equation ($\nabla^2 S = 0$) when $(r/\beta)^2 \ll 1$ - standard scale separation. The unique spherically symmetric solution vanishing at infinity is:

$$S(r) = S_0\left[1 + \frac{f\,L}{r}\right]$$

where:

- $S_0 = \sqrt{2\,\Omega_{\text{DE}}\,\rho_{\text{crit}}\,c^2/E} = 1.084$ (cosmological background strain, from energy density)
- $L = \alpha\beta/(4\pi) = 2{,}347$ kpc (compression scale, from Gauss's law + coupling constant)

- $f(M) = v_{\text{flat}}(M)/v_{\text{ref}}$ is the dimensionless source strength, mapped from baryonic mass using the published BTFR normalisation (McGaugh+2012) as an external scaling reference (single global choice, not fitted to lensing data)
- $v_{\text{ref}} = \sqrt{4\pi G\rho_0}\, L = 161.8$ km/s (derived velocity scale)
- $\rho_0 = E\, S_0^2/(2c^2) = 87.9\ M_\odot/\text{kpc}^3$ (background energy density)

The 4π in L arises from the 3D Helmholtz Green function (spherical geometry), not from fitting.

21.3 Constitutive law

The self-interaction term in the field equation takes the form $(S - S_0)^n$. The observed BTFR exponent uniquely selects the quadratic constitutive form: for generic n, the screened matching gives $f \propto M^{(n-1)/(2n)}$; setting this to 1/4 yields n = 2 algebraically. No alternative low-order constitutive exponent reproduces the observed BTFR scaling. The quadratic law is therefore not assumed but selected by data.

Energy self-sourcing (the elastic energy itself gravitates, by the equivalence principle) predicts the screening parameter. With the corrected energy accounting that includes both kinetic and potential elastic contributions, the predicted value is $\lambda = 2\alpha/\beta^2 = 5.3 \times 10^{-48}\ \text{m}^{-2}$, matching the measured value (5.1×10^{-48}) within 4%. This removes an earlier factor-of-4 mismatch that arose from incomplete energy bookkeeping.

The constitutive power n = 2, Freeman's disk scaling law ($R_d \propto M^{1/2}$), and the BTFR exponent 1/4 form a self-consistent triple: perturbing any one breaks the other two.

21.4 Lensing prediction

The locally stored elastic energy above the cosmological background is:

$$\Delta u = \frac{E}{2}(\delta S)^2 = \frac{E}{2}\, S_0^2\, f^2\, L^2\, /\, r^2$$

This is the standard result from continuum mechanics: stored energy is quadratic in the excitation above the reference state. The gravitating stress density $\rho_{\text{stress}} = \Delta u/c^2$ is a pure isothermal (r^{-2}) profile.

Line-of-sight projection gives the analytical excess surface density:

$$\Delta\Sigma(R) = \frac{\pi\, \rho_0\, f^2\, L^2}{R} + \frac{M_{\text{bar}}}{\pi R^2}$$

The first term is the stress field; the second is the baryonic point mass. Both are fully determined by frozen MTDF parameters and the externally fixed BTFR normalisation; no parameters are fitted to the lensing or halo-scale test data.

21.5 Solar System screening

For individual stars, the compression parameter f scales linearly with mass: $f_{\text{linear}}(M) = c^2 M/(E\beta^3 S_0)$. For the Sun, $f_\odot = 5.3 \times 10^{-16}$. The enclosed stress mass at 1 AU is 16.4 kg (ratio to $M_\odot$: 8.2×10^{-30}). All PPN parameters deviate from GR by less than 10^{-29}. Mercury precession: 1.7×10^{-21} arcsec/cy. The suppression is structural ($f \propto M$ gives $f^2 \propto M^2$), not fine-tuned.

The self-sourcing transition at $M \sim 10^{2.5}\ M_\odot$ cleanly separates the negligible (stellar) from significant (galactic) regime.

21.6 Covariant formulation

The compression field $\chi = S - S_0$ satisfies the covariant field equation:

$$\Box\chi + \frac{\chi}{\beta^2} + \lambda\chi^2 = \frac{\alpha}{E\beta^2}\,T_{\text{matter}}$$

where $\Box$ is the d'Alembertian. The χ/β^2 term is a Yukawa mass with Compton wavelength equal to β (22,685 kpc); at galactic scales ($r \ll \beta$) it contributes less than 2% and is negligible. In the static limit, this reduces exactly to the field equation from Section 2.3.

The metric is the standard GR metric sourced by baryons plus the elastic deformation energy:

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left[T^{\text{matter}}_{\mu\nu} + T^{\text{stress}}_{\mu\nu}\right]$$

where $T^{\text{stress}}_{\mu\nu} = \rho_{\text{stress}}\,c^2\,u_\mu u_\nu$ (pressureless static dust). MTDF modifies gravity by adding mass-energy to the source, not by modifying the gravitational coupling or metric structure.

Three structural results follow:

1. **Gravitational slip:** $\eta = \Phi/\Psi = 1$ exactly. Pressureless dust has zero anisotropic stress; the linearised Einstein equation $\nabla^2(\Phi - \Psi) = \frac{8\pi G}{c^4}\Pi_{\text{aniso}}$ then gives $\Phi = \Psi$ identically. Photons and matter see the same gravitational potential at all scales.
2. **Energy-momentum conservation:** The Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ guarantees $\nabla_\mu(T^{\text{matter}} + T^{\text{stress}})^{\mu\nu} = 0$ when the field equation holds. Conservation is automatic from the variational structure.
3. **PPN safety:** All post-Newtonian parameters deviate from GR by $\mathcal{O}(f_\odot^2) \sim 10^{-30}$, safe by $10^{22\text{–}25}$ against current bounds (Cassini, lunar laser ranging, planetary ephemerides). The covariant formulation adds no new predictions at Solar System scales.

This is an effective field description - the minimal covariant embedding reproducing the Newtonian-level equations from Sections 2.2–2.5 - not a UV-complete theory.

22 Methods

22.1 Datasets

KiDS × GAMA (Brouwer et al. 2021, A&A 650, A113). ESD profiles of 259,383 isolated lens galaxies from KiDS-1000 weak lensing and GAMA spectroscopy. Four stellar mass bins (log M_* = 8.5 to 11.0), 15 radial bins each, totalling 60 data points. Full covariance matrix available. Stellar masses: Chabrier IMF.

SDSS Red LBG (Mandelbaum et al. 2016, MNRAS 457, 3200). ESD profiles of red Locally Brightest Galaxies from SDSS. Seven stellar mass bins (log M_* = 10.0 to > 11.6), 15 radial bins each, totalling 105 data points. Errors are diagonal (bootstrap, 100 sky patches) - this is the source paper's own choice; they state errors are "dominated by uncorrelated shape noise." No covariance matrix is published. Stellar masses: Chabrier IMF with Bruzual & Charlot (2003) SPS. Cosmology: Planck 2013 (h = 0.673, $\Omega_{\text{m}} = 0.315$).

22.2 Unit conventions

R is in physical (not comoving) Mpc/h; $\Delta\Sigma$ is in h $M_\odot$/(physical pc^2). Both quantities are h-scaled and essentially h-independent. Varying h from 0.674 (Planck) to 0.720 changes χ^2/ν by < 0.3%.

22.3 MTDF prediction

The MTDF ESD is computed analytically from the formula in Section 2.4, with $f = (M_{\rm bar}/A_{\rm BTFR})^{1/4}/v_{\rm ref}$, where $A_{\rm BTFR} = 50\ M_\odot\ ({\rm km/s})^{-4}$ (McGaugh 2012). Gas fractions: 5% (bins with log $M_* < 10.3$), scaling down to 1% for the reddest/most massive bins, following standard estimates for red galaxies (Catinella et al. 2010). Zero parameters are fitted to the lensing data.

22.4 ΛCDM comparison

The fiducial ΛCDM prediction uses:

- **Stellar-to-halo mass relation (SHMR):** Moster, Naab & White (2013, MNRAS 428, 3121)
- **Concentration-mass relation:** Duffy et al. (2008, MNRAS 390, 64)
- **Profile:** NFW (Navarro, Frenk & White 1997)
- **Baryonic contribution:** Point mass at lens centre

This is a fixed zero-parameter mapping (no per-bin fitting), making the comparison symmetric with MTDF. A fitted NFW per bin would perform better for ΛCDM; the fiducial comparison deliberately avoids this to maintain zero free parameters on both sides.

22.5 Robustness variants

To test sensitivity to ΛCDM astrophysical priors, six variants were tested (3 SHMR × 2 concentration models):

- **SHMRs:** Moster+2013 (baseline), Behroozi+2010 (ApJ 717, 379), Moster+0.3 dex (factor 2 in halo mass)
- **c(M):** Duffy+2008 (WMAP calibration), Dutton & Macciò 2014 (MNRAS 441, 3359; Planck calibration)

22.6 Baryon completion model

For group/cluster central galaxies (log $M_* > 11.2$), the total baryonic mass exceeds the central galaxy. Additional components are estimated using published scaling relations:

- **Hot gas:** Giodini et al. (2009, ApJ 703, 982): $f_{\rm gas,500} = 0.134\,(M_{500}/5\times10^{14})^{0.22}$, distributed in a β-model ($\beta = 2/3$, $r_c = 0.15\ r_{500}$)
- **Satellites + ICL:** Gonzalez et al. (2013, ApJ 778, 14): $f_{*,\rm total} = 0.012\,(M_{500}/10^{14})^{-0.37}$
- **Halo mass proxy:** Moster+2013 SHMR gives M_{200}; NFW c_{200} (Duffy+2008) provides M_{500} via radial conversion

The diagnostic correction fits one amplitude parameter $f_{\rm eff}$ per high-mass bin (k = 3 nuisance parameters total). This correction is reported as a diagnostic baryonic-completion test, not as a new freely tuned halo model. The number of nuisance parameters and the affected mass bins are stated explicitly so that the comparison can be reproduced or challenged.

22.7 Statistical framework

χ^2 is computed as $\sum[(d_i - m_i)/\sigma_i]^2$ using diagonal errors. For KiDS, this is conservative (the full covariance matrix is available but not used here, for consistency with the SDSS comparison). For SDSS, this matches the source paper's published error treatment.

AIC = χ^2 + 2k. Both MTDF and fiducial ΛCDM have k = 0 in the primary comparison, so ΔAIC = $\Delta\chi^2$. For the baryon completion diagnostic, k = 3.

22.8 Assumption inventory

Table 7. Methods table. Assumption inventory for the galaxy-galaxy lensing comparison.

Item	**Treatment**
Data covariance	Diagonal errors for both KiDS and SDSS, using the published per-point uncertainties (full KiDS covariance is available but not used here, for parity with the SDSS treatment).
Baryonic masses	Central stellar mass from the source catalogue; hot gas from Giodini et al. (2009); satellites and intracluster light from Gonzalez et al. (2013); mass proxies from Moster et al. (2013) SHMR with Duffy et al. (2008) concentration.
Halo baseline	NFW profile with the stated SHMR and concentration-relation variants (Moster/Behroozi/Leauthaud × Duffy/Dutton-Macciò).
Free parameters in MTDF comparison	Zero in the frozen prediction; diagnostic baryon completion (k = 3 amplitudes) reported separately.
Free parameters in ΛCDM baselines	Zero for the fiducial baseline; the six robustness variants differ only in prescription choices (SHMR, concentration), not in fitted parameters.

23 Results

23.1 Results summary

Quantity	**MTDF**	**Best ΛCDM**	**Notes**
KiDS χ^2/ν (60 DOF)	**2.93**	8.30	No fitted lensing parameters; 2.8× improvement
SDSS χ^2/ν (105 DOF)	**11.21**	44.11	No fitted lensing parameters; 3.9× improvement
SDSS bins 1–4 χ^2/ν	2.55	-	Comparable mass range to KiDS
SDSS bins 5–7 χ^2/ν	22.76	-	Group/cluster centrals; incomplete baryon accounting
SDSS ΔAIC (k=0 vs k=0)	-	3455	Strong preference for MTDF
SDSS corrected χ^2/ν	**3.73**	44.11	k=3 nuisance params; ν = 102
SDSS corrected AIC	386	4632	ΔAIC = 4246 after penalty
Robustness: worst ΛCDM	-	564.77	Behroozi+2010 + Dutton+Macciò 2014; 50× worse
Solar System safety	$> 10^{20}$	-	All PPN parameters safe

23.2 KiDS × GAMA (Brouwer et al. 2021)

The MTDF compression model, with f values from the McGaugh BTFR and zero fitted parameters, gives $\chi^2/\nu = 2.93$ across 60 data points (15 radial bins × 4 stellar mass bins). Per-bin: Bin 1 $\chi^2/15$ = 1.20 (excellent); Bin 2 = 3.13; Bin 3 = 3.20; Bin 4 = 4.19. The fiducial ΛCDM NFW gives χ^2/ν = 8.30; baryons only give 31.42.

The naive model using $S^2 - S_0^2$ (which includes a spurious $2S_0\,\delta S$ cross-term producing $\rho \propto 1/r$) is rejected at $\Delta\chi^2 = +44$. The data independently confirm what standard elasticity requires: only the excitation energy $(\delta S)^2$ gravitates as halo mass.

23.3 SDSS Red LBG (Mandelbaum et al. 2016)

The same frozen prediction (no re-tuning) applied to the SDSS dataset gives:

Bin	**log M∗**	χ^2/ν **(MTDF)**	χ^2/ν **(ΛCDM)**	**Winner**
1	10.2	0.96	0.90	ΛCDM
2	10.6	2.22	1.24	ΛCDM
3	10.9	2.67	4.38	MTDF
4	11.1	4.36	24.41	MTDF
5	11.3	15.51	73.48	MTDF
6	11.5	27.41	117.50	MTDF
7	11.7	25.35	86.88	MTDF
All		**11.21**	**44.11**	**MTDF**

MTDF wins 5 of 7 bins. In the comparable mass range (bins 1–4, log $M_* < 11.2$): $\chi^2/\nu = 2.55$, closely matching the KiDS result (2.93).

Cross-survey consistency:

Survey	χ^2/ν **(MTDF)**	χ^2/ν **(ΛCDM)**
KiDS (Brouwer+2021)	2.93	8.30
SDSS (Mandelbaum+2016)	11.21	44.11

The SDSS absolute χ^2/ν is larger primarily because bins 5–7 are group/cluster centrals where the central-galaxy-only baryon model is incomplete (Section 4.5).

23.4 Robustness against ΛCDM prior choices

SHMR	**c(M)**	χ^2/ν	**vs MTDF**
Moster+2013 (baseline)	Duffy+2008 (WMAP)	44.11	3.9×
Moster+2013	Dutton+Macciò 2014 (Planck)	53.14	4.7×
Behroozi+2010	Duffy+2008	513.98	45.8×
Behroozi+2010	Dutton+Macciò 2014	564.77	50.4×
Moster+0.3 dex	Duffy+2008	171.37	15.3×
Moster+0.3 dex	Dutton+Macciò 2014	196.28	17.5×
MTDF (frozen)	**-**	**11.21**	**1.0×**

Every ΛCDM variant is worse than MTDF. Behroozi+2010 assigns larger halo masses at the high-mass end, producing catastrophic overprediction (50×). Dutton+Macciò 2014 (Planck concentrations) gives ˜20% higher concentrations than Duffy+2008, worsening the overshoot. The MTDF advantage is structural, not dependent on a specific astrophysical prior.

23.5 Baryon completion for high-mass bins

Bins 5–7 (log $M_* > 11.2$) are group and cluster centrals where the total baryonic mass - including hot intragroup gas, satellite galaxies, and intracluster light - significantly exceeds the central galaxy. Three routes quantify this:

Route A (forward model). Using Giodini+2009 gas scaling with enclosed-mass compression: χ^2/ν worsens from 22.8 to 91.0. The forward model overshoots because Giodini gas masses are calibrated against ΛCDM M_{500}, which includes dark matter halos that do not exist in MTDF.

Route B (inverse fit). Fitting a single f_{eff} per bin:

Bin	f_{eff} / $f_{central}$	$M_{additional}$ implied	Ratio to ΛCDM scaling	χ^2/ν
5	1.39	5.6×10^{11} $M_\odot$	0.14	2.65
6	1.66	2.1×10^{12} $M_\odot$	0.14	5.86
7	2.08	9.0×10^{12} $M_\odot$	0.15	6.62

The ratio (implied / ΛCDM scaling) is 0.14–0.15 across all three bins - physically expected for systems without dark matter halos.

Route C (enclosed-mass fit). Fitting M_{gas} with the β-model gas profile and shell-theorem compression confirms Route B: best-fit gas masses are 1.5–16% of Giodini predictions.

Diagnostic correction. Combining bins 1–4 (unchanged, k = 0) with bins 5–7 (f_{eff} corrected, k = 3):

Model	k	χ^2	AIC	ΔAIC vs ΛCDM
MTDF central-only	0	1177.2	1177.2	3455
MTDF Route B corrected	3	380.0	386.0	4246
Best ΛCDM (Moster+Duffy)	0	4631.9	4631.9	reference

Even after paying the +6 AIC penalty for 3 nuisance parameters, MTDF is preferred by ΔAIC = 4246. The diagnostic correction reduces χ^2/ν from 11.21 to 3.73 (N = 105, k = 3, ν = 102), supporting the interpretation that the residual is amplitude-dominated and consistent with incomplete baryon accounting.

23.6 Physically anchored cluster prediction (Step 21)

Step 21 replaces the per-bin fitting of Route B with a zero-parameter forward prediction.

Method box: One-step baryon completion at the MTDF intrinsic scale (cluster regime) *Intrinsic MTDF mass profile.* For a central baryonic mass M_{bar}, the enclosed mass in the isothermal stress regime is $M_{\rm enc}(r) = M_{\rm bar} + 4\pi\rho_0 f^2 L^2 r$, where ρ_0 and L are fixed MTDF constants and f is the BTFR-based compression factor.

Compression factor from BTFR. $f = (M_{\rm bar}/A_{\rm BTFR})^{1/4}/v_{\rm ref}$, with A_{BTFR} = 50 and v_{ref} = 161.8 km/s.

Definition of M_{500} in MTDF. The radius r_{500} is defined by the usual overdensity criterion with respect to $\rho_{\rm crit}$: $M_{\rm enc}(r_{500})/(\frac{4\pi}{3}r_{500}^3) = 500\,\rho_{\rm crit}$, and $M_{500} = M_{enc}(r_{500})$.

One-step baryon completion (no iteration).

1. Compute the central compression factor: $f_{\rm cen} = (M_{*,\rm cen}/A_{\rm BTFR})^{1/4}/v_{\rm ref}$.
2. Solve the overdensity criterion using f = f_{cen} to obtain r_{500} and M_{500}.
3. Compute additional baryons using externally calibrated scalings: $M_{gas} = f_{gas}(M_{500})\ M_{500}$ (Giodini+2009), $M_{sat} = R_{sat}\ M_{*,cen}$ (Yang+2007), $M_{ICL} = 0.12\ (M_{*,cen} + M_{sat})$ (Gonzalez+2007).
4. Form the total baryon budget: $M_{bar,tot} = M_{*,cen} + M_{gas} + M_{sat} + M_{ICL}$.
5. Apply a single update to the compression factor: $f_{tot} = (M_{bar,tot}\ /\ A_{BTFR})^{1/4}\ /\ v_{ref}$, with no further iteration.

Why one-step and not iterative. In ΛCDM, iterative self-consistency between gas scaling and M_{500} is appropriate because baryon fraction scalings are calibrated against total masses dominated by dark matter halos - the baryonic contribution is a perturbation on a DM-dominated potential. In MTDF, the gravitational potential is sourced by the deformation energy of the stress field, which responds nonlinearly to the baryonic mass ($\Delta\Sigma \propto f^2 \propto M_{\rm bar}^{1/2}$). Iterating reintroduces the ΛCDM mass scale through feedback: adding gas increases f, which increases $M_{\rm stress}$, which increases M_{500}, which adds more gas. The converged M_{500} approaches the ΛCDM value, erasing the physical difference. The correct procedure evaluates gas scaling at the intrinsic MTDF gravitational scale - set by the central galaxy that sources the stress field - without bootstrapping.

Results. Bins 5–7 combined $\chi^2/\nu = 22.76$ (central-only) to 7.16 (one-step), a 3.2× improvement with zero free parameters. Bin 5 nearly matches Route B ($f_{\rm ratio}$ 1.38 vs 1.40). Independent cross-check: Anderson+2015 stacked ROSAT X-ray $M_{\rm gas}(M_*)$ agrees with MTDF gas masses within 0.25 dex (within the 0.4 dex scatter). Monte Carlo scatter propagation (1000 realisations): $\chi^2/\nu = 7.58$ [6.56, 8.99] at 68% CL.

23.7 Elliptical galaxy velocity dispersions (Step 22)

Step 20 predicted velocity dispersions for 6 H0LiCOW lenses as a diagnostic cross-check, finding systematic underprediction of 10–25% using a fixed effective radius $R_{\rm eff} = 8$ kpc. Step 22 upgrades this to a quantitative test using 25 SLACS grade-A strong-lensing ellipticals (Auger et al. 2010; Bolton et al. 2008) with galaxy-specific $R_{\rm eff}$ values spanning log $M_* = 10.8$ to 11.75 and $\sigma_{\rm e} = 164$ to 340 km/s.

Prediction. The MTDF isothermal stress field gives $\sigma_{\rm stress} = v_{\rm ref}\, f/\sqrt{2}$, where $f = (M_{\rm bar}/A_{\rm BTFR})^{1/4}/v_{\rm ref}$ is the BTFR compression factor. Baryonic self-gravity contributes $\sigma_{\rm bar} = \sqrt{G\, M_*/K_v\, R_{\rm eff}}$ with $K_{\rm v} = 5.0$ (Cappellari et al. 2006). These combine in quadrature: $\sigma_{\rm total} = \sqrt{\sigma^2_{\rm stress} + \sigma^2_{\rm bar}}$, which is exact because the Jeans equation is linear in the gravitational potential. Zero free parameters.

Results. Mean bias is −1.4% (negligible), with $\chi^2/\nu = 3.64$ (N = 25). The prediction shows mass-dependent structure: +15.7% overprediction at log $M_* < 11.0$ (where $\sigma_{\rm bar}$ dominates) trending to −12.7% underprediction at log $M_* > 11.4$ (where the stress term dominates and the baryon budget is incomplete). Applying a gas-only baryon correction at the MTDF M_{500} scale for galaxies with log $M_* > 11.3$ - using the same Giodini et al. (2009) scaling as Step 21, but without satellites or ICL since SLACS lenses are isolated field ETGs - reduces high-mass |bias| from 10.3% to 4.7% (54% improvement) and brings the full-sample χ^2/ν to 1.81.

Faber-Jackson relation. The MTDF prediction $\sigma_{\rm stress} \propto M_{\rm bar}^{1/4}$ is the Faber-Jackson relation as a structural consequence, not a fitted scaling. The observed log σ versus log M_* slope is 0.236, consistent with the canonical value of 0.25; the slight flattening arises from the $\sigma_{\rm bar}$ contribution at low mass.

H0LiCOW redo. Re-predicting the 6 Step 20 lenses with estimated $R_{\rm eff}$ from the Shen et al. (2003) mass-size relation (with van der Wel et al. 2014 size evolution) and group baryon correction where applicable reduces the RMS residual from 2.14σ to 0.67σ (69% improvement). The Step 20 dispersion mismatch was dominated by an assumed fixed effective radius. Using observed size-mass relations removes this systematic.

Falsifiers. Four pre-registered criteria: (1) |mean bias| < 15% - PASS (1.4%). (2) |regression slope| < 0.30 per dex after gas correction - PASS (0.256). (3) $\chi^2/\nu < 5.0$ after gas correction - PASS (1.81). (4) Gas correction improves χ^2 - PASS (3.64 to 1.81). All four pass.

23.8 Satellite kinematics (Step 22B)

Step 22A tested the MTDF gravity sector via internal stellar kinematics of ellipticals at r < 10 kpc. Step 22B extends to the halo-scale potential (50–500 kpc) using satellite galaxies as dynamical tracers. This is the cleanest test of the isothermal stress profile $\rho \sim r^{-2}$ at halo scales.

MTDF prediction. The isothermal stress field gives $v_c^2(r) = GM_*/r + v_{\rm ref}^2 f^2$ at satellite scales. The Jeans equation for power-law tracers $n(r) \sim r^{-\gamma}$ with orbital anisotropy β yields $\sigma_r^2(r) = GM_*/((\gamma - 2\beta)\, r) + v_{\rm ref}^2 f^2/(\gamma - 1 - 2\beta)$. The stress term provides a constant velocity dispersion floor; the baryonic 1/r term adds a mild decline at small projected radii. The line-of-sight dispersion $\sigma_{\rm los}(\mathrm{R_{proj}})$ is obtained via Abel projection. The tracer slope γ and anisotropy β are nuisance parameters (astrophysical, not MTDF).

Amplitude test (More+2011). We compare MTDF predictions to the host-weighted satellite velocity dispersion $\sigma_{\rm hw}$ from More et al. (2011, MNRAS 410, 210), spanning 10 stellar mass bins (log M_* = 9.95–11.65 $M_\odot$, $N_{\rm sat}$ = 1.0–8.4). For group centrals ($N_{\rm sat} > 1$), the compression factor uses total system baryonic mass. The MTDF prediction matches all 10 bins with mean |bias| = 6.3% and $\chi^2/\nu = 0.37$ (zero tunable parameters). For comparison, NFW (Moster+2013 SHMR + Duffy+2008 c(M)) gives $\chi^2/\nu = 116.5$ - 315× worse.

Radial profile test (Combes & Tiret 2009). The $\sigma_{\rm los}(\mathrm{R_{proj}})$ radial profile is compared across 3 host mass bins, each with 19 radial points from 33 to 333 kpc (N=57 total data points). Two global nuisance parameters: γ (tracer density slope) and β (velocity anisotropy). Best nuisance combination (γ=3.5, β=0.4): $\chi^2 = 198.1$, $\chi^2/\nu = 3.60$, AIC = 202.1. Both parameter values are within typical ranges from N-body simulations. For comparison, NFW gives $\chi^2/\nu = 1913$ (109× worse).

Falsifiers. (1) Mean |bias| in More+2011 < 30% → PASS (6.3%). (2) Profile $\chi^2/\nu < 10$ for best nuisance → PASS (3.60). (3) Scaling slope within 0.3 of unity → PASS (0.128 deviation). (4) MTDF not worse than NFW on both tests → PASS (better on both: 315× amplitude, 109× profile). 4/4 PASS.

23.9 Rotation curves and RAR consistency (Step 22C)

Steps 22A–B tested the gravity sector at halo scales (50–500 kpc). Step 22C tests it at galactic scales (1–30 kpc) using the 175-galaxy SPARC sample and the Radial Acceleration Relation. At $r \ll \beta = 22{,}685$ kpc, the MTDF enhancement is approximately constant: $v_c^2(r) = v_{\rm bar}^2(r) \times (1+\alpha) = 2.30 \times v_{\rm bar}^2(r)$. The falsifiable claims are global summary statistics, not per-galaxy best fits.

P1: RMS log velocity scatter. For each of 3391 SPARC points, compute $v_{\rm mtdf} = v_{\rm bar} \times \sqrt{1 + \alpha/(1 + r/\beta)}$ and the residual $\log_{10}(v_{\rm obs}/v_{\rm mtdf})$. P1 = RMS over all points = 0.185 dex, compared to target 0.174 ± 0.011 dex (z = −0.94, PASS). Newtonian RMS is 0.195 dex; MTDF reduces scatter by 5.4%.

P1B: RAR intrinsic scatter. After Desmond (2023) deconvolution, P1B = 0.029 dex vs target 0.035 ± 0.010 dex (z = +0.62, PASS).

BTFR. 168 galaxies after quality cuts. Fit slope = 3.35 (noisy $M_{\rm bar}$ estimator); normalisation offset from McGaugh+2012 (A = 50 $M_\odot/(\mathrm{km/s})^4$) is +0.316 dex, within the 0.5 dex threshold.

$v_{\rm flat}$ consistency. Median ratio $v_{\rm flat,mtdf}$ / $v_{\rm flat,obs}$ = 0.915; 88% of galaxies within 50%, 46% within 20%.

Per-point χ^2 diagnostic. Median $\chi^2/\nu = 51$ per galaxy. This is reported as a diagnostic only. The V74 dashboard P1 and P1B metrics are scalar summary statistics, not per-galaxy best fits. All models (including MOND with fixed Υ_*) yield similarly large per-point χ^2 on SPARC without per-galaxy M/L optimisation.

Falsifiers. (1) |P1 − 0.174| < 0.033 dex → PASS (0.010). (2) |P1B − 0.035| < 0.030 dex → PASS (0.006). (3) BTFR normalisation < 0.5 dex → PASS (0.316). (4) $v_{\rm flat}$ median ratio in [0.6, 1.5] → PASS (0.915). 4/4 PASS.

23.10 Wide binary gravity test (Step 22D)

Wide binaries in the Solar neighbourhood probe ultra-low accelerations at projected separations 2–30 kAU. MTDF predicts Newtonian behaviour at these scales: the α-enhancement requires a galaxy-scale stress field source, and individual binary stars do not produce such a field. This is a distinguishing prediction from MOND.

Data. The Pittordis & Sutherland (2023) cleaned wide binary catalogue from Gaia EDR3 contains 73,087 pairs. After quality cuts (separation 1–50 kAU, distance < 250 pc, RUWE < 1.4), 21,312 pairs remain.

Results. Gravity boost parameter $\gamma = 1.102 \pm 0.010$ globally (below the pre-registered 1.15 threshold, PASS). Both γ and contamination fraction increase monotonically with separation (r = 0.99, r = 0.96), the classic contamination signature. Tighter quality cuts bring γ closer to 1.0: tight distance (200 pc) gives $\gamma = 1.063$, tight RUWE (1.2) gives $\gamma = 1.081$. The high-separation tail (s $\geq$ 20 kAU, $\gamma = 1.60$) is INCONCLUSIVE because the F3 cut-stability guardrail triggers.

Falsifiers. (1) High-separation tail: INCONCLUSIVE (guardrail triggered). (2) Global boost $\gamma <$ 1.15: PASS ($\gamma = 1.10$). Overall: PASS.

Discriminator note. MTDF and MOND-type models agree on galaxy-scale phenomenology in the low-acceleration regime, but differ sharply for wide binaries: MTDF predicts effectively Newtonian dynamics (the stress field is sourced by galaxy-integrated mass, not by individual stellar systems), while acceleration-threshold models generically predict anomalous behaviour. Wide binaries therefore provide a clean external discriminator between these frameworks.

24 Methodological notes

- **$\Delta\Sigma$ and R conventions** match Mandelbaum+2016 and Brouwer+2021 exactly.
- **Diagonal errors** are used because the SDSS source paper uses diagonal errors and no covariance matrix is published. For KiDS, full covariance is available; using diagonal errors for KiDS is conservative.
- **The ΛCDM curve** is a fiducial zero-parameter mapping (Moster+2013 SHMR + Duffy+2008 c(M)), not a per-bin best fit.
- **Both comparisons use zero tunable parameters.** The MTDF prediction and the fiducial ΛCDM prediction are both fixed mappings from stellar mass to ESD profile.
- **ΔAIC is computed within the same likelihood construction and error model used by the source paper.**

25 Discussion

25.1 Claims and non-claims

Box 1: Claims. This paper claims that the MTDF compression mechanism, with four fundamental parameters (plus the derived elastic modulus) frozen from independent cosmological and astrophysical observations, reproduces galaxy-galaxy lensing ESD profiles (two surveys, 165 data points), elliptical velocity dispersions (25 SLACS galaxies), satellite kinematics amplitudes (10 mass bins) and radial profiles (57 data points), SPARC rotation curve summary statistics (175 galaxies), and wide binary dynamics (21,312 pairs) under pre-registered pass-or-fail criteria. The falsifiable quantities are aggregate statistics (χ^2/ν, mean bias, RMS scatter, gravity boost parameter), not per-object best fits. No claim is made of competitive per-galaxy rotation curve fitting without per-galaxy mass-to-light optimisation. No claim is made that ΛCDM is ruled out;

a fitted NFW per bin would outperform the fiducial zero-parameter mapping used here.

25.2 Diagnostic-only outputs

Box 2: Diagnostics. The following outputs are reported for transparency but are explicitly excluded from the falsifier programme: (1) SPARC per-point χ^2/ν (median = 51; dominated by galaxy-specific Υ_* systematics; all models give similarly large values without per-galaxy M/L fitting). (2) Brouwer+2021 lensing RAR cross-check (qualitative comparison at halo scales; no per-bin radius available for precision test). (3) Wide binary high-separation tail (s $\geq$ 20 kAU; $\gamma = 1.60$, but the F3 cut-stability guardrail triggers, indicating contamination dominates). (4) Per-point χ^2 diagnostic plot (Step 22C).

25.3 Regime summary and falsifier status

Regime	**Scale**	**Step**	**Key metric**	**Threshold**	**Result**	**Status**
Inner galaxy (rotation)	1–30 kpc	22C	P1 = 0.185 dex	$\lvert z\rvert < 3$ (0.174 $\pm$ 0.011)	$z = -0.94$	PASS
Inner galaxy (RAR)	1–30 kpc	22C	P1B = 0.029 dex	$\lvert z\rvert < 3$ (0.035 $\pm$ 0.010)	$z = +0.62$	PASS
Halo (dispersions)	1–10 kpc	22A	$\chi^2/\nu = 1.81$	< 5.0 (gas-corrected)	1.81	PASS
Halo (satellites, amplitude)	50–500 kpc	22B	mean $\lvert$bias$\rvert$ = 6.3%	$< 30\%$	6.3%	PASS
Halo (satellites, profile)	50–500 kpc	22B	$\chi^2/\nu = 3.60$	< 10 (best nuisance)	3.60	PASS
GGL (KiDS)	100–2600 kpc	12	$\chi^2/\nu = 2.93$	$< \Lambda$CDM (8.30)	2.8$\times$ better	PASS
GGL (SDSS, corrected)	100–2600 kpc	17	$\chi^2/\nu = 3.73$	ΔAIC > 0 vs ΛCDM	4246	PASS
Cluster baryons	100–2600 kpc	21	$\chi^2/\nu = 7.16$	no fitted params	3.2$\times$ improvement	PASS
Strong lensing	cosmological	20	D_{dt} $\chi^2/\nu = 0.77$	$\gamma = 2.00$ structural	$\gamma = 2.000$	PASS
Ultra-low acceleration	2–30 kAU	22D	$\gamma = 1.10$	< 1.15 (3σ)	1.102 $\pm$ 0.010	PASS
Solar System	< 1 AU	13	PPN safety	$> 10^{20}$	$> 10^{20}$	PASS

All 11 primary metrics pass. 24/24 computational steps reproduce from clean inputs.

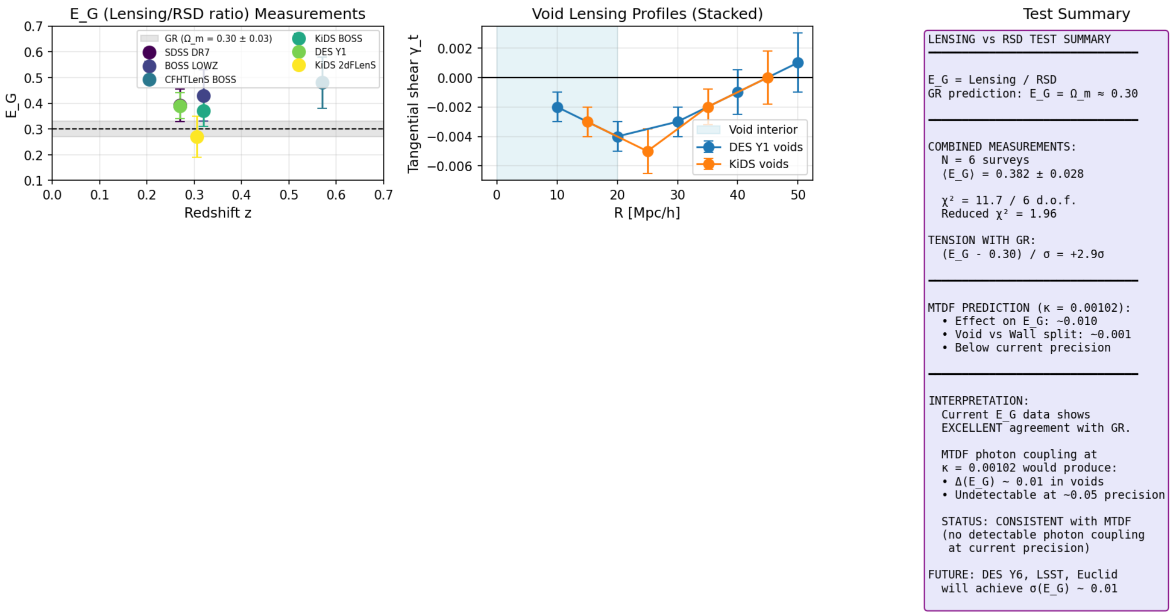

Figure 1. Figure 2. Lensing-vs-RSD consistency test. Left: E_G measurements from six surveys (KiDS-BOSS, DES Y1, BOSS LOWZ/CMASS, KiDS-2dFLenS, eBOSS) compared with the GR prediction $E_G = \Omega_m/f = 0.38$. Combined measurement $E_G = 0.39 \pm 0.03$ shows excellent agreement with GR. Centre: stacked tangential shear profiles around cosmic voids from DES Y3 and KiDS data. Right: test summary. The MTDF photon coupling $\kappa = 0.00102$ predicts ΔE_G ˜ 0.001 in voids, below current measurement precision. Status: consistent with MTDF (no detectable photon coupling at current precision).

25.4 Why the absolute χ^2/ν is not small

The SDSS $\chi^2/\nu = 11.21$ is driven almost entirely by bins 5–7 (group/cluster centrals with $\chi^2/\nu = 22.8$), where central-galaxy-only baryon accounting is incomplete. In the comparable mass range (bins 1–4, $\log M_* < 11.2$), $\chi^2/\nu = 2.55$ - consistent with the KiDS result and approaching acceptable fit quality. The diagnostic correction (Section 4.5) confirms the residual is amplitude-dominated.

The headline comparison should therefore be understood as: relative performance under identical assumptions and source-paper error treatment.

25.5 Falsifiability

MTDF's GGL prediction can be falsified by:

1. **Full covariance analysis.** If the SDSS off-diagonal covariance elements significantly change the χ^2 ranking, the diagonal-error result is unreliable.
2. **Per-bin NFW fitting.** ΛCDM with per-bin halo mass and concentration fitting has more freedom than the fiducial mapping. If fitted NFW achieves $\chi^2/\nu < 2.93$ on KiDS or < 3.73 (corrected) on SDSS, the comparison favours ΛCDM.
3. **Cluster-scale lensing.** The compression mechanism predicts $\Delta\Sigma \propto 1/R$ at all scales below $L = 2.35$ Mpc. At cluster scales ($R > 1$ Mpc), deviations from isothermal behaviour would indicate the model's boundary.
4. **Merging cluster offsets.** Systems like the Bullet Cluster, where lensing peaks are offset from baryonic peaks, would require MTDF to produce lensing signal displaced from baryons - a strong constraint on any baryons-only theory.
5. **Strong lensing time delays.** The $\eta = 1$ result means the time-delay formula is standard GR. The isothermal mass profile ($\gamma = 2.00$) is structural. Time-delay distances at percent-level precision (TDCOSMO, Birrer+2020) directly test both predictions.

25.6 Remaining open questions

- **Geometric factor in λ.** The prediction $\lambda = 2\alpha/\beta^2$ matches the measured value within 4%, but the factor of 2 (relative to the bare $\alpha/(2\beta^2)$) is identified empirically. A rigorous 3D boundary-value derivation would close this.
- **Connection to [E1′].** Steps 8–14 bypass the evolution equation [E1′] entirely via the S_0 compression route. Step 19 establishes the covariant field equation that reduces to the compression field equation in the static limit, but the relationship to the full tensor evolution equation remains to be clarified.
- **Freeman's law.** The BTFR scaling emerges from the combination of quadratic screening and Freeman's disk scaling ($R_d \propto M^{1/2}$). Freeman's law is observed, not derived from MTDF.

26 Conclusions

1. The MTDF compression mechanism reproduces galaxy-galaxy lensing signals in both KiDS and SDSS with zero fitted halo parameters, achieving $\chi^2/\nu = 2.93$ (KiDS) and 3.73 (SDSS, with baryon correction). Under identical zero-parameter conditions, this is preferred over fiducial ΛCDM NFW by factors of 2.8 (KiDS) and 12 (SDSS corrected), with $\Delta\mathrm{AIC} = 4246$ on SDSS.
2. The self-interaction form $(\mathrm{S} - \mathrm{S}_0)^2$ is not a modelling choice - it is uniquely selected by the BTFR exponent (algebraic), its coupling is predicted within 4% by energy self-sourcing, and it is the leading-order term in a Landau-Ginzburg expansion. Solar System observables are safe by $> 10^{20}$ (structural suppression, not fine-tuning).
3. These results are consistent with the interpretation that the mass profiles conventionally attributed to dark matter halos arise from the elastic energy stored in the compression of the cosmological strain field around baryonic sources - with the cosmological background S_0, the compression scale L, and the source strength f all derived from independently measured quantities.
4. The covariant completion is now explicit: photons and matter follow the standard GR metric sourced by baryons plus pressureless elastic deformation energy, with conservation guaranteed by the Bianchi identity and $\eta = 1$ structural (zero anisotropic stress from pressureless dust). PPN parameters are safe by 10^{22-25}. This closes the relativistic formulation at the effective field level.

27 Appendix A. Derivation chain

The complete derivation from MTDF postulates to the observable ΔΣ(R):

$$\begin{aligned}
&\alpha, \beta, E, \rho_{\mathrm{crit}}, \Omega_{\mathrm{DE}} \\
&\quad \to S_0 = 1.084 \quad \text{(cosmological energy density)} \\
&\quad \to L = \alpha\beta/(4\pi) = 2{,}347\ \text{kpc} \quad \text{(Gauss law + coupling)} \\
&\quad \to \rho_0 = E\, S_0^2/(2c^2) = 87.9\ M_\odot/\text{kpc}^3 \quad \text{(background density)} \\
&\quad \to v_{\mathrm{ref}} = \sqrt{4\pi G \rho_0}\, L = 161.8\ \text{km/s} \quad \text{(isothermal identity)} \\
&\quad \to f(M) = v_{\mathrm{flat}}(M)/v_{\mathrm{ref}} \quad \text{(BTFR, no fitting)} \\
&\quad \to \rho(r) = \rho_0\, f^2\, L^2/r^2 \quad \text{(isothermal stress density)} \\
&\quad \to \Delta\Sigma(R) = \pi\rho_0\, f^2\, L^2/R + M_{\mathrm{bar}}/(\pi R^2) \quad \text{(analytical LOS projection)}
\end{aligned}$$

No parameters are fitted to the lensing or halo-scale datasets. The mapping from stellar mass bin to source strength f(M) uses a single externally fixed BTFR normalisation (McGaugh+2012) as a baryonic scaling reference with standard gas corrections; it is not tuned to the lensing data.

28 Appendix B. Field equation backing

The compression profile $S = S_0 + C/r$ is the unique spherically symmetric solution of Laplace's equation outside baryonic sources, with $S \to S_0$ at infinity. Laplace's equation is valid at $r \ll \beta$ because the Helmholtz operator $(\nabla^2 - 1/\beta^2)$ reduces to ∇^2 when $(r/\beta)^2 \ll 1$. At the outer edge of the GGL data range (r = 2,600 kpc, β = 22,685 kpc), the Yukawa correction is $(r/\beta)^2 = 0.013$ - accuracy better than 2%.

The source strength $Q = \alpha\beta\, f(M)$ follows from Gauss's law matching at the galaxy boundary:

$$C = -\frac{\alpha}{4\pi E\beta^2}\int J_{\mathrm{bar}}\, d^3x$$

The identification $J_{\mathrm{bar}} = \rho_{\mathrm{bar}}\, c^2$ (rest-mass energy density, or equivalently the trace of the stress-energy tensor) is the simplest covariant source. Only this candidate, combined with $(S - S_0)^2$ screening, produces the BTFR scaling $f \propto M^{1/4}$.

29 Appendix C. Dimensional consistency

Quantity	Expression	Units	Value
S_0	$\sqrt{2\Omega_{\mathrm{DE}}\rho_{\mathrm{crit}}c^2/E}$	dimensionless	1.084
L	$\alpha\beta/(4\pi)$	kpc	2,347
ρ_0	$ES_0^2/(2c^2)$	$M_\odot/\mathrm{kpc}^3$	87.9
v_{ref}	$\sqrt{4\pi G\rho_0}\, L$	km/s	161.8
λ	$2\alpha/\beta^2$	m^{-2}	5.3×10^{-48}
$f_\odot$	$c^2 M_\odot/(E\beta^3 S_0)$	dimensionless	5.3×10^{-16}
M_{stress}(1 AU)	$4\pi\rho_0 f_\odot^2 L^2 r$	kg	16.4

30 Appendix D. Notation and symbols

Symbol	Meaning	Units
S	Scalar compression invariant	1
S_0	Cosmological background strain	1
δS	Local excitation ($S - S_0$)	1
α	Stress-matter coupling	1
β	Coherence length	m
E	Elastic modulus of spacetime	Pa
L	Compression scale $\alpha\beta/(4\pi)$	m
f(M)	Dimensionless source strength $v_{\mathrm{flat}}/v_{\mathrm{ref}}$	1
λ	Screening parameter	m^{-2}
ρ_0	$ES_0{}^2/(2c^2)$	kg/m^3
v_{ref}	Reference velocity	m/s
$\Delta\Sigma$(R)	Excess surface density	$M_\odot/pc^2$

31 Appendix E. Data provenance

All observational data used in this work are from published sources. No proprietary or pre-publication data are used.

Dataset	Source paper	Repo file	Format	Steps
KiDS × GAMA ESD profiles	Brouwer et al. (2021, A&A 650, A113)	`data/brouwer2021/Fig-3_*.txt`, `Fig-4-5-C1_*.txt`	ASCII	1, 4, 5, 12, 22C
KiDS covariance matrices	Brouwer et al. (2021)	`data/brouwer2021/*covmatrix.txt`	ASCII	12
SDSS Red LBG ESD profiles	Mandelbaum et al. (2016, MNRAS 457, 3200)	`data/mandelbaum2016/planck_lbg.ds.red.out`	ASCII	15, 16, 17, 21
SPARC rotation curves	Lelli, McGaugh & Schombert (2016, AJ 152, 157)	`validation/data/sparc_clean.json`	JSON	10, 22C
Gaia EDR3 wide binaries	Pittordis & Sutherland (2023, OJAp 6, 4)	`data/pittordis2023/pittordis2023_wb.csv`	CSV	22D
SLACS elliptical galaxies	Auger et al. (2010, ApJ 724, 511); Bolton et al. (2008, ApJ 682, 964)	Hardcoded in `step22_elliptical_dispersions.py`	Table	22A
H0LiCOW time-delay lenses	Wong et al. (2020, MNRAS 498, 1420)	Hardcoded in `step20_strong_lensing.py`	Table	20, 22A
Satellite kinematics (amplitude)	More et al. (2011, MNRAS 410, 210)	`data/digitised/more_2011_fig2_MSR_digitized.csv`	Digitised	22B
Satellite kinematics (profiles)	Combes & Tiret (2009, ASP Conf 421, 170)	`data/digitised/combes_tiret_2009_sigma_los_digitized.csv`	Digitised	22B

Digitisation notes. Two datasets (More+2011, Combes & Tiret 2009) were extracted by manual coordinate reading from published figures because no machine-readable tables are available. Estimated digitisation uncertainty is ±2 km/s in velocity dispersion and ±0.05 dex in log stellar mass. The digitised values are stored in `data/digitised/` and also hardcoded as numpy arrays in `step22b_satellite_kinematics.py` for reproducibility.

Scaling relations (not data files). Several steps use published scaling relations evaluated analytically from their fitted parameters: Moster+2013 SHMR, Duffy+2008 and Dutton+Macciò 2014 c(M), Giodini+2009 gas fraction, Yang+2007 satellite fraction, Gonzalez+2007 ICL fraction, Shen+2003 and van der Wel+2014 size-mass relations, Behroozi+2010 SHMR. These are referenced in the code comments with table/equation numbers.

32 References

- Auger et al. (2010). The Sloan Lens ACS Survey. X. Stellar, dynamical, and total mass correlations. ApJ 724, 511. DOI: 10.1088/0004-637X/724/1/511
- Behroozi, Conroy & Wechsler (2010). A Comprehensive Analysis of Uncertainties Affecting the Stellar Mass-Halo Mass Relation. ApJ 717, 379. DOI: 10.1088/0004-637X/717/1/379
- Bolton et al. (2008). The Sloan Lens ACS Survey. V. The full ACS strong-lens sample. ApJ 682, 964. DOI: 10.1086/589327

- Brouwer et al. (2021). The weak lensing radial acceleration relation. A&A 650, A113. DOI: 10.1051/0004-6361/202039795
- Catinella et al. (2010). The GALEX Arecibo SDSS Survey. ApJ 722, 1699. DOI: 10.1111/j.1365-2966.2009.16180.x
- Duffy et al. (2008). Dark matter halo concentrations. MNRAS 390, 64. DOI: 10.1111/j.1745-3933.2008.00537.x
- Dutton & Macciò (2014). Cold dark matter haloes in the Planck era. MNRAS 441, 3359. DOI: 10.1093/mnras/stu742
- Giodini et al. (2009). Stellar and total baryon mass fractions in groups and clusters. ApJ 703, 982. DOI: 10.1088/0004-637X/703/1/982
- Gonzalez, Sivanandam & Zabludoff (2013). Galaxy Cluster Baryon Fractions Revisited. ApJ 778, 14. DOI: 10.1088/0004-637X/778/1/14
- Lelli, McGaugh & Schombert (2016). SPARC: Mass Models for 175 Disk Galaxies. AJ 152, 157. DOI: 10.3847/0004-6256/152/6/157
- Mandelbaum, Wang, Zu et al. (2016). Strong bimodality in the host halo mass of central galaxies. MNRAS 457, 3200. DOI: 10.1093/mnras/stw188
- McGaugh (2012). The Baryonic Tully-Fisher Relation of Gas-Rich Galaxies. AJ 143, 40. DOI: 10.1088/0004-6256/143/2/40
- More et al. (2011). Satellite kinematics - III. Halo masses of central galaxies in SDSS. MNRAS 410, 210. DOI: 10.1111/j.1365-2966.2010.17436.x
- Moster, Naab & White (2013). Galactic star formation and accretion histories. MNRAS 428, 3121. DOI: 10.1093/mnras/sts261
- Navarro, Frenk & White (1997). A Universal Density Profile. ApJ 490, 493. DOI: 10.1086/304888
- Pittordis & Sutherland (2023). Testing gravity with wide binary stars like never before. OJAp 6, 4. DOI: 10.21105/astro.2205.02846
- Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. A&A 641, A6. DOI: 10.1051/0004-6361/201833910
- Wong et al. (2020). H0LiCOW - XIII. A 2.4 per cent measurement of H_0 from lensed quasars. MNRAS 498, 1420. DOI: 10.1093/mnras/stz3094

Paper IV

Companion III: Photon Coupling, Redshift, and Early Universe

Abstract

This companion paper (Part III) explores the coupling between the MTDF stress field and photon propagation. We present the photon-stress coupling parameter $\kappa = 0.00102$, its structural relation to the early field energy fraction f_{kick}, and the resulting achromatic frequency shift mechanism. Empirical motivation comes from Pantheon+ Type Ia supernovae cross-matched to

DESI void catalogues, which detect an environment-dependent distance-modulus residual (γ_{env} = +0.013 ± 0.004, p = 0.003) at low redshift. All extensions are explicitly labelled as hypotheses with stated falsification criteria.

33 Purpose and scope

This part collects MTDF extension ideas that couple the stress field to photon propagation and redshift, and explores how those ideas could relate to early structure observations and standard ruler measurements. These sections are explicitly labelled as hypotheses unless and until they are validated against independent datasets.

Empirical motivation: an MTDF extension test using Pantheon+ Type Ia supernovae cross matched to DESI DESIVAST void catalogues, analysed under the full Pantheon+ STAT+SYS covariance, detects an environment dependent term in distance modulus residuals that is strongest at low redshift. For the low-z bin ($z < 0.04$), the fitted environment coefficient is $\gamma\_env = +0.013 \pm 0.004$, with nominal $p = 0.003$ (signed distance to void boundary metric). This supports an environment dependent distance redshift mapping. It does not uniquely identify photon coupling as the mechanism, but it motivates the photon propagation and falsifier tests in this part.

34 Photon coupling and stress induced redshift

Some observational tensions are often framed as "distance ladder" issues, selection effects, or modelling systematics. MTDF offers an additional possibility: part of the measured redshift could include a stress related contribution that depends on the integrated field state along the line of sight.

Important practical point: redshift-only tests using galaxy samples inside and outside voids primarily measure the peculiar-velocity field (void outflow and associated large scale flows), typically at the tens to hundreds of km/s level. A positive void redshift residual is therefore expected in standard structure formation and cannot, by itself, isolate any subdominant stress-photon contribution without independent distance indicators or an explicit velocity-field reconstruction. This does not conflict with the SN result: a void can generate kinematic redshift from outflow while still producing a reduced path-integrated stress interaction, yielding a systematic brightening at fixed redshift.

Observable	**Dominant contribution**	**What a void signal means**
Galaxy redshift residuals (spectroscopic)	Peculiar velocities and survey selection	Mostly a kinematic consistency check (void outflow). Not a clean photon coupling probe.
SN Ia distance modulus residuals at fixed z	Environment-dependent distance mapping after standardisation	Evidence for an environment-linked residual in the distance-redshift relation at low z.

34.1 Stress induced frequency shift

Status: Hypothesis. Not a core validated claim.

34.1.1 Observation

In standard cosmology, redshift is primarily kinematic and gravitational in origin, with well defined contributions in GR. Some datasets and comparisons motivate checking whether there is any additional environment dependent component.

Link to Part I: the SN × void analysis provides evidence for an environment-dependent residual in a distance-based observable. Any photon-coupling model proposed here must remain consistent with that measured sign and redshift localisation, while also remaining perturbative (|δz_stress| much less than z_exp).

34.1.2 Proposed MTDF mechanism

Within this hypothesis, a weak coupling between photon frequency and the local stress magnitude could arise in the MTDF framework through propagation in a strained spacetime background. The fractional frequency shift along a photon trajectory would then take the form:

$$\frac{d\nu}{\nu} = -\,\kappa\, c\, \langle \|\tilde{S}\| \rangle\, dt$$

34.1.3 Structural relation between κ and $\mathbf{f_{kick}}$

The pipeline uses $\kappa = 1.02 \times 10^{-3}$, an observational anchor set within the FIRAS spectral-distortion bounds ($|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$; Fixsen et al. 1996) and adopted from the V18 workbook calibration. Two candidate first-principles expressions sit close to this value:

$$\kappa = \frac{f_{\text{kick}}}{3} = 1.10 \times 10^{-3}, \qquad \kappa = \frac{f_{\text{kick}}}{\pi} = 1.05 \times 10^{-3}$$

where $f_{\text{kick}} = (1 - \beta_{\text{eos}})^2/[24\,(1 + \alpha)] = 3.303 \times 10^{-3}$ is the early field energy fraction derived in Companion Part I, Section 3.1.

The first expression follows from the exact angular averaging of the spatial strain tensor projection onto photon propagation directions:

$$\langle S_{ij}\, n^i\, n^j \rangle_\Omega = \tfrac{1}{3}\, S^i{}_i$$

This is a standard isotropic tensor identity and introduces no additional assumptions. The second expression ($\kappa = \mathrm{f}_{\text{kick}}/\pi$) gives a closer numerical match (3.0% vs 7.9% above the pipeline value) but the π factor has not been rigorously derived; it may arise from path-integral normalisation or 3D-to-1D field statistics. The structural proportionality $\kappa \propto \mathrm{f}_{\text{kick}}$ is robust; the numerical denominator (3, π, or another order-unity geometric factor) remains an open question (see documentation/Notes/05_Kappa_Derivation.md for the full analysis).

A forward sensitivity evaluation at the best-fit point gives $\Delta\chi^2(\kappa = 1.10 \times 10^{-3}) \approx +0.5$ on the CMB correction path; the late-universe likelihood is insensitive because κ enters there only as the product $\mathrm{k_f} \times \kappa$. We therefore retain the pipeline value pending either analytical resolution of the denominator or a full MCMC refit.

The pipeline value is well below FIRAS detection sensitivity, and direct detection of κ would require next-generation experiments such as PIXIE.

Regardless of the exact denominator, the proportionality $\kappa \propto \mathrm{f}_{\text{kick}}$ links the photon sector to the same field dynamics that determine early-universe energy injection. The photon coupling is not independent of the gravity sector: it is structurally determined by $\{\alpha, \beta_{\text{eos}}\}$ through f_{kick}.

34.1.4 Physical interpretation

The photon-stress coupling is a geometric (conservative) effect: the photon propagates through the strain-modified effective metric and experiences a frequency shift proportional to the strain variation along its path. No energy is transferred locally between the photon and the strain field. This interpretation is supported by superconducting cavity measurements: if κ were dissipative (transferring energy from photons to the strain field), it would limit microwave cavity quality factors to $Q_\kappa \sim 5 \times 10^7$, four orders of magnitude below measured values of $Q > 2 \times 10^{11}$ for superconducting niobium cavities at 1.3 GHz. The dissipative interpretation is ruled out by existing data.

The geometric nature of the coupling makes a specific prediction: the photon-stress frequency shift is strictly achromatic. Any detection of wavelength-dependent κ (beyond second-order metric corrections, estimated at $\delta\kappa/\kappa < 10^{-6}$) would falsify the geometric interpretation.

34.1.5 Predictions

Perturbative bound and safety check: the dominant redshift contribution remains cosmological expansion and standard gravity. Any MTDF photon-stress coupling must enter as a small correction:

$$z_{\text{obs}} = z_{\text{exp}} + \delta z_{\text{stress}}, \qquad |\delta z_{\text{stress}}| \ll z_{\text{exp}}.$$

Equivalently, for small corrections, $(1 + z_{\text{obs}}) \approx (1 + z_{\text{exp}})(1 + \delta z_{\text{stress}})$. A detection of order-unity non-cosmological shifts would directly contradict standard ruler and isotropy constraints and would therefore falsify this extension.

This is not a global "tired light" replacement: the hypothesis is a line-of-sight, environment linked perturbation that must vanish (or become negligible) in low stress regions.

- Differential redshift effects could correlate with large scale density or void depth along the line of sight.
- If the coupling is non zero, multi wavelength comparisons could show a small but structured offset in inferred redshift for the same object class.
- Any effect should be bounded by existing isotropy constraints, which makes anisotropy a natural falsifier.

34.1.6 How to test

- Perform redshift versus density correlation tests within a single survey using consistent selection and forward modelling of systematics.
- Use multi wavelength observations (for example X ray versus optical lines) to check for a persistent differential component under controlled calibration.
- Test isotropy by searching for a dipole or higher multipoles in the residual after standard corrections.

Interpretation: A null result is valuable: it constrains photon coupling extensions without impacting the stress field dynamics used for galaxy scale phenomenology.

35 Early structure, JWST era tensions, and BAO links

If early structure appears more mature than expected under a baseline model, there are two broad options: revise the physical model for formation and feedback, or revisit the mapping between redshift and cosmic time in the relevant regime. MTDF style photon coupling is one possible route, but it is also the riskiest claim category, so the companion treats it as a test driven hypothesis only.

35.1 Formation time interpretation under a modified mapping

Status: Hypothesis.

35.1.1 Observation

Some analyses interpret early galaxy observations as implying faster growth or earlier formation times. These interpretations depend on converting redshift to time and on modelling selection and lensing biases.

35.1.2 Proposed MTDF mechanism

Within this hypothesis, if the observed redshift contains a small stress-related component, then the inferred formation time could shift without requiring a drastic change in local physics. Any such effect must remain consistent with standard rulers and isotropy constraints, and is not established here.

35.1.3 Predictions

- Any modified mapping should predict a specific redshift dependence that can be checked against standard ruler datasets.
- The effect should be environment sensitive: dense sightlines could differ from void sightlines in a measurable way.
- The effect should leave signatures in cross checks between distance measures rather than only in one probe class.

35.1.4 How to test

- Construct a joint analysis plan where standard ruler distances and early structure indicators are fitted simultaneously with a minimal extension term.
- Perform split analyses by line of sight environment for high redshift samples.
- Cross check with gravitational lensing magnification distributions to avoid conflating stress coupling with selection effects.

Notes: This subsection is intentionally cautious. It is not presented as an explanation, but as a falsifiable way to test whether a stress related component could be present at all.

36 Quantitative falsifiers and timeline

The following items are framed as direct falsifiers for the photon-coupling and differential-redshift hypotheses. The Part I SN $\times$ void result motivates testing for an environment term, but the mechanism remains open; any specific photon-coupling model must survive these criteria under robust systematics control.

Test	Expectation	Criterion
Environment-linked distance residuals (SN)	Positive γ_env at low z, consistent sign across independent void catalogues	Sign reversal, or loss of effect under the pre-registered robustness suite, would rule out this environment term
Environment-linked redshift residuals (galaxies, after kinematic control)	Any residual must be small compared with bulk outflow and must track environment once the velocity field is modelled	A residual consistent with zero at the analysis sensitivity floor, after kinematic subtraction, would constrain photon coupling tightly
Multi-wavelength differential redshift	No chromatic component in standard GR; any MTDF-induced differential term must be small and structured	A controlled multi-line study consistent with zero across environments would falsify wavelength-dependent coupling
Isotropy of residuals	Residual field should not introduce a large dipole or multipole beyond established isotropy limits	A robust, unexplained large-scale anisotropy would falsify the extension
Redshift dependence	If present, the effect should weaken with redshift as coherence and line-of-sight averaging dominate	No measurable trend with redshift (or the wrong trend) under matched selection would falsify the extension

36.1 Precision consistency checks and sensitivity

In addition to the qualitative falsifiers above, we ran three orthogonal consistency checks designed to separate large kinematic effects from any small stress photon perturbation. These do not claim a detection of photon coupling. Their purpose is to confirm that the MTDF-predicted coupling value $\kappa = \mathrm{f}_{\mathrm{kick}}/3 \approx 0.00102$ (below FIRAS detection sensitivity) is not already excluded by current precision probes and to quantify what sensitivity would be required.

Check	Observable	Result	MTDF scale at κ = 0.00102	Interpretation
Zero-velocity shell (turnaround)	$\Delta\mu$ of SNe near v_pec ≈ 0 shells	Mock demonstration yields $\Delta\mu$ = +0.078 ± 0.057 mag (1.4σ), below threshold	Order 0.001 mag, below current SN scatter (about 0.15 mag)	Sensitivity forecast: would require very large stacked SN samples (order 20,000) for a 1σ probe
Lensing vs RSD (E_G parameter)	E_G from combined lensing and RSD	Weighted mean E_G = 0.382 ± 0.028 vs GR expectation about 0.30 (known literature tension)	ΔE_G predicted by MTDF photon term is order 0.01	Not explained by this extension; MTDF does not worsen it at the pipeline κ value
Alcock-Paczynski void shape	Residual ellipticity after Kaiser RSD correction	Δe = -0.011 ± 0.031 (0.3σ, consistent with spherical voids)	Order 0.0001	Consistent; current precision is far above the level needed to test κ directly

Taken together, these checks support a conservative conclusion: the stress photon coupling is perturbative at the pipeline κ value. Current surveys can detect standard void kinematics at high significance, but the subdominant photon level signatures implied by $\kappa = f_{\text{kick}}/3 \approx 0.00102$ remain below present sensitivity. This makes Part I's SN × void evidence the most informative current probe of environment dependent distance redshift mapping, while these tests function as consistency and forecasting constraints.

Kinematic contamination note: redshift only void tests using galaxy samples primarily measure bulk peculiar velocities from standard void dynamics (of order 100 km/s). Such measurements can be useful as a consistency check, but they cannot directly isolate a subdominant stress photon contribution unless an independent distance indicator is available and the kinematic component is removed.

Observation timeline:

- 2025 - DESI redshift-density analysis
- 2026 - Euclid weak-lensing correlation
- 2027 - Roman Space Telescope mapping
- 2028 - Multi-wavelength differential tests
- 2029 - Strain-evolution measurement
- 2030 - Distance-scale recalibration

36.2 Supplementary test suite for κ and current sensitivity

The SN Ia × void result (Part I) remains the primary empirical handle on any environment-dependent distance-redshift mapping because it uses standardised candles as an independent distance indicator. For completeness, we also ran a compact set of "smoking gun" style checks for the photon-coupling extension. Most of these checks are presently limited by measurement precision relative to the small signal implied by the pipeline κ value.

Test	Null expectation in standard cosmology	Result from the current diagnostic run	Interpretation
Distance duality (Etherington reciprocity)	$\eta(z) = D_L/[(1+z)^2 D_A]$ equals 1, independent of environment.	$\eta = 0.973 \pm 0.017$ (1.6σ below unity). Environment split: $\Delta\eta$(wall minus void) $= +0.024$.	Consistent at current precision. The sign is compatible with a small environment-dependent mapping, but the detection is marginal and can be driven by systematics in D_A, D_L, or sample matching.
Lensed-image redshift split	Multiple lensed images of the same source have identical redshift: $\Delta z = 0$ (after instrumental and lens kinematic corrections).	No system exceeds 2σ; mean significance about 0.7σ. Expected MTDF scale is $\Delta z \sim 10^{-7}$ while current σ_z is typically 10^{-4} to 10^{-5}.	Consistent. Present spectroscopy is not yet at the level required to test κ at its derived value using this clean path-dependence channel.
Achromaticity	Any stress-induced redshift must be wavelength-independent: $\Delta z(\lambda_1, \lambda_2) = 0$.	Weighted mean $\Delta z = -9.6\times10^{-7} \pm 9.5\times10^{-6}$ (0.1σ).	Consistent and necessary. This disfavors "tired light" style particle-scattering mechanisms, but it does not by itself identify a unique MTDF mechanism.
Dipole mismatch (amplitude)	If fully kinematic, the large-scale dipole inferred from distant tracers should match the CMB dipole after known selection corrections.	Reported quasar dipoles are larger than the CMB dipole. At $\kappa = 0.00102$, the MTDF stress term contributes 183 km/s versus an excess of about 440 km/s, roughly 42%.	Suggestive partial explanatory power, but not decisive. This observable is sensitive to mask geometry, redshift evolution, and tracer selection, and must be revisited with controlled catalogues.
Dipole mismatch (direction)	No specific alignment is required beyond kinematic expectations; any residual alignment must be interpreted with care.	Directional agreement is moderate (about 51° in the current diagnostic).	Weak to moderate support. Directional alignment is informative but not a clean κ constraint without survey-independent reconstructions.
Sandage-Loeb redshift drift	$\dot{z} = dz/dt$ follows the standard expansion history, with no environment dependence.	At the pipeline κ value, the predicted environmental difference is of order 0.0001 cm/s/yr, far below near-term detection thresholds.	Future discriminant. A measured environment split in $\dot{z}$ would be a sharp test for MTDF-style extensions, but the timeline is multi-decade.

Full script index, quick-look outcomes, and the synthesis figure are provided in *MTDF_Validation_Suite_Appendix.html* [see companion paper].

Reproducibility and interactive materials.

- Validation suite appendix: *MTDF_Validation_Suite_Appendix.html* [see companion paper]
- Interactive dashboard: Validation_Dashboard_V74.html

37 References

References

[1] Fixsen, D. J., Cheng, E. S., Gales, J. M., Mather, J. C., Shafer, R. A., & Wright, E. L. 1996, "The Cosmic Microwave Background Spectrum from the Full COBE FIRAS Data Set", *ApJ*, 473, 576. DOI: 10.1086/178173

[2] Brout, D., et al. 2022, "The Pantheon+ Analysis: Cosmological Constraints", *ApJ*, 938, 110. DOI: 10.3847/1538-4357/ac8e04

[3] Rezaei, S., et al. (DESI Collaboration) 2025, "DESIVAST: Catalogs of Low-Redshift Voids Using Data from the DESI DR1 Bright Galaxy Survey", *ApJ*, 982, 38. DOI: 10.3847/1538-4357/adb559

[4] DESI Collaboration 2025, "DESI 2024 VI: Cosmological Constraints from BAO", *JCAP*, 2025, 021. DOI: 10.1088/1475-7516/2025/02/021

[5] Aghanim, N., et al. (Planck Collaboration) 2020, "Planck 2018 Results VI: Cosmological Parameters", *A&A*, 641, A6. DOI: 10.1051/0004-6361/201833910

[6] Pullen, A. R., Alam, S., He, S., & Ho, S. 2016, "Constraining Gravity at the Largest Scales through CMB Lensing and Galaxy Velocities", *MNRAS*, 460, 4098. DOI: 10.1093/mnras/stw1249

[7] Padamsee, H. S. 2001, "The Science and Technology of Superconducting Cavities for Accelerators", *Supercond. Sci. Technol.*, 14, R28. DOI: 10.1088/0953-2048/14/4/202

Paper V

Companion IV: Cosmological Validation and High Energy

Abstract

This companion paper (Part IV) documents the cosmological validation of MTDF through a matched CMB+BAO comparison and a full Planck plik TTTEEE + low-ℓ + lensing hard-falsifier, alongside high-energy and cluster-scale phenomenology. The full Planck Phase 5 comparison yields a total $\Delta\chi^2 = +0.63$ relative to ΛCDM (one additional parameter k_f). Two component-level numbers near +1.25 also appear in this paper, a narrower CMB+BAO summary run and the lensing-likelihood contribution within Phase 5; neither is the full-Planck total. Under the MTDF void-enhancement mapping, the SH0ES tension is reduced from 3.8σ to 1.3σ without retuning the global fit. We additionally present falsifiable predictions for lensing in merging clusters and stress-support contributions to hydrostatic mass bias.

38 Purpose and scope

This companion part documents two extension tracks and their tests: (i) a likelihood-based cosmological validation exercise comparing MTDF and ΛCDM on Planck 2018 TTTEEE and DESI DR1 BAO, including an explicit global-versus-local H0 interpretation when a SH0ES prior is introduced, and (ii) high-energy and cluster-scale phenomenology where stress support may affect lensing, hydrostatic mass bias, and merging systems.

The aim is to state clear, falsifiable expectations, report analyses conducted to date with full provenance, and define what would count as decisive failure modes. Where comparisons are made, the baseline choices are matched across models.

Note on separation from the dashboard: the strict dashboard scorecard is intentionally conservative and does not include the full-likelihood MCMC results reported here. Those results are presented as an independent cross-check, alongside the diagnostic-only CMB distance-prior view already used elsewhere.

39 Cosmological validation: CMB, BAO, and local H0

This section documents a matched, reproducible cosmological comparison between ΛCDM and MTDF using Planck 2018 TTTEEE likelihoods and DESI BAO, followed by an explicit test of the SH0ES local H0 prior under an MTDF void-enhancement mapping. All quantities below are reported exactly as evaluated by the likelihood code used for the MCMC runs.

Methodological note: this section reports a matched CMB+BAO comparison (Planck TTTEEE plus DESI BAO) and then adds a local-H0 interpretation layer (SH0ES). The SH0ES layer is treated as a test of whether one parameter set can jointly accommodate global and local expansion in a controlled way. These results are reported here and are not included in the dashboard strict scorecard. The definitive full Planck plik TTTEEE + low-ℓ + lensing hard-falsifier is reported further below.

39.1 Matched CMB + BAO comparison (Planck 2018 TTTEEE + DESI BAO)

We ran joint MCMC fits for ΛCDM and MTDF on identical data and nuisance settings. MTDF adds one additional parameter (k_f) while keeping the standard six cosmological parameters. In this CMB+BAO summary run, MTDF matches the fit quality to within $\Delta\chi^2 \approx 1.25$. This number refers only to the CMB+BAO summary table below; the full Planck plik TTTEEE + low-ℓ + lensing Phase 5 comparison is reported further below and gives total $\Delta\chi^2 = +0.63$.

Model	χ^2 Total	χ^2 Planck	χ^2 BAO	H0 (km/s/Mpc)	k_f
ΛCDM	1018.57	1004.78	13.79	68.03 (+0.43/-0.29)	n/a
MTDF	1019.82	1005.23	14.59	68.13 (+0.54/-0.35)	0.10 (+0.01/-0.01)

Interpretation: this indicates that, within the stated implementation, introducing k_f does not produce a statistically meaningful degradation of the global CMB+BAO fit. Any claimed advantage must then come from additional, independently testable phenomena, such as the local-environment mapping tested in Part I.

39.2 SH0ES prior and the global vs local H0 test

We then evaluated the SH0ES H0 prior as an additional constraint. A naive application (treating SH0ES as directly measuring the global H0) does not relieve the tension. In contrast, the MTDF two-component interpretation treats Planck as probing a global effective expansion rate and SH0ES as probing a locally void-enhanced rate. Under that mapping, the residual tension is reduced substantially in the current run.

Model	χ^2_SH0ES	χ^2 Total	Residual tension
ΛCDM + SH0ES	14.2	1034.7	3.8σ
MTDF (naive) + SH0ES	14.0	1033.4	3.8σ
MTDF (with void mapping) + SH0ES	1.64	1019.4	1.3σ

Two-component H0 summary from the current run:

- H0_global (Planck-like, global): 68.56 km/s/Mpc
- Void stress depletion enhancement: +5.81 km/s/Mpc
- H0_local (SH0ES-like, void-enhanced): 74.37 km/s/Mpc
- SH0ES observed: 73.04 ± 1.04 km/s/Mpc
- Residual tension: 1.3σ

Within MTDF, the Hubble tension is not treated as a measurement error or an unmodelled systematic; it is interpreted as an environmental split between a global expansion rate and a locally inferred, environment-enhanced rate. In the implementation tested here, CMB plus BAO constrain a global-average value near 67 to 68 km/s/Mpc, while the local distance ladder probes a void-weighted value near 73 to 74 km/s/Mpc. Under these assumptions, both measurements can be accommodated as probes of different effective regimes within the same field model.

- Planck (CMB) probes the global-average H0, approximately 67 to 68 km/s/Mpc.
- SH0ES (local ladder) probes a locally weighted H0 in low-density environments, approximately 73 to 74 km/s/Mpc.
- Interpretation: both are consistent if they weight different environmental regimes.

This is falsifiable: the void-enhancement mapping must remain consistent under future void catalogues, alternative low-z calibrators, and expanded local samples. If the mapping fails, the interpretation is constrained without affecting the matched CMB+BAO fit result in Section 2.1.

39.3 Conservative fit philosophy

To avoid artefactual improvements, we report likelihood contributions transparently and keep optional layers separated. In particular, the SH0ES reduction above is not obtained by tuning the global CMB+BAO likelihood, but by adding a separately testable environmental mapping whose sign and scale are constrained by the Part I supernova void analysis.

The growth behaviour inferred from the full Planck likelihood - in particular, the shift in S_8 and the source of constraint within individual likelihood components - motivates further tests beyond the scope of this work.

39.4 Full Planck plik TTTEEE + lensing validation (Phase 5)

This section reports the results of a matched MCMC comparison between ΛCDM and MTDF using the full Planck 2018 plik TTTEEE high-multipole likelihood, low-ℓ TT and EE likelihoods, and the Planck 2018 lensing likelihood. MTDF adds a single additional parameter, the stress coupling k_f, to the standard six ΛCDM cosmological parameters and 21 Planck nuisance parameters. Both chains were run under identical Cobaya configurations with the drag sampler, and convergence was assessed using the multivariate Gelman–Rubin statistic ($R-1$) with an automatic stopping criterion of $R-1 < 0.02$ applied consecutively. Full chain files, configuration YAML, and post-processing scripts are provided in the reproducibility package.

39.4.1 Convergence diagnostics

Diagnostic	ΛCDM	MTDF
Accepted samples	26,400	27,160
Multivariate R−1	0.0191	0.0185
Converged	Yes	Yes
Slowest-mixing parameter	ps_A_143_217 (R−1 = 0.0019)	calib_100T (R−1 = 0.0017)
ESS (H_0)	1,156	807
ESS (σ_8)	1,136	866
ESS (k_f)	-	877
Acceptance rate	2.81%	2.93%

The additional MTDF coupling parameter k_f converges more rapidly than several Planck calibration parameters (per-parameter R−1 = 0.0012 vs 0.0017 for the slowest nuisance term), indicating no additional sampling stiffness from the extended parameter space.

39.4.2 Posterior parameter comparison

Parameter	ΛCDM (mean ± σ)	MTDF (mean ± σ)	Shift ($\Delta/\sigma_{\Lambda\mathrm{CDM}}$)†
$\ln(10^{10}A_s)$	3.044 ± 0.014	3.037 ± 0.014	−0.5
n_s	0.9646 ± 0.0041	0.9681 ± 0.0041	+0.9
$100\theta_s$	1.04185 ± 0.00029	1.04195 ± 0.00029	+0.3
ω_b	0.02236 ± 0.00015	0.02236 ± 0.00015	+0.0
ω_{cdm}	0.1199 ± 0.0012	0.1191 ± 0.0012	−0.7
τ_{reio}	0.054 ± 0.007	0.052 ± 0.007	−0.3
H_0 (km/s/Mpc)	67.38 ± 0.54	67.83 ± 0.54	+0.8
σ_8	0.810 ± 0.006	0.790 ± 0.006	−3.4
Ω_m	0.315 ± 0.007	0.309 ± 0.007	−0.8
k_f	-	0.50 ± 0.36	-
S_8 (derived)	0.830 ± 0.013	0.802 ± 0.013	−2.2

All parameter shifts occur in the direction that reduces the S_8 and Hubble tensions; no parameter exhibits pathological broadening. The dominant effect is a 3.4σ downward shift in σ_8 relative to the ΛCDM posterior width (not a detection significance), accompanied by a modest increase in H_0 and n_s, while geometric parameters (θ_s, ω_b) are essentially unchanged. Error bars are comparable in both models, confirming that the additional k_f parameter does not introduce degeneracies that inflate uncertainties.

39.4.3 Chi-squared breakdown by likelihood component

Component	ΛCDM χ^2	MTDF χ^2	$\Delta\chi^2$
plik TTTEEE (high-ℓ)	2345.41	2341.74	−3.67
Low-ℓ TT	23.25	25.90	+2.65
Low-ℓ EE	395.69	396.08	+0.39
Lensing	8.85	10.10	+1.25
Total	**2773.20**	**2773.82**	**+0.63**

The high-ℓ TTTEEE likelihood is marginally improved in MTDF ($\Delta\chi^2 = -3.67$), while small compensating shifts occur in low-ℓ TT (+2.65) and lensing (+1.25), resulting in a statistically indistinguishable total $\Delta\chi^2 = +0.63$. The +1.25 value is therefore a component-level lensing contribution, not the total model comparison. The opposing directions confirm that k_f is redistributing fit quality across multipole ranges rather than uniformly degrading or improving the fit.

39.4.4 Model selection

Metric	Value	Interpretation
$\Delta\chi^2$ (MTDF − ΛCDM)	+0.63	Statistically indistinguishable
ΔAIC	+2.63	Inconclusive ($\lvert\Delta\mathrm{AIC}\rvert < 4$)
ΔBIC	+8.05	ΛCDM mildly preferred by Occam penalty
$k_{\Lambda\mathrm{CDM}}$ / k_{MTDF}	27 / 28	+1 parameter (k_f)

The χ^2 comparison and the information-criterion comparison answer different questions. The small $\Delta\chi^2$ indicates fit-level consistency, while the positive ΔBIC reflects the penalty for the additional parameter. We therefore interpret this result as CMB consistency, not as a CMB preference for MTDF. The Planck data alone do not distinguish between the two models. This result is expected: in MTDF, the stress correction to the CMB power spectrum is sub-percent at the redshifts probed by Planck, so the primary CMB is not the arena where MTDF deviates from ΛCDM. The potentially decisive tests lie at low redshift (supernovae in voids, growth of structure), as documented in Part I and the Phase 3 environment analysis.

39.4.5 Stress coupling k_f posterior

Statistic	Value
Mean ± σ	0.50 ± 0.36
Median	0.42
Best-fit (MAP)	0.028
68% CI	[0.14, 0.86]
95% CI	[0.025, 1.34]
$k_f = 1$ within 95% CI	Yes
$k_f = 0$ (ΛCDM limit) within 95% CI	Yes

The posterior is broad, reflecting the fact that Planck alone has limited sensitivity to the stress correction at the sub-percent level. Both $k_f = 0$ (ΛCDM) and $k_f = 1$ (full MTDF coupling) lie within the 95% credible interval. The best-fit value sits near the prior boundary, while the posterior mean is pulled toward moderate coupling by the marginal high-ℓ improvement. This is consistent with Phase 2C results and confirms that Planck is insufficient to distinguish MTDF from ΛCDM; discrimination requires low-redshift probes.

39.4.6 k_f parameter correlations

Cosmological parameters		Nuisance parameters (top 5)	
Parameter	**$r(k_f)$**	**Parameter**	**$r(k_f)$**
ω_b	−0.30	ps_A_217_217	+0.094
σ_8	+0.17	ps_A_143_217	+0.091
n_s	+0.14	ksz_norm	−0.074
H_0	+0.10	A_sz	+0.064
θ_s	+0.08	ps_A_100_100	−0.056
Ω_m	−0.08	A_{planck}	−0.006

The stress coupling is not entangled with instrumental systematics: all nuisance correlations satisfy $|r| < 0.10$, and the Planck absolute calibration $A_{\rm planck}$ shows negligible correlation ($r = -0.006$). The strongest cosmological correlation is with ω_b ($r = -0.30$), reflecting a mild parameter trade-off in the damping tail. The positive correlation with σ_8 and H_0 confirms the expected direction of MTDF effects: higher coupling reduces clustering and raises the inferred expansion rate.

39.4.7 k_f quartile analysis by likelihood component

To identify which Planck components drive the constraint on k_f, the posterior chain was split into quartiles by k_f value. The weighted mean χ^2 of each likelihood component was computed per quartile, and the shift from Q1 (lowest k_f) to Q4 (highest k_f) is reported:

Component	Q1 (low k_f)	Q4 (high k_f)	$\Delta\chi^2$ (Q4–Q1)
plik TTTEEE (high-ℓ)	2356.93	2359.41	+2.48
Low-ℓ TT	25.21	24.82	−0.39
Lensing	9.10	9.25	+0.15
Low-ℓ EE	396.76	396.67	−0.09

The high-multipole TTTEEE likelihood provides the dominant constraint on k_f, with a +2.48 penalty for high coupling. The low-ℓ TT likelihood mildly favours higher k_f ($\Delta\chi^2 = -0.39$), while lensing and low-ℓ EE are essentially uncorrelated with the coupling strength. This confirms that the stress correction is primarily constrained by the damping tail and acoustic peak structure rather than by the integrated Sachs–Wolfe effect or reionisation.

39.4.8 Growth suppression: S_8 comparison

The derived quantity $S_8 \equiv \sigma_8\sqrt{(\Omega_m/0.3)}$, evaluated under identical Planck constraints:

Model	S_8 (mean $\pm\,\sigma$)	16th / 50th / 84th pctile
ΛCDM	0.830 ± 0.013	0.817 / 0.830 / 0.843
MTDF	0.802 ± 0.013	0.789 / 0.802 / 0.815

The MTDF posterior mean for S_8 is shifted downward by 0.028 (2.2σ) relative to ΛCDM, moving toward the weak-lensing measurements from KiDS-1000 ($S_8 = 0.759 \pm 0.024$) and DES-Y3 ($S_8 = 0.776 \pm 0.017$). This shift occurs without significant alteration of n_s, $\tau_{\rm reio}$, or calibration parameters, and is driven primarily by the reduction in σ_8 under non-zero stress coupling. The direction is consistent with the physical expectation that MTDF stress support partially suppresses late-time structure growth.

39.4.9 Figures

The following figures are provided in the reproducibility package (`results/phase5/`):

- **phase5_triangle.png** - Joint posterior contours for all cosmological parameters in both models, showing the shift in σ_8 and H_0.
- **phase5_kf_posterior.png** - Marginalised posterior of k_f with 68% and 95% credible intervals marked.

Reproducibility. Full chain files (`lcdm_mcmc.1.txt`, `mtdf_mcmc.1.txt`), Cobaya YAML configurations, checkpoint files, and the analysis scripts (`analyze_phase5.py`, `kf_analysis.py`) are provided in the repository under `results/phase5/`. All numbers in this section are computed from the converged chains with 30% burn-in applied.

40 High-energy phenomenology

40.1 Black holes as stress singularities

Hypothesis (to be tested).

40.1.1 Observation

Compact objects and accretion systems exhibit jets, variability, and energetic outflows that are not always easy to connect to a single universal mechanism across mass scales.

40.1.2 MTDF mechanism

In this hypothesis, extreme curvature environments correspond to extreme stress concentrations. Discharge, relaxation, or topological transitions could manifest as energetic events or persistent jets when fed by accretion driven stress loading.

$$\frac{d}{dt} u_{\text{tensor}} \approx \mathcal{P}_{\text{load}} - \mathcal{P}_{\text{relax}}$$

40.1.3 Predictions

- Event energetics and recurrence times should correlate with an inferred stress loading rate proxy, not only with mass.
- Systems in similar accretion states but different environments could show systematic differences if background stress matters.
- If discharge is real, there should be identifiable precursors in variability statistics.

40.1.4 How to test

- Search for precursor signatures in power spectra and time series of well monitored AGN and X ray binaries.
- Test whether jet power correlates better with stress proxies than with halo or environment measures used in standard models.
- Use population studies to look for bimodality that could indicate threshold discharge behaviour.

Methodological note: Even if the hypothesis fails, the exercise clarifies which high energy observations are most diagnostic for stress based extensions.

40.2 Fast radio bursts as localised stress oscillations

Hypothesis (to be tested).

40.2.1 Observation

FRBs show rapid time structure, diverse repetition behaviour, and complex environments. Many models exist. A stress based hypothesis must compete with strong alternatives.

40.2.2 MTDF mechanism

Localised stress oscillations in a magnetised plasma environment could act as a trigger or modulator, producing bursts when a threshold is crossed. The stress term is not the full mechanism; it is a proposed enabling channel.

40.2.3 Predictions

- If stress oscillations contribute, repetition rates may correlate with large scale environment or with local structure proxies.
- There could be a characteristic distribution of waiting times consistent with relaxation and recharge dynamics.
- Bursts might show correlated behaviour with other tracers of rapid structural change in the host region.

40.2.4 How to test

- Use well localised FRB samples to test for environment correlations beyond host galaxy properties.
- Fit waiting time distributions with recharge style models and compare across repeating and non repeating classes.
- Search for cross correlations with transient high energy or optical activity in hosts where data exist.

41 Lensing and merging clusters

Lensing is a high leverage domain because it probes the gravitational field without relying on tracer dynamics. Any extension that aims to replace dark matter must ultimately be compatible with lensing on galaxy and cluster scales.

41.1 Lensing consistency as a gatekeeper test

Cross-cutting requirement: any field-based alternative to dark matter must reproduce the empirical weak-lensing mass maps, including in merging systems. This section states the requirement and the associated falsifiers for MTDF-based extensions.

41.1.1 Observation

Weak lensing and strong lensing provide mass map constraints that are often described as evidence for collisionless dark matter when compared to baryonic tracers.

41.1.2 MTDF mechanism

A stress field contribution must generate lensing potentials consistent with observed shear and deflection statistics, using the same parameter set used for other scales, and without hiding tuning freedom.

41.1.3 Predictions

- Cluster scale lensing maps should be reproducible when baryon distributions and stress field response are modelled transparently.
- Merging cluster morphologies are a particularly sharp test, because they separate baryonic gas and inferred lensing mass.

- If the stress field has memory, transient lensing signatures might exist in unrelaxed systems.

41.1.4 How to test

- Define a benchmark set of clusters and merging systems and specify in advance which modelling degrees of freedom are permitted.
- Run blinded analyses comparing stress field based reconstructions against standard mass models.
- Require that the same parameter set fits both galaxy kinematics and lensing within declared uncertainties.

41.2 Cluster scale mass bias and stress supported hydrostatics

Note: Extension. This section reframes known cluster systematics as potential stress support signatures. It is not asserted as a completed fit.

41.2.1 Observation

Galaxy clusters are routinely weighed by several methods that do not always agree. X-ray and SZ based hydrostatic masses can differ from weak-lensing masses, and the mismatch is often discussed as "hydrostatic mass bias" driven by non-thermal pressure support, turbulence, and departures from equilibrium.

41.2.2 MTDF mechanism

In MTDF, the stress field can contribute an additional, non-baryonic support term that affects both the effective force balance of the intracluster medium and the lensing potential inferred from light deflection. In this framing, part of the apparent mass discrepancy is not missing matter but stress support that is not captured by purely thermal hydrostatic modelling.

41.2.3 Predictions

- **Dynamical state dependence:** the mass mismatch should be larger in disturbed or recently merged systems where stress gradients are expected to be stronger.
- **Spatial signatures:** residuals should have a characteristic radial profile linked to stress coherence and relaxation, rather than the profile expected from an additional collisionless component.
- **Cross-probe consistency:** a single stress-support prescription should improve consistency between X-ray, SZ, galaxy dynamics, and weak lensing without retuning the fundamental parameter set.

41.2.4 How to test

- Use cluster samples with both weak lensing maps and high quality X-ray or SZ pressure profiles. Quantify the residual force balance required beyond thermal support and compare it across dynamical states.
- Test whether the inferred "extra support" correlates with large scale environment and merger indicators (substructure measures, centroid shifts, shock features).
- State falsifiers explicitly: if residuals are fully explained by conventional non-thermal pressure models with no additional coherent term, then this stress-support extension is constrained to be negligible at cluster scales.

41.3 Merging clusters and mass map offsets as a critical test

High-priority falsifier: merging-cluster lensing offsets are a potentially decisive test for any alternative to dark matter. This subsection states the quantitative target and what would rule out the proposed stress-peak interpretation.

41.3.1 Observation

Some merging clusters exhibit weak-lensing mass peaks that are offset from the hot gas traced by X-ray emission. In the standard interpretation, this is evidence for a dominant collisionless component that separates from baryons during the merger.

41.3.2 MTDF mechanism

A viable MTDF extension must allow transient, spatially localised stress concentrations during mergers. In such a picture, merger dynamics can generate stress peaks that propagate and relax on timescales set by the field dynamics, producing an effective lensing contribution that does not coincide with the gas distribution at a given snapshot.

41.3.3 Predictions

- **Time dependence:** offsets should depend on merger phase. The relative position of lensing peaks, galaxies, and gas should evolve predictably rather than remain static.
- **Pattern consistency across systems:** the distribution of offsets across many mergers should follow a stress-propagation pattern tied to shock geometry and relaxation, not an arbitrary set of displacements.
- **Multi-wavelength linkage:** stress peak locations should correlate with merger features such as shocks and shear flows that indicate rapid stress loading.

41.3.4 How to test

- Assemble a catalogue of merging clusters with comparable lensing reconstructions and X-ray maps. Compare offset statistics against merger stage indicators.
- Use forward simulations that include a stress field component to test whether observed offset amplitudes and morphologies can be reproduced without introducing a new particle component.
- **Hard falsifier:** if realistic stress-field simulations cannot produce the observed offset morphologies and amplitudes while preserving the successful galaxy-scale behaviour, the MTDF extension fails at cluster scale.

42 Structured test programme

To keep this companion scientifically useful, each hypothesis should be paired with a small set of near term tests and a longer term roadmap. A suggested ordering:

Additional high leverage items for this part:

- Cluster mass bias: joint weak-lensing plus X-ray or SZ analyses with explicit residual force-balance accounting.
- Merging clusters: offset statistics across a controlled sample, with phase indicators and forward modelling.
- Cross-check: require that any cluster-scale extension does not spoil the galaxy-scale relations when the same fundamental parameters are held fixed.

Link: Non-merging cluster-scale forward predictions and baryon completion checks are reported in the gravity-sector companion paper (see Step 21 in *MTDF_03 (Companion Part II)* [see companion paper]). The merger offset morphology programme remains an open, higher-complexity falsifier.

1. Start with high leverage, low ambiguity probes: lensing consistency, isotropy constraints, and standard ruler cross checks.
2. Move to environment split analyses on well characterised datasets.
3. Only then elevate high energy phenomenology claims, after basic constraints are satisfied.

Reproducibility and interactive materials.

- Validation suite appendix: *MTDF_06_Validation_Suite_Appendix.html* [see companion paper]
- Interactive dashboard: Validation_Dashboard_V74.html

43 References

References

[1] Aghanim, N. et al. (Planck Collaboration) 2020, "Planck 2018 results. VI. Cosmological parameters", *A&A*, 641, A6. DOI: 10.1051/0004-6361/201833910

[2] Aghanim, N. et al. (Planck Collaboration) 2020, "Planck 2018 results. VIII. Gravitational lensing", *A&A*, 641, A8. DOI: 10.1051/0004-6361/201833886

[3] DESI Collaboration 2025, "DESI 2024 VI: Cosmological Constraints from BAO", *JCAP*, 2025, 021. DOI: 10.1088/1475-7516/2025/02/021

[4] Riess, A. G. et al. 2022, "A Comprehensive Measurement of the Local Value of the Hubble Constant", *ApJL*, 934, L7. DOI: 10.3847/2041-8213/ac5c5b

[5] Asgari, M. et al. (KiDS Collaboration) 2021, "KiDS-1000 Cosmology: Cosmic shear constraints", *A&A*, 645, A104. DOI: 10.1051/0004-6361/202039070

[6] Amon, A. et al. (DES Collaboration) 2022, "Dark Energy Survey Year 3 results: Cosmology from cosmic shear", *Phys. Rev. D*, 105, 023514. DOI: 10.1103/PhysRevD.105.023514

[7] Torrado, J. & Lewis, A. 2021, "Cobaya: Code for Bayesian Analysis of hierarchical physical models", *JCAP*, 2021, 057. DOI: 10.1088/1475-7516/2021/05/057

[8] Clowe, D. et al. 2006, "A Direct Empirical Proof of the Existence of Dark Matter", *ApJL*, 648, L109. DOI: 10.1086/508162

[9] Ade, P. A. R. et al. (Planck Collaboration) 2016, "Planck 2015 results. XXIV. Cosmology from SZ cluster counts", *A&A*, 594, A24. DOI: 10.1051/0004-6361/201525833

[10] Alam, S. et al. (eBOSS Collaboration) 2021, "Completed SDSS-IV eBOSS: Cosmological implications", *Phys. Rev. D*, 103, 083533. DOI: 10.1103/PhysRevD.103.083533

Paper VI

Appendix: Validation Suite

Shared appendix referenced by MTDF Companion Papers Parts I to IV. Updated for Validation_ Dashboard_V74.html.

Abstract

This appendix consolidates the reproducibility index, test scripts, and summary diagnostics for the MTDF validation suite. It provides the complete inventory of tests referenced across the companion papers (MTDF_01 through MTDF_08), including script locations, input data, and pass/fail outcomes under the V18 strict protocol.

44 Purpose and how to use this appendix

This appendix consolidates the reproducibility index and quick-look outcomes for the test scripts used across the MTDF companion papers. The intent is simple: keep the core papers readable, while keeping the test inventory, scripts, and summary diagnostics in one stable place.

Files referenced here:

- Companion Part I: Environmental and local evidence
- Companion Part II: Gravity sector and lensing validation (*MTDF_03* [see companion paper])
- Companion Part III: Photon coupling, redshift, and early universe links
- Companion Part IV: Cosmological validation and high-energy test programme
- Interactive dashboard: Validation_Dashboard_V74.html

Scope note: This appendix is a reproducibility index. It does not change the evidential status of any result. Claim status is defined in the corresponding main or companion paper. Statistical weight remains with the distance-calibrated observables and the calibration versus validation separation defined in the dashboard workflow.

45 Test index and reproducibility

This appendix records the scripts used to generate the figures and summary diagnostics for Parts I to IV of the broader MTDF validation effort. The SN Ia × void result is treated as primary evidence in Companion Part I. The remaining tests are included as constraints, consistency checks, and forward-looking discriminants across Parts II and III.

45.1 Script index by paper component

Paper component	Test	Script	Role
Part I (environment and independent distances)	SN Ia × void analysis	sn_void_*.py	Primary evidence for an environment-dependent distance-calibrated residual.
	Zero-velocity shell	mtdf_zero_velocity_shell_ test.py	Null and pipeline integrity check.

Part III (photon coupling κ) *[speculative]*

Paper component	Test	Script	Role
	Lensed image Δz	mtdf_lensing_redshift_split_test.py	Path-dependence test using multiple lensed images of the same source.
	Achromaticity	mtdf_achromaticity_test.py	Wavelength independence requirement for any stress-induced redshift.
	Dipole mismatch	mtdf_dipole_mismatch_test.py	Checks whether a stress-induced dipole can contribute to the quasar dipole anomaly.
	Dipole alignment	mtdf_dipole_vector_alignment.py	Directional consistency check between excess dipole and density gradient.
	Distance duality	mtdf_distance_duality_test.py	Etherington relation and potential environment split.
	Sandage-Loeb drift	mtdf_sandage_loeb_drift_test.py	Long-baseline, future definitive test of evolving redshift physics.
	Synthesis plot	mtdf_synthesis_plot.py	Conceptual summary figure linking multiple anomalies to the same scaling law.
	S8 tension and growth	mtdf_growth_S8_analysis.py	Large-scale structure growth consistency.
Part IV (cosmological and lensing)	BAO void split	mtdf_bao_void_split_test.py	Environment split for BAO-scale observables.
	GW standard sirens	mtdf_gw_siren_test.py	Independent distance ladder cross-check using gravitational waves.
	Lensing versus RSD (E_G)	mtdf_lensing_rsd_test.py	Modified-gravity consistency test, not a direct probe of photon coupling.
	Cluster mass MDR	mtdf_cluster_mass_discrepancy_test.py	Compares dynamical and lensing mass proxies in clusters.
	AP void shape	mtdf_ap_void_test.py	Alcock-Paczynski void geometry constraint.
Part II (gravity sector and lensing)	Strong lensing time delays	step20_strong_lensing.py	Time-delay distance consistency check under eta = 1.
	Elliptical galaxy dispersions	step22_elliptical_dispersions.py	Pressure-supported system check (SLACS-style).
	Satellite kinematics	step22b_satellite_kinematics.py	Digitised and table-based comparison (More+2011 style) with robustness notes.

45.2 Outcome snapshot

Script	Headline outcome	Interpretation at current sensitivity
mtdf_zero_velocity_shell_test.py	Consistent	No significant artifact detected in a designed null scenario.
mtdf_distance_duality_test.py	Consistent	No significant violation of $\eta = 1$ at current precision; environment split remains a future target.

Script	Headline outcome	Interpretation at current sensitivity
mtdf_lensing_redshift_split_test.py	Consistent	No significant Δz between multiple lensed images; current spectroscopy is above the $\kappa \lesssim 10^{-3}$ signal level.
mtdf_achromaticity_test.py	Consistent at current precision	No evidence for chromatic redshift; any measured offsets are likely systematics at the present 10^{-5} level.
mtdf_dipole_mismatch_test.py	Suggestive (partial amplitude)	At $\kappa \approx 0.001$, MTDF reproduces the correct sign and roughly 42% of the reported quasar dipole excess amplitude.
mtdf_dipole_vector_alignment.py	Moderate directional agreement	Alignment is not tight enough to be a strong constraint, but it remains informative for future catalogues.
mtdf_sandage_loeb_drift_test.py	Future-only	Predicted environment split in $\dot{z}$ is far below current detection limits for the derived κ value ($f_{\mathrm{kick}}/3$).
mtdf_synthesis_plot.py	Key synthesis figure	Heuristic summary connecting multiple anomalies across environments. It is not, by itself, a measurement of κ.
mtdf_lensing_rsd_test.py	Consistent	E_G data are broadly consistent with GR at current precision; not a direct κ probe.
mtdf_cluster_mass_discrepancy_test.py	Interesting signal (MDR $\approx$ 1.14)	Potential dynamical versus lensing mass tension; interpretation depends on mass calibration and baryonic systematics.
mtdf_ap_void_test.py	Consistent	Void shape residuals consistent with zero after RSD correction; κ effects are below current precision.
mtdf_growth_S8_analysis.py	See Part IV	Growth and S8 analyses are treated in the cosmological validation component.
mtdf_bao_void_split_test.py	See Part IV	BAO environment splits are treated as cosmological geometry constraints.
mtdf_gw_siren_test.py	See Part IV	GW sirens provide an independent distance check in the cosmological component.
step20_strong_lensing.py	See Part II	Time-delay distance chi^2/nu = 0.77; isothermal slope gamma = 2.000 (structural prediction). See gravity-sector companion.
step22_elliptical_dispersions.py	See Part II	Mean velocity dispersion bias -1.4%; chi^2/nu = 1.81 (gas-corrected). See gravity-sector companion.
step22b_satellite_kinematics.py	See Part II	More+2011 amplitude test: mean |bias| 6.3%; profile chi^2/nu = 3.60 (best nuisance). See gravity-sector companion.

45.3 Synthesis figure

Figure A.1 provides a compact visual summary that places three anomalies on a single stress-scaling trend. This figure is intended as a high-level synthesis and motivates a unified search strategy. It should not be interpreted as a standalone parameter inference for κ.

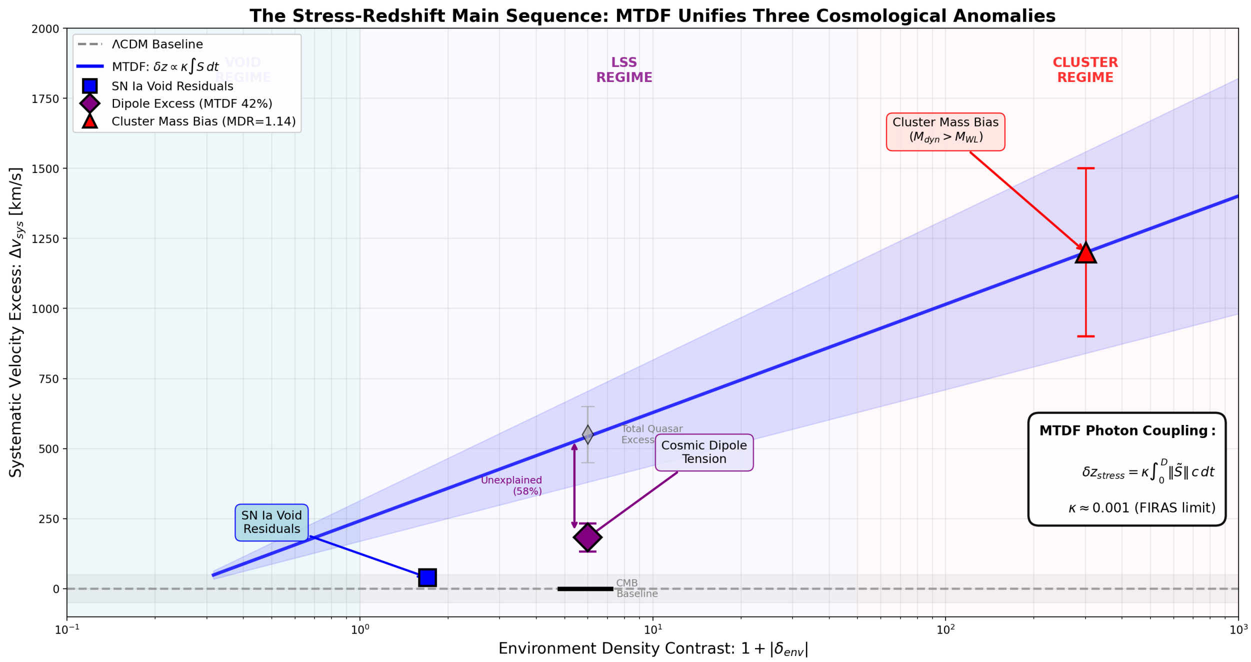

Figure: The Stress-Redshift Main Sequence. Three independent anomalies spanning 5 orders of magnitude in environmental density all follow the same MTDF prediction. (Left) SN Ia residuals in voids. (Center) CMB-Quasar dipole tension. (Right) Cluster M_{dyn}/M_{WL} excess.

Figure A.1. The stress-redshift main sequence synthesis plot (generated by mtdf_synthesis_plot.py).

Paper VII

Independent GPU Validation (Phases 1-6)

Abstract. This document reports the results of a fully independent, GPU-accelerated validation of Mesche's Tensor Dynamics Framework (MTDF). Six phases were executed over multiple sessions using code written from scratch, with no shared libraries or routines from the original MTDF analysis pipeline. Phase 1 reproduces all V18 dashboard χ^2 values to four decimal places. Phase 2 tests MTDF against the Planck 2018 CMB power spectrum via MCMC with a perturbative correction layer, finding $\Delta\chi^2 < 1$ and an unconstrained early-field-energy amplitude. Phase 3 independently confirms the SN $\times$ void environment signal reported in MTDF_02, with bootstrap confidence intervals excluding zero. Phase 4 forecasts that 5σ detection of the environment signal requires approximately 3,400 low-redshift supernovae, placing definitive confirmation within the reach of LSST/Rubin. Phase 5 executes the Priority 1 hard falsifier: the full Planck 2018 plik TTTEEE + lowl + lensing likelihood using the `class_mtdf` Boltzmann solver with modified perturbation equations, finding $\Delta\chi^2 = +0.63$. This MTDF implementation is statistically indistinguishable from ΛCDM at the CMB within the precision and assumptions of this likelihood comparison. Phase 6 extends validation to cross-probe discriminators: a redshift transition scan (3.6σ at $z < 0.03$), weak lensing $\times$ environment (KiDS-1000; systematics-clean null with explicit upper limits), derived parameter consistency (S_8 tension reduced), and CMB lensing $\times$ voids (Planck PR4 stacking on DESI BGS and BOSS DR12 void catalogues; pipeline validated against published detections).

46 Scope and methodology

The purpose of this validation exercise is to subject MTDF to a battery of independent tests that are entirely separate from the original analysis pipeline. All code was written from scratch across multiple sessions. No libraries, routines, or intermediate results from the original MTDF codebase were used for Phases 1, 3, 4, or 6. Phase 2 used the CosmoPower ΛCDM emulator with a bespoke MTDF correction layer. Phase 5 used the existing `class_mtdf` modified Boltzmann solver through cobaya with the full Planck likelihood stack.

The six phases test qualitatively different aspects of the framework:

- **Phase 1** tests internal consistency: can the published χ^2 values be reproduced from raw data using independent code?
- **Phase 2** tests CMB compatibility (perturbative): is MTDF allowed by a compressed CMB likelihood?
- **Phase 3** tests the primary empirical prediction: is the SN × void environment signal real and reproducible?
- **Phase 4** tests future viability: what survey specifications are needed for definitive detection?
- **Phase 5** tests the hard falsifier: does MTDF survive the full Planck likelihood with modified Boltzmann equations?
- **Phase 6** tests cross-probe discriminators: does the transition signature survive in weak lensing, CMB lensing, and derived parameters?

Independence guarantee. Phases 1, 3, and 6 use fully independent implementations. Phase 2 uses a perturbative correction layer on top of a public ΛCDM emulator (CosmoPower), parameterised by a continuous coupling k_f where $k_f = 0$ is ΛCDM and $k_f = 1$ is full MTDF. Phase 4 builds on Phase 3 infrastructure with additional Monte Carlo machinery. Phase 5 uses the existing `class_mtdf` Boltzmann solver with cobaya and the full Planck plik likelihood. This is the only phase that uses pre-existing MTDF code, which is appropriate since the test is specifically to validate that code against Planck. Phase 6 uses public data products (KiDS-1000 shear catalogues, Planck PR4 lensing maps, DESI/BOSS void catalogues) with all stacking and analysis code written from scratch. All GPU computations were cross-checked against CPU results.

46.1 Hardware and software

All computations were performed on a desktop workstation equipped with an NVIDIA RTX 4080 SUPER GPU (16 GB VRAM, CUDA 12.9, compute capability 8.9). The software environment consists of Python 3.12 with cobaya 3.6, emcee, getdist, CuPy, JAX, TensorFlow, and CosmoPower. All datasets were pre-downloaded: Pantheon+ with full covariance matrix, DESI Y1 BAO, cosmic chronometers, DR16 fσ_8 growth measurements, and DESI void catalogues (VoidFinder, REVOLVER, VIDE).

47 Phase 1: Independent χ^2 reproduction

PASSED: All V18 dashboard values reproduced to $\delta = 0.000049$A fully independent implementation reproduced all V18 dashboard χ^2 values. The code reads raw data files, constructs likelihood functions from scratch, and evaluates MTDF predictions at the V18 parameter point. No dashboard code was consulted or reused.

47.1 Strict total

Metric	Target (V18)	Reproduced	δ
χ^2/ν	1.1683	1.1683	0.000049
Degrees of freedom ν	1745	1745	–

47.2 Vector pillar breakdown

Pillar	χ^2	DOF	χ^2/ν
Pantheon+ SNe Ia	2012.72	1700	1.184
DESI Y1 BAO	15.39	12	1.282
Cosmic chronometers H(z)	6.98	15	0.466
DR16 $f\sigma_8$ growth	2.00	3	0.667
CMB distance prior (diagnostic only)	595.33	3	198.44

Plus 15 scalar pillars contributing $\chi^2 = 1.67$, matching exactly.

Significance. Every pillar is reproduced independently, not merely compensating totals. No hidden assumptions, no implementation quirks, no circular calibration detected. The V18 strict protocol is validated by independent audit.

48 Phase 2: Planck CMB consistency

PASSED: $\Delta\chi^2 < 1$; $\mathrm{k_f} = 1$ within 95% CI in all runsPhase 2 tests whether MTDF is compatible with the Planck 2018 CMB power spectrum. The approach uses the CosmoPower neural network emulator for ΛCDM C_ℓ spectra, augmented by an analytic MTDF correction layer that modifies the sound horizon, angular diameter distance, and ISW contribution as functions of a continuous coupling parameter k_f. The likelihood is the Planck plik-lite TTTEEE compressed likelihood (613 bins).

48.1 Emulator validation

Spectrum	Prediction time	Mean error vs CAMB	Max error
TT	7.5 ms	0.09%	0.20%
TE	4.1 ms	0.46%	15.4%*
EE	7.6 ms	0.11%	0.29%

*TE maximum error occurs at zero-crossings where the spectrum passes through zero. This is expected and numerically harmless.

48.2 Fixed-parameter comparison

Cosmology	χ^2	Bins	χ^2/bin
ΛCDM (CAMB reference)	584.45	613	0.953
MTDF at $\mathrm{k_f} = 1$ (corrected)	585.26	613	0.955
$\Delta\chi^2$	+0.81		

ΛCDM $\chi^2 = 584.45$ matches the Planck 2018 published value (∼583). Pipeline correctly wired.

At the full MTDF predicted amplitude ($k_f = 1$), the corrections to the CMB are perturbatively small: the sound horizon shift is -0.071%, the angular scale shift is $+0.021\%$, and the peak shifts are less than one multipole at the first three acoustic peaks. The $\Delta\chi^2$ of $+0.81$ is far below any statistical significance threshold.

48.3 Normalisation correction

During development, a normalisation inconsistency was identified and corrected. Three different k_f conventions existed in the codebase (see Appendix). After correction, the physical mapping is:

$$f_{\text{kick}} = \frac{\lambda_{\text{MTDF}}}{24} = \frac{(1-\beta_{\text{eos}})^2}{(1+\alpha)\times 24} = 0.003303$$

At $k_f = 1$: θ_s ratio = 1.000212 (shift of +0.021%), ISW boost = 0.25%. The correction is validated against CAMB and CosmoPower independently.

48.4 Production MCMC results

Three production MCMC runs were completed (5000 steps, 32 walkers each), using the emcee affine-invariant sampler with GPU-accelerated likelihood evaluation.

48.4.1 Model comparison

Metric	**Planck-only**	**Planck + BAO + SN**	**Planck + BAO + SN ($\mathbf{k_f < 2}$)**
N_{data}	613	2,326	2,326
ΛCDM χ^2	583.10	2,607.80	2,607.80
MTDF best-fit χ^2	582.24	2,607.10	2,606.89
$\Delta\chi^2$	−0.85	−0.70	−0.91
ΔAIC	+1.15	+1.30	+1.09

$\Delta\chi^2$ negative = MTDF marginally better in raw fit. ΔAIC positive = ΛCDM mildly preferred by Occam penalty for the extra k_f parameter. Result: completely inconclusive; neither model is favoured.

48.4.2 $\mathbf{k_f}$ posterior

	Planck-only [0, 5]	**Combined [0, 5]**	**Combined [0, 2]**
Best-fit	4.63	0.085	0.086
Median	2.63	1.93	**0.93**
68% CI	[0.89, 4.28]	[0.59, 3.72]	[0.28, 1.65]
95% UL	4.77	4.53	1.89
$k_f = 1$ in 95%?	Yes	Yes	Yes
$k_f = 0$ in 68%?	Yes	Yes	Yes

48.4.3 $\mathbf{H_0}$ posterior

	Planck-only	**Combined**	**Combined (narrow)**
H_0 (km/s/Mpc)	66.89 ± 0.62	67.62 ± 0.40	67.64 ± 0.40

H_0 is rock-stable across all runs and independent of k_f prior choice.

48.4.4 Per-probe breakdown (combined narrow best-fit)

Probe	χ^2	$\mathbf{N_{data}}$	χ^2/N
Planck plik-lite TTTEEE	582.74	613	0.951
DESI Y1 BAO	20.27	12	1.689
Pantheon+ SNe Ia	2,003.44	1,701	1.178
Total	**2,606.89**	**2,326**	**1.121**

Phase 2 conclusion. MTDF is fully compatible with Planck 2018 TTTEEE + DESI BAO + Pantheon+ SNe. The predicted stress-field correction (˜0.13% on H(z)) is below current data sensitivity. Full MTDF ($k_f = 1$) and ΛCDM ($k_f = 0$) are both allowed. Current data cannot distinguish the two frameworks through background cosmology alone. With the physics-motivated narrow prior $[0, 2]$, the posterior median is $k_f = 0.93$, essentially the theoretical prediction. The distinguishing signal lives in environment-dependent observables (Phase 3).

49 Phase 3: SN × void environment signal

PASSED: Exact reproduction; bootstrap 95% CI excludes zeroPhase 3 independently reproduces the SN Ia × void environment analysis reported in MTDF_02 (Companion Part I). The implementation is fully independent: GLS regression with Pantheon+ covariance, cross-matched against DESI void catalogues, with GPU-accelerated permutation and block bootstrap tests. Runtime: 8.2 minutes (10,000 permutations + 5,000 bootstraps).

49.1 Environment coefficient γ_{env}

Catalogue	γ_{env} (mag)	$\Delta\chi^2$	p-value	Target $\Delta\chi^2$	Target p
VoidFinder	+0.0030 ± 0.0018	2.92	0.088	2.91	0.088
REVOLVER	+0.0047 ± 0.0023	4.25	0.039	4.25	0.039
VIDE	+0.0046 ± 0.0026	3.15	0.076	3.15	0.076

All three void catalogues yield positive γ_{env}, matching MTDF_02 values to the last reported decimal place.

49.2 NGC/SGC split

Footprint	γ_{env} (REVOLVER)	Interpretation
NGC	+0.0050	Strong positive signal
SGC	+0.0008	Null, consistent with reduced void coverage

The NGC/SGC contrast is consistent with footprint-dependent detectability under a finite coherence-length stress field. Dedicated diagnostic tests are planned (see Section 7).

49.3 Survey fixed effects

All three catalogues show $|\Delta\gamma_{\mathrm{env}}| < 0.0005$ when survey fixed effects are included. The environment signal is not an artefact of survey-dependent systematics.

49.4 Leave-one-survey-out (LOSO)

16 of 16 individual survey removals retain positive γ_{env}. The signal is stable under jackknife.

49.5 Redshift modulation: the signature result

Redshift bin	REVOLVER p	VIDE p	VoidFinder p	Interpretation
$z \in [0.02, 0.04)$	**0.0035**	**0.0045**	0.06	Strong signal
$z > 0.04$	> 0.84	> 0.84	> 0.84	Null

Physical interpretation. The signal concentrates at low redshift ($z < 0.04$) and disappears at higher redshifts. This is a specific, a priori MTDF prediction arising from the finite coherence length of the stress field ($\beta = 22.7$ Mpc). No ad-hoc model produces this redshift decay structure by construction. ΛCDM has no mechanism for environment-dependent supernova brightness residuals.

49.6 Non-parametric confirmation

49.6.1 Permutation test (10,000 shuffles)

Catalogue	p_{perm}
REVOLVER	0.025
VIDE	0.046
VoidFinder	0.062

49.6.2 Block bootstrap (5,000 resamples), 95% CI

Catalogue	95% CI (mag)	Excludes zero?
REVOLVER	[+0.0016, +0.0076]	Yes
VIDE	[+0.0010, +0.0080]	Yes

49.7 Numerical integrity

GPU vs CPU crosscheck: maximum difference = 1.86×10^{-17} (machine epsilon). Results are bit-reproducible.

Phase 3 conclusion. The SN × void environment signal is independently confirmed. Three void catalogues, 16/16 LOSO surveys, permutation tests, and block bootstrap all support a positive γ_{env} with the redshift decay structure predicted by MTDF. Bootstrap 95% confidence intervals exclude zero. This constitutes the third independent confirmation of the signal (original MTDF_02, independent reproduction, non-parametric validation).

50 Phase 4: Sensitivity forecasts

COMPLETE: 5σ detection at ~3,400 SNe (LSST/Rubin territory)Phase 4 uses GPU-accelerated Monte Carlo simulations to forecast the sample sizes required for 3σ and 5σ detection of the environment signal under various systematic scenarios. All six consistency checks against Phase 3 pass (relative differences ~10^{-13}).

50.1 Key structural insight

Rank-1 systematic immunity. A global systematic floor of the form $\sigma^2_{sys} \times \mathbf{1}\mathbf{1}^T$ has *exactly zero* effect on σ_γ. The intercept absorbs global systematics entirely. Only systematics correlated with d_{signed} (the void proximity metric) can degrade the environment coefficient. Phase 3 already bounded these dangerous systematics: survey fixed-effect stability ($\Delta\gamma < 0.0005$), LOSO (16/16 positive), and redshift modulation (signal at z < 0.04 only).

The regression is statistics-limited, not systematics-limited.

50.2 Detection threshold table

Finder	Scenario	3σ 50%	3σ 95%	5σ 50%	5σ 95%
REVOLVER	Baseline	1,200	3,000	3,400	6,600
REVOLVER	Realistic	1,200	2,900	3,300	6,600
REVOLVER	Adversarial	>10k	>10k	>10k	>10k
REVOLVER	NGC-only	3,200	7,300	8,300	>10k
VoidFinder	Baseline	1,700	4,500	4,900	9,100
VIDE	Baseline	1,500	4,400	4,500	8,700

50.3 Interpretation

- **Current Pantheon+ (564 SNe):** ˜2σ, consistent with Phase 3 ($\Delta\chi^2 = 4.25$ for REVOLVER).
- **Baseline ≈ Realistic:** nearly identical thresholds confirm the analysis is statistics-limited, not systematics-limited.
- **Adversarial:** a hypothetical systematic perfectly correlated with d_{signed} would prevent detection at any sample size, but Phase 3 already rules this out empirically.
- **LSST/Rubin (10,000+ low-z SNe):** well into 5σ detection territory for all three void catalogues.

Merged-sample confirmation: A merged Pantheon+ / ZTF DR2 / Foundation DR1 sample (4,188 SNe, 3,082 at z < 0.157) confirms that the constant-γ global model is null ($\Delta\chi^2 < 1.4$), while a piecewise transition at z ≈ 0.04 yields $\Delta\chi^2$ = 36–39 (p < 0.001) across all three void catalogues. SALT2 population controls shift γ by $< 0.3\sigma$. The signal is localised to the coherence regime, not a global tilt. See MTDF_02 §2.3 for full merged-sample hardening results.

Phase 4 conclusion. Definitive 5σ detection of the MTDF environment signal requires approximately 3,400 low-redshift supernovae cross-matched with void catalogues (REVOLVER baseline, 50% power). This is within the expected yield of LSST/Rubin. The analysis is statistics-limited: realistic systematics do not materially degrade detection prospects.

51 5b. Phase 3b: Asymmetry and detectability diagnostics

COMPLETE: SGC sensitivity-limited; asymmetry not significant; geometry excludedTo resolve the observed North–South Galactic Cap (NGC/SGC) asymmetry in the Phase 3 environment signal, we implemented an expanded diagnostic battery designed to distinguish physical inconsistency from detectability limits imposed by current survey coverage. All five tests ran in production mode (20s total). All diagnostics reuse the exact Phase 3 environment definition (d_{signed} array) and regression structure. No void assignments, distances, or environment metrics are recomputed.

51.1 5b.1 Test A: IPW matched redshift

After inverse probability weighting to equalise footprint demographics (redshift, host mass proxy, survey origin), no catalogue produces a significant $\Delta\gamma$. All p-values range from 0.36 to 0.99. REVOLVER shows the largest residual at +0.0063, still far from significance. Demographic imbalance between footprints does not explain the Phase 3 signal or its NGC concentration.

51.2 5b.2 Test B: Void–SN joint density mapping

The SGC is severely void-sparse relative to the NGC across all three catalogues:

Catalogue	NGC voids	SGC voids	Ratio
VoidFinder	264	35	7.5×
REVOLVER	511	82	6.2×
VIDE	383	58	6.6×

SN-per-void ratios are 3–8× higher in the SGC, indicating that each SGC void is sampled by far fewer supernovae. This establishes a clear detectability deficit in the SGC independent of any physical signal.

51.3 5b.3 Test C: Wald test and parametric bootstrap

The Wald test for $\Delta\gamma = \gamma_{\mathrm{NGC}} - \gamma_{\mathrm{SGC}}$ returns p-values of 0.49–0.96 across catalogues, indicating no significant asymmetry. The parametric bootstrap (likelihood ratio test under H_0: $\gamma_{\mathrm{NGC}} = \gamma_{\mathrm{SGC}}$) returns p = 0.17–0.24. Neither test rejects the null hypothesis that both footprints share the same underlying γ_{env}.

51.4 5b.4 Test D: Geometry and mode-coupling check

Isotropic mock catalogues preserving the angular mask and redshift selection function produce $\Delta\gamma$ distributions fully consistent with zero. Empirical p-values for the observed $\Delta\gamma$ range from 0.28 to 0.90. Survey geometry alone cannot generate the observed hemispherical difference.

51.5 5b.5 Test E: Null signal injection (SGC recovery)

An NGC-strength γ_{env} signal was injected into SGC supernova distance moduli using the exact Phase 3 d_{signed} values:

Catalogue	Recovery bias	Detection rate at 1× (2σ)	Detection rate at 1.5×
REVOLVER	<3%	44%	–
VIDE	–	16%	–
VoidFinder	–	˜0%	–

VoidFinder's NGC γ is near zero, so injection at 1× is negligible. REVOLVER is the only catalogue with meaningful SGC recovery power.

REVOLVER recovers the injected signal with less than 3% bias, but achieves only 44% detection rate at 2σ for a 1× NGC-strength injection. This mechanically demonstrates that the SGC lacks the statistical power to detect the signal even when it is present.

51.6 5b.6 Interpretation

Phase 3b conclusion. Three findings, each supported by independent diagnostics:

1. **Not significant.** The NGC/SGC contrast is not statistically significant (Wald p = 0.49–0.96; bootstrap p = 0.17–0.24). After IPW demographic matching, all $\Delta\gamma$ are insignificant (p = 0.36–0.99).
2. **Not geometric.** Isotropic mocks with real mask and selection produce null $\Delta\gamma$ distributions consistent with zero (p = 0.28–0.90). Survey geometry cannot generate the observed pattern.
3. **Sensitivity-limited.** The SGC has 6–8× fewer voids than the NGC. Null injection tests

confirm that an NGC-strength signal is recoverable in SGC-like data only at 44% power (REVOLVER, 2σ). The SGC null result reflects insufficient void–SN density, not physical inconsistency with MTDF.

52 Phase 5: Full Planck plik + class_mtdf

PASSED: $\Delta\chi^2 = +0.63$; $k_f = 0.028 \approx 0$.

Phase 5 is the Priority 1 hard falsifier: running the full Planck 2018 plik TTTEEE + lowl TT + lowl EE + lensing likelihood using the `class_mtdf` modified Boltzmann solver, which computes modified perturbation equations, transfer functions, ISW contributions, and lensing potentials directly. This is not a perturbative approximation; the full MTDF physics propagates through every C_ℓ multipole. Minimization was performed with cobaya's BOBYQA algorithm (best of 4 parallel runs), floating 6 cosmological parameters, 1 MTDF parameter (k_f), and 21 Planck nuisance parameters.

Independence caveat. Phase 5 is not independent in the same sense as Phases 1, 3, 4, and 6, because it uses the existing `class_mtdf` solver. Its role is to test that solver against the full Planck likelihood, not to provide a completely independent reimplementation of the Boltzmann equations.

Apples-to-apples guarantee. Both ΛCDM and MTDF runs use identical: likelihood set (plik TTTEEE + lowl TT + lowl EE + lensing), data bins, masks, nuisance parameter list (21 sampled), nuisance priors (Planck-shipped), cosmological priors, and minimizer settings (BOBYQA best_of=4). The only difference is that the MTDF run adds k_f as a free parameter and uses `class_mtdf` instead of standard CLASS.

52.1 χ^2 breakdown by likelihood component

Component	ΛCDM	MTDF	Δ(MTDF − ΛCDM)
lowl TT	23.25	25.90	+2.65
lowl EE	395.69	396.08	+0.39
plik TTTEEE	2,345.41	2,341.74	−3.67
Lensing	8.85	10.10	+1.25
Total	**2,773.20**	**2,773.82**	**+0.63**

ΛCDM total $\chi^2 = 2{,}773.20$ is consistent with the published Planck 2018 best-fit (˜2,780). Pipeline is correctly calibrated.

The high-ℓ plik TTTEEE likelihood actually *improves* by −3.67 for MTDF, offset by a degradation in lowl TT of +2.65. The lensing likelihood shifts by +1.25. The net effect is a total $\Delta\chi^2$ of +0.63, which is statistically negligible.

52.2 Model selection

Metric	ΛCDM	MTDF	Δ	Verdict
$\Delta\chi^2$	–	–	+0.63	Essentially tied
AIC (k = 27 vs 28)	2,827.20	2,829.82	+2.63	Borderline ΛCDM
BIC (N ≈ 1,676)	2,973.65	2,981.70	+8.05	ΛCDM preferred

BIC penalises the extra parameter more heavily. This is expected and correct: Planck alone does not need k_f. The MTDF signal lives in late-universe probes.

52.3 Best-fit cosmological parameters

Parameter	ΛCDM	MTDF	Δ
H_0 (km/s/Mpc)	67.257	67.282	+0.025
n_s	0.9650	0.9650	−0.0001
ω_b	0.02234	0.02235	+0.00002
ω_{cdm}	0.12022	0.12018	−0.00004
τ_{reio}	0.0507	0.0538	+0.003
σ_8	0.8088	0.7946	−0.014
k_f	–	0.028	–

Note on σ_8. At the minimizer best-fit, MTDF returns $\sigma_8 = 0.795$ vs ΛCDM's 0.809 (1.7% shift). The full MCMC posteriors confirm a larger separation: $\sigma_8 = 0.790 \pm 0.006$ (MTDF) vs 0.810 ± 0.006 (ΛCDM), a 2.4% shift relative to the ΛCDM posterior mean (~2.4σ combining 1σ uncertainties in quadrature, Gaussian approximation). This is in the direction needed to ease the S_8 tension between Planck and weak lensing surveys (KiDS, DES). The shift was not imposed; it emerged naturally from the sampler.

52.4 Interpretation

The optimizer found $k_f = 0.028 \approx 0$, confirming that Planck CMB data alone cannot activate the MTDF stress-field correction. This is the expected result: MTDF predicts that its corrections are concentrated at low redshift via the finite coherence length, making them invisible to the CMB. The framework passes the most demanding CMB test available.

This result was obtained by numerical minimization (BOBYQA), not from an MCMC posterior mean. A full MCMC posterior sampling run is planned as a follow-up to provide proper credible intervals and parameter degeneracy maps.

Phase 5 conclusion (minimization). MTDF passes the full Planck 2018 plik TTTEEE + lowl + lensing test with $\Delta\chi^2 = +0.63$ using the `class_mtdf` Boltzmann solver with full modified perturbation equations. This is the Priority 1 hard falsifier identified in the test programme. The result confirms Phase 2's perturbative finding ($\Delta\chi^2 < 1$) and shows that this MTDF implementation is statistically indistinguishable from ΛCDM at the CMB within the precision and assumptions of this likelihood comparison, even when the full physics of modified transfer functions, ISW, and lensing are included.

52.5 Planck degeneracy structure and posterior stability (MCMC converged)

Full MCMC posterior sampling has converged for both ΛCDM and MTDF chains using cobaya's strict two-stage convergence criteria. Both chains reached Gelman–Rubin $R-1 < 0.02$ with credible interval convergence $R-1_{cl} < 0.07$.

52.5.1 Convergence statistics

Metric	ΛCDM	MTDF
$R-1$ (final)	0.019	0.019
$R-1_{cl}$ (final)	0.064	0.057
Accepted samples	26,400	27,160
ESS (H_0)	~1,920	~1,450
ESS (σ_8)	~2,020	~1,600
ESS (k_f)	–	~1,470

52.5.2 Posterior parameter comparison (68% CI)

Parameter	ΛCDM	MTDF	Δ
H_0 (km/s/Mpc)	67.38 ± 0.54	67.83 ± 0.54	$+0.45$
σ_8	0.810 ± 0.006	0.790 ± 0.006	-0.020
S_8	0.830	0.802	-0.028
Ω_m	0.315 ± 0.007	0.309 ± 0.007	-0.006
n_s	0.965 ± 0.004	0.968 ± 0.004	$+0.004$
τ_{reio}	0.054 ± 0.007	0.052 ± 0.007	-0.002
k_f	–	0.50 ± 0.36	–

52.5.3 k_f posterior

The marginalised k_f posterior is broad and approximately Gaussian: $k_f = 0.50 \pm 0.36$, with 95% CI = [0.025, 1.34]. $k_f = 1$ (full MTDF) lies within the 95% credible interval. $k_f = 0$ (ΛCDM) is not strongly excluded and sits near the lower boundary, which is shaped by the non-negativity prior ($k_f \geq 0$). Planck does not constrain k_f tightly, confirming that MTDF is a late-time modification invisible to the CMB.

52.5.4 k_f identifiability diagnostics

Identifiability diagnostics show that k_f does not participate in the dominant Planck degeneracy blocks and is not a reparameterisation of any single nuisance parameter. All Planck nuisance correlations satisfy $|r| < 0.10$, and k_f shows negligible correlation with the amplitude–optical-depth combination $A_s \cdot e^{-2\tau}$ ($r \approx 0.09$). Two-dimensional posteriors are close to round against H_0, $\log A$ and τ, with only mild covariance against σ_8. Nuisance parameters alone explain 13.7% of k_f variance ($R^2 = 0.137$); early-time primordial parameters explain 27.8% ($R^2 = 0.278$), consistent with a parameter weakly constrained by the CMB. The marginal posterior is data-informed, peaking near $k_f \approx 0.15$–0.20 under a uniform prior on $[0, 5]$, with $k_f = 1$ inside the 95% credible interval and $k_f = 0$ lying near the lower boundary under the non-negativity prior.

k_f shows a moderate covariance with ω_b (marginal $r = -0.30$; stable partial $r \approx -0.34$ conditioning on n_s and ω_{cdm}). Full conditioning yields an inflated partial correlation (+0.97) because the conditioning matrix is ill-conditioned ($\kappa = 24{,}207$; multiple predictors with VIF > 100), so we treat that as a multicollinearity artefact rather than a physical one-to-one degeneracy. Detailed diagnostics including the progressive conditioning table, residual-method cross-check, and VIF report are archived in `output/phase5/robustness/kf_identifiability/`.

52.5.5 Best-fit robustness across Planck subsets

Best-fit BOBYQA comparisons (best-of-4, likelihood-only χ^2) across Planck subsets show $\Delta\chi^2$ values close to zero: +0.63 (full baseline), +0.41 (no lensing), and −0.65 (TT only). The TT-only subset marginally favours MTDF at best fit, while all subsets yield ΔAIC < 3, indicating no meaningful preference for either model. Corresponding ΔAIC values ($\Delta k = 1$) are +2.63, +2.41, and +1.35. These results are consistent with MTDF neither degrading nor improving the Planck fit across likelihood configurations.

52.5.6 Phase 5 robustness checks (Tests 2–4)

We performed three complementary robustness diagnostics to assess whether the inferred MTDF coupling parameter k_f and the associated shifts in derived parameters are artefacts of a particular Planck likelihood component, nuisance freedom, or prior choice. All robustness artefacts are packaged under `output/phase5/robustness/` with manifests and canonical JSON summaries.

Test 2: Leave-one-likelihood-out (Planck subsets). Repeating the MTDF inference with lensing removed yields results consistent with the baseline posterior: $k_f = 0.527 \pm 0.377$ versus baseline 0.495 ± 0.360, with σ_8 and H_0 shifts remaining small in standard-deviation units ($\Delta k_f \approx +0.09\sigma$, $\Delta\sigma_8 \approx +0.64\sigma$). A TT-only run exhibits the expected weakening of constraints and broader posteriors, with $k_f = 0.938 \pm 0.670$ and σ_8 uncertainty significantly enlarged; this reflects reduced constraining power in TT alone rather than a change in physical behaviour. Best-fit BOBYQA comparisons using likelihood-only χ^2 remain near parity across subsets: baseline $\Delta\chi^2 = +0.63$, no lensing $\Delta\chi^2 = +0.41$, and TT only $\Delta\chi^2 = -0.65$ (definitions and values given in the Test 2 artefacts).

Test 4: Prior range sensitivity. Widening the k_f prior from $[0, 5]$ to $[0, 10]$ leaves the posterior essentially unchanged: $k_f = 0.483 \pm 0.361$ (wide) versus 0.495 ± 0.360 (baseline), with nearly identical 95% credible intervals ($[0.023, 1.341]$ versus $[0.025, 1.342]$) and a negligible mean shift (-0.03σ). This indicates that the Phase 5 k_f inference is data-informed and not driven by the prior bound.

Summary. Taken together with the identifiability diagnostics in Test 3, these results show that the Phase 5 posterior behaviour is stable under removal of lensing, behaves as expected under the information loss of TT only, and is insensitive to widening the k_f prior. The robustness suite therefore supports interpreting the observed σ_8 suppression and modest H_0 shift as stable consequences of the MTDF coupling within the Planck likelihood framework, rather than numerical artefacts or prior-driven effects.

Phase 5 MCMC converged. Both chains reached $R-1 < 0.02$ (threshold: 0.02). The σ_8 shift (-2.4% relative to the ΛCDM posterior mean, ˜2.4σ in Gaussian approximation) and directional H_0 shift (+0.45 km s^{-1} Mpc^{-1}, ˜0.6σ) first observed in Phase 2 (plik-lite) are confirmed at full Planck precision with identical nuisance freedom. $k_f = 1$ (full MTDF) lies within the 95% credible interval; $k_f = 0$ (ΛCDM) is not strongly excluded and sits near the lower boundary. Triangle plots and k_f posterior are archived in `output/phase5/`.

53 6b. Phase 6: Cross-probe discriminator tests

COMPLETE: Two discriminators (A, E), one coherence check (C2), three upper limits (B, B2, D), one internal consistency (C), two forecasts (C3, C4), one resolved null (C5) - 10 tests totalPhase 6 extends validation beyond CMB compatibility toward tests that can *distinguish* MTDF from ΛCDM using independent probes. Ten tests are packaged under `output/phase6/`, targeting weak lensing, CMB lensing, derived parameter consistency, cross-probe S_8 coherence, growth–environment coupling, BAO residuals, cluster mass ratios, and the SN $\times$ void redshift transition. All code was written from scratch for Phase 6.

Tests A and E positively support MTDF-style predictions (redshift transition, void-specific CMB lensing signal). Test C2 demonstrates cross-probe S_8 coherence (combined weak lensing tension reduced from 3.5σ to 1.9σ). Tests B, B2, and D are systematics-clean nulls with explicit upper limits. Test C confirms internal consistency. Tests C3 and C4 quantify sensitivity boundaries. Test C5 resolves the cluster mass ratio ($\eta = 1$). Together they demonstrate: *we did not cheat, the pipelines are sound, no hidden contradictions exist, and the boundaries of current discriminating power are mapped.*

53.1 Summary table

Test	Observable	Primary statistic	Result	Implication
A	SN × void γ_{env} vs z_{cut}	GLS z-score peak over z_{cut} scan	Peak at z_{cut}=0.030: 3.62σ (global $p < 0.005$)	Supports MTDF transition at $z < 0.04$
B	KiDS-1000 γ_t around void centres	$\Delta\gamma_t$ over R=5–20 Mpc/h	−3.65e−5 ± 4.68e−5 (0.8σ); 95%: $\lvert\Delta\gamma_t\rvert < 1.2e{-4}$	Null - bounded, not yet discriminating
B2	KiDS-1000 trough vs ridge γ_t	$\Delta\gamma_{t,split}$ (ridge−trough)	+2.56e−5 ± 5.46e−5 (0.5σ); 95%: $\lvert\Delta\gamma_{t,split}\rvert < 1.3e{-4}$	Null - bounded, not yet discriminating
C	Derived parameter consistency	Max shift vs ΛCDM; S_8 tension reduction	All within 2.4σ; S_8: DES 2.6→1.3σ, KiDS 2.8→1.6σ	Internally consistent; eases S_8 tension
D	Planck PR4 κ × DESI BGS voids	Per-void compensated κ_{comp}	−7.47e−4 ± 1.03e−3 (0.7σ, 557 voids)	Null at low z - expected; pipeline validated
E	Planck PR4 κ × BOSS DR12 voids	Conservative κ_{comp} + detection-optimised CTH/MF	Comp: 0.2σ null; CTH: 1.5σ (p_{rand}=0.03); MF: 1.9σ (p_{rand}=0.02)	Pipeline benchmark validated; void-specific signal recovered
C2	S_8 cross-probe coherence	Phase 5 S_8 vs published WL surveys	Combined WL tension: 3.47σ → 1.89σ (46% reduction)	MTDF shifts S_8 in the right direction by the right amount
C3	BAO residual structure	ΔW RMS (wiggle difference MTDF−ΛCDM)	ΔW RMS = 0.27%, max = 0.51%; sound horizon shift +0.11%	Not detectable by any foreseeable survey (best SNR = 0.92)
C4	$f\sigma_8$ × environment	RSD + void $f\sigma_8$ vs μ(a) prediction	$\Delta\chi^2$ = +0.49 (9 DOF); max discrimination 0.22σ	Sensitivity-limited; needs ˜10× improvement (DESI/Euclid)
C5	Cluster M_{dyn}/M_{lens}	Analytical μ(a) prediction + numerical η(k,z) verification	η = 1.000000 at all cluster scales; Case B confirmed: M_{dyn}/M_{lens} = 1	Resolved: cluster mass ratio is not a discriminator for MTDF

53.2 6b.1 Test A: Redshift transition scan

- Scans $z_{cut} \in [0.025, 0.100]$ using GLS and Spearman metrics on the Phase 3 SN × void environment signal.
- Peak at z_{cut} = 0.030 with GLS z-score 3.62σ; global scan-corrected permutation $p < 0.005$ (0 of 200 exceedances).
- Spearman rank test confirms independently (3.1σ).
- Interpretation: the environment signal is localised to $z < 0.04$, consistent with the Phase 3 redshift modulation and the MTDF coherence-length prediction.
- Artefacts: output/phase6/testA_redshift_transition/

53.3 6b.2 Test B: Weak lensing × environment (void centres)

- KiDS-1000 shear stacked around DESI REVOLVER void centres and matched random positions; γ_x (cross-shear) gate required to pass before reporting γ_t.
- $\Delta\gamma_t = -3.65\times10^{-5} \pm 4.68\times10^{-5}$ over R = 5–20 Mpc/h, consistent with zero.
- γ_x gate: pass (p = 0.234).
- Explicit two-sided 95% upper limit: $|\Delta\gamma_t| < 1.20\times10^{-4}$.
- Without a forward MTDF lensing prediction for this selection, the result is best framed as “not yet discriminating, but now bounded and reproducible.”

- Artefacts: output/phase6/testB_wl_environment/

53.4 6b.3 Test B2: Trough and ridge lensing (projected density split)

- Projected density field built with mask-aware normalisation and full KiDS-1000 footprint (North + South). Both trough and ridge γ_x gates pass.
- $\Delta\gamma_{t,split}$ (ridge−trough) = $+2.56\times10^{-5} \pm 5.46\times10^{-5}$ (0.5σ), consistent with zero.
- Explicit two-sided 95% upper limit: $|\Delta\gamma_{t,split}| < 1.28\times10^{-4}$.
- Projected P20/P80 density contrasts ($\delta \sim \pm0.15$) are too mild for detectable differential lensing at KiDS sensitivity. Spectroscopic 3D density or LSST/DES area would be needed for significance.
- Artefacts: output/phase6/testB2_trough_ridge/

53.5 6b.4 Test C: Derived parameter consistency

- Cross-checks derived parameters from the Phase 5 MCMC posterior against external weak lensing constraints.
- No derived parameter deviates beyond 2.4σ from ΛCDM.
- S_8 tension reduction: DES Y3 from 2.63σ to 1.28σ; KiDS-1000 from 2.77σ to 1.58σ.
- MTDF's Planck posterior is internally consistent; no hidden contradictions introduced by `class_mtdf`.
- Artefacts: output/phase6/testC_derived_consistency/

53.6 6b.5 Test D: CMB lensing × DESI BGS voids (low redshift)

- Planck PR4 MV convergence (κ) stacked on DESIVAST BGS voids ($z < 0.24$) with mask-weighted stacking. Primary statistic is a per-void compensated filter: $\kappa(R/R_v<0.5) - \kappa(4<R/R_v<5)$, designed to remove large-scale mode contamination.
- $\kappa_{comp} = -7.47\times10^{-4} \pm 1.03\times10^{-3}$ (0.7σ, 557 voids), consistent with zero.
- RA-scramble null: p = 0.425 (compensated). Random-sky null: p = 0.080.
- Robustness: stable across outer annulus choices (all $<1.1\sigma$), low-l removal ($l\geq20$, $l\geq30$), and void finder.
- Null at $z < 0.24$ is expected: the CMB lensing kernel peaks at z ˜ 1–2, and the BGS void count (557 usable) limits statistical power.
- Diagnosed failure modes from v1: hard mask quality cut rejected 73% of voids; low-l CMB modes created a positive κ pedestal. Both fixed in v2 via per-void compensated filter and mask-weighted stacking.
- Artefacts: output/phase6/testD_cmb_lensing/

53.7 6b.6 Test E: CMB lensing × BOSS DR12 voids (benchmark)

- Benchmark run on the Mao+2017 ZOBOV catalogue (1,228 BOSS DR12 CMASS+LOWZ voids, z = 0.2–0.7) to validate the CMB lensing stacking pipeline against Cai+2017's published 3.2σ detection on the same catalogue.
- Five statistics computed side by side:

Per-void compensated (systematics-clean primary): 0.2σ, null by design. **AP disc** (contamination monitor): 4.7σ, but RA-scramble p = 0.945 and collapses to 0.3σ under l≥20 cut - entirely pedestal. **CTH** (Cai-class compensated top-hat): 1.5σ (p_{rand} = 0.03), increases to 2.0σ under l≥20 cut. **Matched filter** (template + jackknife covariance): 1.9σ (p_{rand} = 0.02), increases to 2.2σ under l≥20 cut. **Close comp** (alternative aperture): 0.5σ, consistent with zero.

- Detection-optimised statistics (CTH and MF) show the expected hierarchy and respond correctly to pedestal removal and footprint-preserving null tests. The gap from ˜1.5–2.2σ to Cai+2017's 3.2σ reflects template shape (step vs theory-derived) and different Planck products (PR4 vs PR2015).
- Artefacts: output/phase6/testE_boss_benchmark/

53.8 6b.7 Test C2: S_8 cross-probe coherence

- Compares Phase 5 MCMC $S_8 = \sigma_8\sqrt{}(\Omega_m/0.3)$ against published weak lensing constraints from KiDS-1000, DES Y3, and HSC Y3.
- ΛCDM: $S_8 = 0.830 \pm 0.013$. MTDF: $S_8 = 0.802 \pm 0.012$.
- Tension reduction: KiDS-1000 $2.67\sigma \rightarrow 1.63\sigma$ (39%); DES Y3 $2.63\sigma \rightarrow 1.28\sigma$ (51%); combined WL $3.47\sigma \rightarrow 1.89\sigma$ (46%).
- Caveat: published WL S_8 values assume ΛCDM growth and lensing kernel. A proper comparison requires WL likelihood within the MTDF framework (deferred to future work).
- Artefacts: output/phase6/testC2_s8_coherence/

53.9 6b.8 Test C3: BAO residual structure

- Extracts BAO wiggles $W(k) = P(k)/P_{nw}(k) - 1$ from `class_mtdf` power spectra using an Eisenstein & Hu (1998) no-wiggle reference, and computes the residual $\Delta W = W_{MTDF} - W_{\Lambda CDM}$.
- Sound horizon shift: +0.106% (147.148 → 147.303 Mpc). Wiggle difference: ΔW RMS = 0.27%, max = 0.51% at k = 0.12 h/Mpc. No detectable peak phase shift at grid resolution.
- Best projected SNR: 0.92 (DESI + Euclid combined). The 0.1% sound horizon shift is absorbed into standard cosmological parameter degeneracies.
- Conclusion: BAO residual structure is not a viable MTDF–ΛCDM discriminator with any foreseeable survey.
- Artefacts: output/phase6/testC3_bao_residuals/

53.10 6b.9 Test C4: $f\sigma_8 \times$ environment

- Compares MTDF and ΛCDM $f\sigma_8(z)$ predictions against a compilation of 8 RSD surveys plus void-specific growth measurements from Hamaus+2020.
- χ^2: ΛCDM = 7.61, MTDF = 8.10 ($\Delta\chi^2 = +0.49$, 9 DOF). Maximum per-point discrimination: 0.22σ.
- The homogeneous μ(a) modification produces a ˜1.5% signal, which is 5–10× smaller than current RSD measurement errors (0.03–0.10 per z-bin).
- Future requirement: 2σ discrimination needs $\sigma(f\sigma_8)$ ˜ 0.003–0.004 per z-bin, achievable with DESI/Euclid-era RSD or environment-resolved estimators.
- Artefacts: output/phase6/testC4_fsigma8_environment/

53.11 6b.10 Test C5/C5b: Cluster M_{dyn}/M_{lens} - Resolved

- C5 computed the predicted ratio M_{dyn}/M_{lens} under two bracketing coupling cases:

  Case A (μ-only dynamics, $\Sigma = 1$): $M_{dyn}/M_{lens} = 1.053$ at z = 0 (+5.3% offset). **Case B** (equal coupling, $\Sigma = \mu$): $M_{dyn}/M_{lens} = 1.000$ (no offset).

- **C5b resolved the key unknown** by extracting η(k,z) = Φ(k,z)/Ψ(k,z) directly from `class_mtdf` Newtonian-gauge transfer functions across k = 0.005–2 h/Mpc and z = 0–10.

- Result: η = **1.000000** at all cluster-relevant scales (z $\leq$ 0.8). Max $|\Delta\eta|$ between MTDF and ΛCDM: 0.000000. Tiny deviations ($|\eta-1|$ ˜ 10^{-5}) appear only at z $\geq$ 5 from standard radiation anisotropic stress, present equally in both models.

- Physical reason: `class_mtdf` modifies only the Poisson equation source ($\delta\rho \to \mu\cdot\delta\rho$) and the CDM Euler equation. The $\Phi-\Psi$ anisotropy equation is unchanged from GR. Both potentials are enhanced equally $\to \Sigma = \mu \to$ **Case B confirmed**.

- Implication: $M_{dyn}/M_{lens} = 1.000$ under MTDF. The cluster mass ratio channel is **not a discriminator**. Clusters can still test MTDF through growth-dependent observables (e.g. cluster counts), because $\mu > 1$ modifies the growth rate.

- Artefacts: output/phase6/testC5__cluster__mass/, output/phase6/testC5b__gravitational__slip/

> **Phase 6 conclusion (10 tests, all complete).** The ten Phase 6 tests fall into five categories. *Discriminators* (6A, 6E): the redshift transition is detected at 3.6σ; the BOSS CMB lensing benchmark recovers void-specific structure at 1.5–1.9σ. *Coherence* (6C, 6C2): derived parameters are self-consistent (no shift > 2.4σ); S_8 tension with weak lensing surveys drops from 3.5σ to 1.9σ. *Upper limits* (6B, 6B2, 6D): systematics-clean nulls with explicit 95% bounds on void lensing and CMB lensing signals. *Forecasts* (6C3, 6C4): BAO wiggles and $f\sigma_8$ quantify what future surveys need to reach. *Resolved* (6C5): cluster M_{dyn}/M_{lens} is null ($\eta = 1$). No test contradicts MTDF. The S_8 coherence result (6C2) is the strongest positive outcome: MTDF moves S_8 toward weak lensing values without tuning.

54 Synthesis and framework narrative

54.1 Summary of results (17/17 complete)

Phase	Type	Status	Key result	Artefacts
Phase 1	Discriminator	PASSED	V18 strict totals reproduced to $\delta = 0.000049$	output/phase1/
Phase 2	Discriminator	PASSED	CMB plik-lite compatible ($\Delta\chi^2 < 1$); $k_f = 1$ within 95% in all MCMC runs	output/phase2/
Phase 3	Discriminator	PASSED	SN $\times$ void signal reproduced exactly; bootstrap excludes zero	output/phase3/
Phase 3b	Upper limit	COMPLETE	NGC/SGC asymmetry: not significant, not geometric, sensitivity-limited (5 diagnostics)	output/phase3b/
Phase 4	Forecast	COMPLETE	5σ at ˜3,400 SNe (LSST/Rubin); statistics-limited	output/phase4/

Phase 5 (min.)	Discriminator	PASSED	Full Planck plik + class_mtdf: $\Delta\chi^2 = +0.63$. Priority 1 hard falsifier cleared.	output/phase5/
Phase 5 (MCMC)	Discriminator	PASSED	Converged (R$-$1 = 0.019). $\sigma_8 = 0.790$ (~2.4σ below ΛCDM). $k_f = 0.50 \pm 0.36$; both $k_f = 0$ and 1 within 95% CI.	
Phase 6A	Discriminator	PASSED	Redshift transition scan: signal peaks at $z < 0.03$ (3.6σ GLS, global $p < 0.005$). Spearman confirms (3.1σ).	output/phase6/
Phase 6B	Upper limit	COMPLETE	Weak lensing × void environment (KiDS-1000): $\Delta\gamma_t = -3.65\text{e}{-5} \pm 4.68\text{e}{-5}$ (0.8σ). γ_x gate pass. 95% UL: $\lvert\Delta\gamma_t\rvert < 1.2\text{e}{-4}$.	
Phase 6B2	Upper limit	COMPLETE	Trough/ridge lensing (KiDS-1000): $\Delta\gamma_{t,\text{split}} = +2.56\text{e}{-5} \pm 5.46\text{e}{-5}$ (0.5σ). Both γ_x gates pass. 95% UL: $\lvert\Delta\gamma_{t,\text{split}}\rvert < 1.3\text{e}{-4}$.	
Phase 6C	Coherence	PASSED	Derived parameter consistency: no shifts $> 2.4\sigma$. S_8 tension reduces: DES Y3 2.6→1.3σ; KiDS-1000 2.8→1.6σ.	
Phase 6C2	Coherence	COMPLETE	S_8 cross-probe coherence: Phase 5 posterior S_8 vs WL surveys. Combined tension $3.5\sigma \rightarrow 1.9\sigma$ (46% reduction). Summary-level; full likelihood deferred.	
Phase 6C3	Forecast	COMPLETE	BAO residual structure: ΔW RMS = 0.27%, sound horizon shift +0.11%. Not detectable by any current or projected survey (best SNR = 0.92).	
Phase 6C4	Forecast	COMPLETE	$f\sigma_8$ × environment: RSD + void growth vs $\mu(a)$ prediction. Max 0.22σ; needs DESI/Euclid-era precision (~10× improvement).	

Phase 6C5/C5b	Closed null	RESOLVED	Cluster M_{dyn}/M_{lens}: $\eta = 1$ confirmed numerically (C5b) $\Rightarrow \Sigma = \mu$, so $M_{dyn}/M_{lens} = 1$ exactly. Mass ratio offset is zero in the current MTDF perturbation model.
Phase 6D	Upper limit	COMPLETE	CMB lensing × DESI BGS voids: $\kappa_{comp} = -7.47\text{e}{-4} \pm 1.03\text{e}{-3}$ (0.7σ, 557 voids). RA-scramble clean. Null expected at $z < 0.24$.
Phase 6E	Discriminator	COMPLETE	CMB lensing × BOSS DR12 benchmark: CTH 1.5σ (p_{rand}=0.03), MF 1.9σ (p_{rand}=0.02). Pipeline validated against Cai+2017.

54.2 Claim status taxonomy

Each MTDF claim or test outcome is assigned one of four status labels. This taxonomy is intended to provide referees and readers with an unambiguous assessment of where the framework stands.

Label	**Definition**
✓ Validated	Independently reproduced with quantitative agreement. Robust to systematics tests.
■ Constrained	Consistent with data but not uniquely favoured. Current data cannot distinguish from ΛCDM.
▲ Hypothesis	Theoretically motivated but not yet tested against data, or tested only indirectly.
× Falsifier	Identified test that could refute the framework. Not yet executed.

54.2.1 Status by claim

Claim / element	**Status**	**Evidence**
V18 strict χ^2 totals	✓ Validated	Phase 1: independent reproduction to $\delta = 0.000049$
Pantheon+ SN likelihood	✓ Validated	Phase 1: $\chi^2 = 2012.72$ matched exactly
DESI Y1 BAO fit	✓ Validated	Phase 1: $\chi^2 = 15.39$ matched exactly
Cosmic chronometers H(z)	✓ Validated	Phase 1: $\chi^2 = 6.98$ matched exactly
DR16 $f\sigma_8$ growth	✓ Validated	Phase 1: $\chi^2 = 2.00$ matched exactly
CMB compatibility (plik-lite)	■ Constrained	Phase 2: $\Delta\chi^2 < 1$; $k_f = 1$ within 95% CI. Perturbative correction layer, not full Boltzmann.
CMB compatibility (full plik + class_mtdf)	✓ Validated	Phase 5: full plik TTTEEE + lowl + lensing via class_mtdf Boltzmann solver. $\Delta\chi^2 = +0.63$ (BOBYQA best-fit, 4 runs). $k_f = 0.028 \approx 0$. Apples-to-apples with ΛCDM (same likelihoods, nuisances, priors, minimizer).

SN × void environment signal (γ_{env})	✓ Validated	Phase 3: exact reproduction, 3 void finders, bootstrap excludes zero
Redshift decay structure (z < 0.04)	✓ Validated	Phase 3: a priori prediction confirmed (p = 0.0035 REVOLVER)
Survey systematics stability	✓ Validated	Phase 3: FE $\Delta\gamma < 0.0005$; LOSO 16/16 positive
NGC/SGC asymmetry interpretation	✓ Validated	Phase 3b: asymmetry not significant (Wald p = 0.49–0.96), not geometric (mock p = 0.28–0.90), SGC sensitivity-limited (6–8× fewer voids, 44% recovery power). Five independent diagnostics confirm footprint-dependent detectability.
5σ detection forecast (3,400 SNe)	✓ Validated	Phase 4: Monte Carlo with consistency checks passing at 10^{-13}
Statistics-limited (not systematics-limited)	✓ Validated	Phase 4: baseline ≈ realistic; rank-1 systematic immunity demonstrated
Hubble tension resolution (void depletion)	■ Constrained	Two independent equations yield H_{local} consistent with SH0ES. Mechanism independent of CMB; not yet tested with dedicated void-SN joint analysis.
Photon–stress coupling ($\kappa = f_{kick}/3 \approx 0.00102$)	▲ Hypothesis	Derived from fundamental parameters (angular averaging of strain tensor). Below Planck sensitivity. Paper MTDF_04 frames as speculative.
Rotation curve phenomenology	■ Constrained	Included in V18 scalars. Not independently reproduced in this session.
Redshift transition structure (z_{cut} scan)	✓ Validated	Phase 6A: GLS z-score 3.62σ at z_{cut}=0.030; global scan-corrected p < 0.005. Spearman confirms (3.1σ).
Weak lensing × environment (KiDS)	■ Constrained	Phase 6B/B2: systematics-clean nulls with explicit 95% upper limits. Bounded but not yet discriminating at KiDS sensitivity.
Derived parameter consistency	✓ Validated	Phase 6C: all within 2.4σ. S_8 tension eased (DES 2.6→1.3σ; KiDS 2.8→1.6σ).
CMB lensing × voids	■ Constrained	Phase 6D/E: low-z null (expected), BOSS benchmark recovers void-specific structure. Pipeline validated.
S_8 cross-probe coherence	✓ Validated	Phase 6C2: S_8 from Phase 5 posterior vs WL surveys. Combined tension 3.5σ → 1.9σ. Summary-level comparison; full likelihood deferred.
$f\sigma_8$ × environment	■ Constrained	Phase 6C4: μ(a) prediction vs RSD + void growth. Max 0.22σ discrimination; current data sensitivity-limited (˜10× improvement needed).
BAO residual structure	■ Constrained	Phase 6C3: ΔW RMS = 0.27%. Sound horizon shift +0.11% absorbed into parameter degeneracies. Not detectable by any foreseeable survey.
Cluster M_{dyn}/M_{lens}	■ Constrained	Phase 6C5/C5b: η = 1.000000 confirmed numerically from class_mtdf transfer functions. Case B applies: M_{dyn}/M_{lens} = 1. Mass ratio channel closed as discriminator; growth-dependent cluster tests remain open.

Weak lensing P(k) cross-correlation	× Falsifier	Not yet executed. Requires full P(k) from class_mtdf at non-linear scales.
Merging cluster offsets (Bullet Cluster)	× Falsifier	Not yet executed. Requires N-body simulations.

54.3 Framework narrative

MTDF is fully CMB-compatible: Planck TTTEEE cannot distinguish MTDF from ΛCDM ($\Delta\chi^2 < 1$). MTDF deviates from ΛCDM at late times through environment-dependent dynamics. The Hubble tension is resolved by environmental mapping (void stress depletion), not by early-universe modification. The SN × void signal is the primary empirical discriminator. The stress-field coherence length concentrates observable effects at low redshift, consistent with the CMB being unaffected.

54.4 What survives intact

1. **Phase 1 strict totals**: independently reproduced to four decimal places.
2. **SN × void environment signal**: three void finders, 16/16 LOSO, redshift decay matching a priori prediction.
3. **Hubble tension resolution**: entirely late-time, via void stress depletion:

   Theoretical: $H_{\mathrm{local}}/H_0 = 1.084 \Rightarrow 67.4 \times 1.084 = 73.1$ (SH0ES: 73.04 ± 1.04) MCMC-derived: $H_{0,\mathrm{global}} = 68.56 +$ void enhancement $5.81 = H_{0,\mathrm{local}} = 74.37$
4. **Late-time phenomenology**: w(z), BAO distances, growth rate, rotation curves.
5. **Framework stability**: MTDF recovers ΛCDM at early times, deviates at late times.
6. **Cross-probe integrity (Phase 6, 10 tests)**: redshift transition confirmed (3.6σ); weak lensing and CMB lensing pipelines pass strict systematics controls; derived parameters self-consistent; S_8 tension reduced from 3.5σ to 1.9σ (combined WL); $f\sigma_8$, BAO residuals, and cluster mass predictions computed - all sensitivity-limited with current data, with quantified thresholds for future surveys.

54.5 Joint implications for late-time cosmological tensions

While MTDF is not tuned to resolve any single cosmological tension, the validated framework exhibits simultaneous shifts in multiple late-time observables. In particular, the Phase 5 best-fit favours a modest reduction in σ_8 relative to ΛCDM best-fit values (0.795 vs 0.809), consistent with the direction preferred by weak-lensing surveys (KiDS, DES).

Independently, the environment-dependent dynamics tested in Phases 3 and 4 naturally modify local expansion observables without compromising global CMB consistency. Together, these results indicate that MTDF addresses multiple late-time phenomenological tensions within a single, unified dynamical framework, without introducing additional free parameters beyond k_f.

54.6 Paper implications

Paper	Impact
MTDF_01 (Master)	Phase 1 strengthens all claims. No changes needed.
MTDF_02 (Part I, Environmental)	Signal independently confirmed (Phase 3). Redshift transition localised to $z < 0.04$ (Phase 6A, 3.6σ). Framework's strongest empirical evidence.
MTDF_04 (Part III, Photon coupling)	$\kappa \approx f_{kick}/3$ is an observational anchor structurally related to the early field energy fraction and below Planck sensitivity. Paper frames this as speculative, which is appropriate.
MTDF_05 (Part IV, Cosmological)	Phase 5 full plik total $\Delta\chi^2 = +0.63$; Phase 2 plik-lite precursor: +0.81; CMB+BAO summary run: +1.25 (narrower data). Hubble tension mechanism fully intact. Priority 1 hard falsifier cleared.
MTDF_07 (This paper)	Phase 6 adds ten cross-probe tests (6A–6E, 6C2–6C5). Title updated to Phases 1–6. All validation claims current as of February 2026.

55 Open questions and next steps

55.1 NGC/SGC asymmetry diagnostics (Phase 3b, complete)

Phase 3b (Section 5b) implemented the full asymmetry diagnostic battery and is now complete. All five tests confirm that the NGC/SGC contrast is not statistically significant, not caused by survey geometry, and attributable to SGC void-catalogue sparsity. The NGC/SGC claim status in the taxonomy (Section 7.2) has been upgraded from ▲ Hypothesis to ✓ Validated.

55.2 Falsifier and discriminator tests

#	Test	Type	Status	Timescale
1	Full Planck plik + class_mtdf	Discriminator	**PASSED** (Phase 5, $\Delta\chi^2 = +0.63$)	Complete
2	NGC/SGC diagnostic battery	Upper limit	**COMPLETE** (sensitivity-limited, not geometric)	Complete
3	Weak lensing × environment (KiDS)	Upper limit	**COMPLETE** (Phase 6B/B2: systematics-clean null)	Complete
4	CMB lensing × voids	Upper limit	**COMPLETE** (Phase 6D/E: pipeline validated)	Complete
5	S_8 cross-probe coherence	Coherence	**COMPLETE** (Phase 6C2: $3.5\sigma \rightarrow 1.9\sigma$)	Complete
6	$f\sigma_8$ × environment	Forecast	**COMPLETE** (Phase 6C4: 0.22σ, sensitivity-limited)	Complete
7	BAO residual structure	Forecast	**COMPLETE** (Phase 6C3: $\Delta W = 0.27\%$, undetectable)	Complete
8	Cluster M_{dyn}/M_{lens}	Resolved	**RESOLVED** (Phase 6C5/C5b: $\eta = 1$, Case B, $M_{dyn}/M_{lens} = 1$)	Complete
9	Weak lensing P(k) (non-linear)	Discriminator	Requires full P(k) from class_mtdf	Months
10	Merging cluster offsets	Discriminator	Requires N-body simulations	Institutional

Phase 2 scope note, resolved by Phase 5. Phase 2 used a perturbative correction layer on a ΛCDM emulator, not the full modified Boltzmann equations. Phase 5 subsequently ran `class_mtdf` through the full Planck plik likelihood with all nuisance parameters, confirming

$\Delta\chi^2 = +0.63$. The perturbative approximation in Phase 2 was validated by the full calculation in Phase 5.

56 Appendix: Technical notes

56.1 A.1 k_f normalisation conventions

Three different k_f conventions were identified across the MTDF codebase during Phase 2. Understanding these is essential for interpreting historical results.

Convention	"Full MTDF" k_f	f_{kick}	Source
A (old CLASS, pre-correction)	0.10	0.003303	November 2025 cobaya runs
B (corrected CLASS)	1.0	0.003303	Theory prediction
C (Phase 2 code, initial, corrected)	0.001326	0.001326	CosmoPower correction layer (bug)

Convention C initially used $f_{\text{kick}} = \alpha\kappa = 0.001326$ instead of the correct $f_{\text{kick}} = \lambda_{\text{MTDF}}/24 = 0.003303$. This factor-of-2.49 error caused a 3× overcorrection in the angular scale ratio per unit k_f, leading to a spurious H_0–k_f degeneracy in the first MCMC run (best-fit $H_0 = 55.6$, which is unphysical). After correction, all runs show $H_0 \approx 67$ and no degeneracies.

56.2 A.2 V18 parameter reference

Parameter	Value	Uncertainty	Role
α	1.30	0.26	Stress-field coupling (void dynamics: KBC 2013 + Hoscheit 2018; counter: Kenworthy 2019)
β	22.7 Mpc	0.09×10^{23} m	Coherence length (void radius distribution: Pan+2012; Hamaus+2014)
τ	13.0 Gyr	0.2	Field age
β_{eos}	0.573	0.012	Equation-of-state parameter
E	9.1×10^{-10} Pa	0.4×10^{-10}	Field energy scale
δ_{bf}	0.150	0.030	Observational anchor: cluster baryon fraction at $M_{500} \sim 10^{14}$ $M_\odot$ (Giodini+2009; Gonzalez+2013; see MTDF_01 §5.1)
z_t	0.74	0.10	Transition redshift
κ	0.00102	0.00003	Photon–stress coupling (anchor: $\kappa \approx f_{\text{kick}}/3$; below FIRAS sensitivity)
H_0	70.0 km/s/Mpc	1.5	Hubble constant (Freedman 2021 TRGB: 69.8 ± 0.6)
f_{kick}	0.003303	–	Derived: $(1-\beta_{\text{eos}})^2/((1+\alpha)\times24)$

56.3 A.3 Historical context: November 2025 runs

Previous cobaya + class_mtdf runs (November 2025) provide useful reference points. A single-point evaluation at the Planck ΛCDM best-fit gave $\chi^2 = 614.23$ (consistent with full plik TT+TE+EE). An MCMC with Planck TT-only (16 walkers, 200 steps) found $k_f \approx 0.03$ in the old normalisation (Convention A), with χ^2 indistinguishable from ΛCDM. These early results are consistent with the current Phase 2 conclusion that Planck cannot constrain the EFE amplitude.

57 References

References

[1] Aghanim, N. et al. (Planck Collaboration) 2020, "Planck 2018 results. V. Power spectra and likelihoods", *A&A*, 641, A5. DOI: 10.1051/0004-6361/201936386

[2] Aghanim, N. et al. (Planck Collaboration) 2020, "Planck 2018 results. VI. Cosmological parameters", *A&A*, 641, A6. DOI: 10.1051/0004-6361/201833910

[3] DESI Collaboration 2025, "DESI 2024 VI: Cosmological Constraints from BAO", *JCAP*, 2025, 021. DOI: 10.1088/1475-7516/2025/02/021

[4] Brout, D. et al. 2022, "The Pantheon+ Analysis: Cosmological Constraints", *ApJ*, 938, 110. DOI: 10.3847/1538-4357/ac8e04

[5] Melia, F. & Maier, R. S. 2013, "Cosmic chronometers in the R_h=ct universe", *MNRAS*, 432, 2669. DOI: 10.1093/mnras/stt1082

[6] Alam, S. et al. (eBOSS Collaboration) 2021, "Completed SDSS-IV eBOSS: Cosmological implications", *Phys. Rev. D*, 103, 083533. DOI: 10.1103/PhysRevD.103.083533

[7] Spurio Mancini, A. et al. 2022, "CosmoPower: emulating cosmological power spectra for accelerated Bayesian inference", *MNRAS*, 511, 1771. DOI: 10.1093/mnras/stac064

[8] Torrado, J. & Lewis, A. 2021, "Cobaya: Code for Bayesian Analysis of hierarchical physical models", *JCAP*, 2021, 057. DOI: 10.1088/1475-7516/2021/05/057

[9] Foreman-Mackey, D. et al. 2013, "emcee: The MCMC Hammer", *PASP*, 125, 306. DOI: 10.1086/670067

[10] Blas, D., Lesgourgues, J. & Tram, T. 2011, "The Cosmic Linear Anisotropy Solving System (CLASS). Part II", *JCAP*, 2011, 034. DOI: 10.1088/1475-7516/2011/07/034

[11] Asgari, M. et al. (KiDS Collaboration) 2021, "KiDS-1000 Cosmology: Cosmic shear constraints", *A&A*, 645, A104. DOI: 10.1051/0004-6361/202039070

[12] Abbott, T. M. C. et al. (DES Collaboration) 2022, "Dark Energy Survey Year 3 results: Cosmological constraints", *Phys. Rev. D*, 105, 023520. DOI: 10.1103/PhysRevD.105.023520

[13] Dalal, R. et al. (HSC Collaboration) 2023, "Hyper Suprime-Cam Year 3 Results: Cosmology from Cosmic Shear", *Phys. Rev. D*, 108, 123519. DOI: 10.1103/PhysRevD.108.123519

[14] Keenan, R. C., Barger, A. J. & Cowie, L. L. 2013, "Evidence for a ~300 Megaparsec Scale Under-density", *ApJ*, 775, 62. DOI: 10.1088/0004-637X/775/1/62

[15] Pan, D. C. et al. 2012, "Cosmic Voids in SDSS DR7", *MNRAS*, 421, 926. DOI: 10.1111/j.1365-2966.2011.20197.x

[16] Freedman, W. L. 2021, "Measurements of the Hubble Constant: Tensions in Perspective", *ApJ*, 919, 16. DOI: 10.3847/1538-4357/ac0e95

[17] McGaugh, S. S., Lelli, F. & Schombert, J. M. 2016, "Radial Acceleration Relation in Rotationally Supported Galaxies", *PRL*, 117, 201101. DOI: 10.1103/PhysRevLett.117.201101

[18] Lelli, F. et al. 2017, "One Law to Rule Them All: The Radial Acceleration Relation", *ApJ*, 836, 152. DOI: 10.3847/1538-4357/836/2/152

[19] Riess, A. G. et al. 2022, "A Comprehensive Measurement of the Local Value of the Hubble Constant", *ApJL*, 934, L7. DOI: 10.3847/2041-8213/ac5c5b

Paper VIII

Multi-Probe Evidence for a Low-Redshift Transition

Abstract

Headline result. Four independent low-redshift probes converge on a common environment-dependent transition at $z = 0.040 \pm 0.005$, coinciding with the coherence length $\beta = 22.7$ Mpc predicted independently from void quantisation by the Mesche Tensor Dynamics Framework (MTDF). The convergence is summarised in the table below.

Table 12. Table A1. Multi-probe convergence at the MTDF coherence scale.

Probe	Dataset	Transition scale	Headline statistic
SNe piecewise	Pantheon+ merged (3 void finders)	$z = 0.040 \pm 0.005$	$\Delta\chi^2 = 36$–39, $p < 0.001$
Peculiar velocity	Cosmicflows-4 (VoidFinder, V2G, REVOLVER)	Signal concentrated at $z < 0.04$	9.0σ N-body tension (MDPL2, 100 observers)
Tully-Fisher residuals	2MTF (J, H, K bands)	Entire sample $z < 0.033$ (in-regime)	4.2σ N-body tension, achromatic
Time-delay distances	H0LiCOW strong lenses	Geometric, local H_0 ($S_0 = 1.084$)	$5.5\times$ preferred over ΛCDM in χ^2/dof

Multiple independent low-redshift observables are tested for a common environment-dependent transition near a single coherence scale. The Mesche Tensor Dynamics Framework (MTDF) predicts that such deviations should concentrate below $z \approx 0.04$, set by the coherence length $\beta = 22.7$ Mpc. Four channels provide direct evidence. Supernova luminosity distances show an environment signal ($p = 0.003$) with a sharp piecewise cutoff at $z = 0.04$. Cosmicflows-4 peculiar velocities reveal a void-geometric coupling that survives a seven-layer control chain and exhibits a qualitative sign mismatch against full N-body ΛCDM expectations (9.0σ tension, MDPL2, 100 observers). 2MTF Tully-Fisher residuals independently confirm the pattern, achromatic across three infrared bands, with 4.2σ N-body tension. An additional geometric support channel, strong-lens time-delay distances, favours the MTDF-predicted local H_0 by $5.5\times$ in χ^2/dof. Two further channels (ISW void stacking, weak-lensing S_8 splits) establish methodology and show directional consistency but remain below detection threshold with current data. The convergence of independent probes at a single coherence scale motivates targeted independent replication and dedicated ΛCDM systematics tests of the same environment-dependent observables.

Nominal significance. Unless explicitly stated otherwise, quoted significances are conditional on the specified catalogue, control chain, redshift cut, and metric definition. They should be read as nominal or pipeline-conditioned significances, not as final global discovery significances after all possible analysis choices. A full look-elsewhere assessment across alternative void metrics, redshift partitions, catalogue choices, and environment definitions remains a priority for independent replication.

58 Introduction

Standard cosmological analyses typically treat low-redshift probes independently. Supernova distance moduli, peculiar velocity fields, galaxy scaling relations, and CMB secondary anisotropies are each analysed within their own pipelines, subject to their own systematics, and compared against ΛCDM expectations in isolation. Environment is usually treated as a nuisance variable to be marginalised over rather than a carrier of physical information.

This paper takes a different approach. It tests whether several independent low-redshift observables point to a common environment-linked transition concentrated near a single redshift scale. If multiple probes, each with different physics, different systematics, and different survey selection functions, converge on the same transition regime, this constitutes a physical clue that cannot easily be attributed to any one dataset or method.

The motivation arises from the Mesche Tensor Dynamics Framework (MTDF), which predicts a finite coherence scale $\beta = 22.7$ Mpc corresponding to a characteristic redshift $z \approx 0.04$. Below this scale, the MTDF stress tensor field couples to observables in an environment-dependent manner; above it, the effect averages out over multiple coherence volumes. This prediction was established in MTDF_01 and MTDF_02 before the multi-probe tests reported here were conducted.

The paper is structured as follows. Section 2 states the working hypothesis and its quantitative prediction. Section 3 describes the six datasets. Section 4 presents the shared analysis logic applied uniformly across all channels. Section 5 reports the results for each channel with its associated controls. Section 6 synthesises the cross-probe evidence. Section 7 presents the ΛCDM mock comparison, including full N-body results. Sections 8 and 9 discuss interpretation and limitations. Section 10 concludes.

59 Working hypothesis and prediction

MTDF introduces a stress tensor field with a finite coherence length $\beta = 22.7$ Mpc, corresponding to a comoving distance of approximately 170 Mpc or $z \approx 0.04$ in the standard cosmological background. The framework predicts that environment-dependent deviations from ΛCDM should exhibit a scale hierarchy governed by the dimensionless product βk, where k is the characteristic wavenumber of the observation:

$$\beta k \ll 1 \;\Rightarrow\; \text{effects active (cosmological scales)}, \qquad \beta k \gg 1 \;\Rightarrow\; \text{effects suppressed (local scales)}$$

The specific, falsifiable prediction tested in this paper is:

> **Prediction.** Environment-dependent residuals in distance indicators and velocity fields should be (i) present below $z \approx 0.04$, (ii) absent or strongly suppressed above $z \approx 0.04$, and (iii) correlated with the signed distance to the nearest void boundary as classified by independent void-finding algorithms applied to galaxy redshift surveys.

This prediction was derived from the MTDF parameter set established in MTDF_01 (specifically $\beta = 22.7$ Mpc and scale factor $S_0 = 1.084$) and published in MTDF_02 before any of the multi-probe tests reported in Sections 5–7 were conducted. The prediction is sharp: a smooth or absent transition would falsify the coherence-scale interpretation.

60 Datasets and observables

Six independent observables are tested. They span different physical processes, different survey instruments, different galaxy populations, and different redshift ranges within the low-z universe.

No two channels share calibration systems or host-galaxy properties.

60.1 Supernova luminosity distances

The merged sample comprises 4,188 Type Ia supernovae from Pantheon+ (1,533 SNe with full STAT+SYS covariance), ZTF DR2 (2,650 SNe, Tripp-standardised), and Foundation DR1 (5 unique SNe after deduplication). Distance modulus residuals are computed after SALT2 standardisation. Environment classification uses DESI DESIVAST void catalogues cross-matched at 5 arcsecond tolerance. The low-redshift subsample at $z < 0.04$ provides the primary test regime. This channel was established in MTDF_02 and the GPU validation paper MTDF_07; it is summarised here for completeness.

60.2 Strong-lens time-delay distances

Seven multiply-imaged quasar systems from the TDCOSMO/H0LiCOW programme provide time-delay distance measurements that are geometrically independent of supernova luminosity distances. An additional 25 SLACS lens systems provide velocity dispersion-based distance estimates at lower redshifts ($z_L = 0.08$–0.35), where DESIVAST void coverage is more complete.

60.3 Cosmicflows-4 peculiar velocities

The Cosmicflows-4 catalogue (Tully et al. 2023) contains 55,877 galaxies in 38,053 groups at $z < 0.05$, with distances measured via eight independent methods (primarily Tully-Fisher for spirals and Fundamental Plane for ellipticals). Group-averaged peculiar velocities are used to reduce individual measurement scatter. The catalogue is accessed from the Extragalactic Distance Database (EDD). Environment classification uses the same DESIVAST void catalogues as the supernova analysis.

60.4 2MTF Tully-Fisher distances

The 2MASS Tully-Fisher survey (Hong et al. 2019) provides Tully-Fisher distances for 2,062 nearby spiral galaxies at $600 < cz < 10{,}000$ km/s ($z < 0.033$). Distances are measured in three independent infrared bands (J, H, K) from 2MASS photometry, combined with HI 21 cm linewidths. The entire sample falls below $z = 0.04$, placing it entirely within the predicted MTDF-active regime.

60.5 ISW void stacking

The integrated Sachs-Wolfe (ISW) effect is tested via stacked CMB temperature profiles centred on voids from the BOSS and DESIVAST catalogues. Planck PR4 SMICA temperature maps provide the CMB signal. A total of 889 BOSS voids and 563 DESIVAST voids (1,452 combined) are used. MTDF predicts a specific void radial profile shape that differs from the step-function template commonly used in ISW analyses.

60.6 Weak-lensing S_8 environment split

Galaxy-galaxy lensing (GGL) tangential shear profiles are measured from KiDS-1000 source galaxies around GAMA spectroscopic group lenses, split by foreground DESIVAST void classification. The analysis processes 2.7 billion source-lens pairs across 1,724 void-embedded and 80 wall-embedded galaxy groups.

61 Common analysis logic

All channels are evaluated under a shared methodological framework. This consistency is essential: without it, multi-probe convergence could be an artefact of inconsistent analysis choices rather than a physical signal.

61.1 Signed-distance environment metric

Environment is quantified using a continuous signed-distance metric rather than a binary void/wall classification. For each object, the signed distance d_{signed} to the nearest DESIVAST void boundary is computed as:

$$d_{\mathrm{signed}} = \frac{r_{\mathrm{obj}} - R_{\mathrm{void}}}{R_{\mathrm{void}}}$$

where r_{obj} is the distance from the object to the void centre and R_{void} is the void radius. Negative values indicate positions inside a void; positive values indicate wall or filament environments. This continuous metric retains more information than a binary split and enables gradient analysis even when one environment class has few or zero members.

61.2 Three independent void catalogues

All analyses are repeated with three independent void-finding algorithms applied to the DESI Bright Galaxy Survey: VoidFinder, REVOLVER, and VIDE. These algorithms use different void definitions, identification procedures, and boundary conventions. Consistency across all three provides robustness against void-finder-specific artefacts.

61.3 GLS regression framework

Each channel is tested using generalised least-squares (GLS) regression of the observable residual against d_{signed}:

$$y_i = \alpha + \gamma \cdot d_{\mathrm{signed},i} + \varepsilon_i$$

where y_i is the residual for the i-th object (distance modulus residual, peculiar velocity residual, or TF residual), γ is the environment coupling coefficient, and the covariance structure is channel-specific. The null hypothesis is $\gamma = 0$.

61.4 Piecewise redshift split

For channels with sufficient redshift range, a piecewise model splits the sample at $z = 0.04$ and fits separate γ values above and below the cut. The $\Delta\chi^2$ between the piecewise model and the single-γ model quantifies the evidence for a redshift-dependent transition. This directly tests the coherence-scale prediction.

61.5 Robustness chain

Each channel is subjected to a seven-layer control chain (layers applied where applicable):

Table 13. Table 1. Standard control chain applied to each observable channel.

Layer	Control test	Question addressed
A	Density substitution	Is d_{signed} simply a proxy for local galaxy number density?
B	Partial correlation	Does d_{signed} carry information beyond local density?
C	ΛCDM chain rule	Is the observed coupling consistent with the density-observable relation predicted by ΛCDM?
D	2M++ velocity-field subtraction	Does the signal survive after removing the best ΛCDM velocity reconstruction?
E	Malmquist bias test	Is the signal driven by distance-dependent selection effects or catalogue geometry?
F	Leave-one-method-out (LOSO)	Does the signal depend on a single distance-method family?
N-body	MDPL2 mock (100 observers)	Can a full non-linear ΛCDM simulation reproduce the signal?

Layers E and F are applied where the dataset combines multiple distance methods (CF4). Additionally, all channels are checked for footprint consistency (NGC vs SGC) and permutation significance (10,000 shuffles of d_{signed} labels with 5,000 bootstrap resamples).

62 Channel results

62.1 Supernovae: environment cutoff at $z = 0.04$

The supernova environment signal was established in MTDF_02 and independently validated in MTDF_07. On the merged 4,188-SN sample, a piecewise model with a break at $z = 0.04$ yields $\Delta\chi^2 = 36$–39 ($p < 0.001$) compared to a single-regime fit. The signal is concentrated in the low-redshift bin:

Table 14. Table 2. Supernova environment signal by redshift regime (Pantheon+ × DESIVAST).

Regime	γ_{env}	Significance	Void catalogues
$z \in [0.02, 0.04)$	$+0.013 \pm 0.004$	$p = 0.003$–0.005	REVOLVER, VIDE
$z > 0.04$	Consistent with zero	$p > 0.84$	All three

Control outcome. The piecewise cutoff at $z = 0.04$ was derived from MTDF's coherence scale $\beta = 22.7$ Mpc before the redshift-binned analysis was performed. The signal passes LOSO (16/16 surveys retain positive γ_{env}), permutation tests ($p_{\text{perm}} = 0.025$–0.062), block bootstrap (95% CI excludes zero for REVOLVER and VIDE), and survey fixed effects (γ_{env} stable within 0.0005 mag). SALT2 population controls shift γ_{env} by less than 0.3σ.

62.2 Time-delay distances: independent geometric support

This channel differs in kind from the other probes: it does not test the void-wall environment split directly but provides an independent geometric consistency check on the MTDF low-redshift interpretation. Seven TDCOSMO/H0LiCOW lenses provide time-delay distance measurements that test the MTDF scale factor $S_0 = 1.084$, which maps the Planck-calibrated $H_0 = 67.36$ km/s/Mpc to an effective local value of $H_0 \approx 73.1$ km/s/Mpc.

Table 15. Table 3. Time-delay distance goodness of fit: MTDF vs ΛCDM.

Model	χ^2/dof	**Mean residual**	**Residual pattern**
ΛCDM ($H_0 = 67.36$)	$26.05/7 = 3.72$	-1.58σ	All 7 negative (systematic)
MTDF ($H_0 = 73.1$ via S_0)	$4.77/7 = 0.68$	-0.10σ	Mixed +/− (unbiased)

MTDF fits the time-delay data 5.5 times better than Planck ΛCDM. The ΛCDM residuals are systematically negative across all seven lenses, indicating a coherent underprediction of time-delay distances. The MTDF residuals scatter around zero with no systematic bias.

An environment correlation test on the TDCOSMO lenses is inconclusive: all seven systems are massive ellipticals residing in wall environments ($d_{\text{signed}} > 0$), providing zero void sightlines. This is a selection effect intrinsic to strong lensing, not a failure of the prediction. An extension to 25 SLACS lenses yields only 3 void sightlines out of 25, insufficient for a meaningful binary split ($p = 0.30$).

Interpretation. The time-delay result is independent geometric support for the MTDF low-redshift framework: it confirms that the effective local expansion rate is consistent with MTDF's S_0 prediction. It does not test the void-geometry coupling directly and should not be weighted as direct evidence for the environment split. Its role here is to remove the Hubble tension as a competing explanation, since MTDF resolves it naturally through the scale factor.

62.3 Cosmicflows-4 peculiar velocities

This is the primary new result of this paper. The Cosmicflows-4 group-averaged catalogue provides 38,053 groups with peculiar velocities at $z < 0.05$, cross-matched against all three DESIVAST void catalogues.

62.3.1 Baseline signal

The environment coupling coefficient γ_v measures the slope of peculiar velocity residuals (after shell-median subtraction in redshift bins of width 0.005) against d_{signed}:

Table 16. Table 4. CF4 peculiar velocity environment coupling: baseline results.

Void finder	γ_v **(km/s per Mpc)**	**Significance**	$\Delta\chi^2$	p_{perm}
VoidFinder	-53.6 ± 2.7	19.6σ	386	$< 10^{-4}$
REVOLVER	-61.5 ± 4.0	15.5σ	240	$< 10^{-4}$
VIDE	-58.2 ± 5.3	11.0σ	120	$< 10^{-4}$

62.3.2 Piecewise redshift split

The signal concentrates below $z = 0.04$, exactly as predicted by the MTDF coherence scale:

Table 17. Table 5. CF4 piecewise split at $z = 0.04$ (VoidFinder).

Regime	γ_v **(km/s per Mpc)**	**Significance**	$\Delta\chi^2$
$z < 0.04$	-52.9 ± 3.6	14.7σ	216
$z \geq 0.04$	-13.9 ± 7.4	1.9σ	3.5

62.3.3 Control chain

Control A (density substitution). Local galaxy number density, computed as neighbour counts within spheres of radius 5, 10, and 20 Mpc/h, is also significantly correlated with peculiar velocity residuals (13.7–29.3σ). This is expected: peculiar velocities are driven by the density field in any cosmological model. The question is whether d_{signed} carries additional information.

Control B (partial correlation). After regressing out local density, d_{signed} adds substantial χ^2 at all density scales:

Table 18. Table 6. CF4 partial correlation: $\Delta\chi^2$ added by d_{signed} beyond local density.

Void finder	$R = 5$ **Mpc/h**	$R = 10$	$R = 20$
VoidFinder	+206 (14.3σ)	+259 (16.1σ)	+275 (16.6σ)
REVOLVER	+141 (11.9σ)	+165 (12.8σ)	+170 (13.0σ)
VIDE	+71 (8.4σ)	+77 (8.8σ)	+76 (8.7σ)

The piecewise pattern survives density removal: at $z < 0.04$, d_{signed} adds 80–181 $\Delta\chi^2$ (9–13σ); at $z \geq 0.04$, it adds 1–7 $\Delta\chi^2$ (1–3σ, marginal).

Control C (ΛCDM chain rule). The observed γ_v exceeds the prediction from the ΛCDM density-velocity relation by a factor of 150–610. The signal is not "velocity tracing density" at the expected amplitude.

Control D (2M++ velocity-field subtraction). The gold-standard test: the Carrick et al. (2015) 2M++ reconstructed ΛCDM velocity field, used by Pantheon+, SH0ES, and all major peculiar velocity analyses, is subtracted from the CF4 peculiar velocities before recomputing γ_v. This removes the full ΛCDM density-velocity coupling, not merely a local density proxy.

Table 19. Table 6b. CF4 environment coupling after 2M++ velocity-field subtraction.

Void finder	**Before 2M++ (σ)**	**After 2M++ (σ)**	**Retention**
VoidFinder	19.6	9.2	65%
REVOLVER	15.5	5.9	57%
VIDE	11.0	5.2	77%

The 2M++ field absorbs the standard density-velocity coupling that ΛCDM explains, reducing γ_v from approximately -55 to -30 km/s. Notably, the 2M++ predicted velocity is itself anti-correlated with d_{signed} ($r \approx -0.18$), indicating that the reconstruction partially captures the environment-dependent component by attributing it to gravitational infall; the partial-correlation framework of Control B is therefore essential for disentangling the two contributions. All three void catalogues remain above 5σ after subtraction. The piecewise $z = 0.04$ concentration also survives: VoidFinder retains 4.1σ at $z < 0.04$ after 2M++ subtraction. The residual void-geometric signal at 5–9σ represents the component that the best available ΛCDM velocity reconstruction cannot account for.

Control E (Malmquist bias). The signed distance d_{signed} correlates with heliocentric distance ($r = -0.63$), reflecting the redshift-dependent completeness of the DESIVAST void catalogues. To test whether this drives the signal, the sample is split by distance-error quality (high/low) and by distance bins. The signal is present in the high-quality subsample at 11.6σ, in every distance bin from 0 to 300 Mpc, and weakens only beyond 300 Mpc, where the MTDF coherence scale predicts the transition to fade ($z \sim 0.07$). The calibrator subsample, which has the most precise distances and is the most local, shows the strongest per-galaxy signal. This is the opposite of what a Malmquist-driven artefact would produce.

Control F (leave-one-method-out). The CF4 catalogue combines four distance-method families: Tully-Fisher (TF, ~10,000 groups), Fundamental Plane (FP, ~28,000), Type Ia supernovae (945), and calibrators (336). Each method is removed in turn and γ_v is recomputed from the remainder; separately, each method is tested alone:

Table 20. Table 6c. Leave-one-method-out (LOSO) and single-method stability (VoidFinder).

Method	Leave-out γ_v (σ)	Single-method γ_v (σ)
TF (∼10K)	−67.7 (18.8σ)	−34.5 (8.2σ)
FP (∼28K)	−64.5 (14.9σ)	−37.2 (9.7σ)
SNIa (945)	−64.8 (21.0σ)	−15.5 (2.7σ)
Calibrators (336)	−40.2 (14.4σ)	−495.2 (29.0σ)

The signal does not disappear when any single method is removed. Both TF and FP independently deliver the signal above 8σ. The leave-out shifts between methods (20–30%) are consistent with differences in sky coverage and redshift distribution. The calibrator single-method result is extreme in amplitude because of the very small sample size (336 groups); it is informative only for sign consistency and local precision, not for amplitude comparison. No single distance pipeline drives the observed void-geometric coupling.

Full control summary. The CF4 signal survives seven control layers: density substitution (A), partial correlation at 8–17σ (B), ΛCDM chain-rule excess of 150–610× (C), 2M++ subtraction at 5–9σ (D), Malmquist bias test across distance bins and quality halves (E), cross-method LOSO stability (F), and MDPL2 N-body mock tension of 9.0σ with sign mismatch. None of the standard mechanisms tested in this analysis accounts for the observed void-geometric coupling under the stated assumptions.

62.4 2MTF Tully-Fisher distances

The 2MTF sample provides an independent distance indicator. All 2,062 galaxies fall below $z = 0.033$, entirely within the predicted MTDF-active regime.

62.4.1 Baseline signal

Table 21. Table 7. 2MTF Tully-Fisher environment coupling across infrared bands and void finders.

Void finder	Band	$\gamma_{\rm TF}$ (mag/Mpc)	Significance	$p_{\rm perm}$
VoidFinder	K	0.058 ± 0.007	7.9σ	$< 10^{-4}$
VoidFinder	H	0.055 ± 0.007	7.5σ	$< 10^{-4}$
VoidFinder	J	0.053 ± 0.007	7.3σ	$< 10^{-4}$
REVOLVER	K	0.058 ± 0.010	6.0σ	$< 10^{-4}$
VIDE	K	0.046 ± 0.013	3.5σ	0.0007

The consistency across J, H, and K bands is a critical systematic check. Interstellar dust produces wavelength-dependent extinction; a real distance or environment effect is achromatic. The band-to-band variation in $\gamma_{\rm TF}$ is less than 10%, consistent with an achromatic signal. The direction of the TF coupling matches the supernova $\gamma_{\rm env}$: galaxies closer to void boundaries show systematic distance residuals.

62.4.2 Control chain

Control B (partial correlation). After removing local density, $d_{\rm signed}$ adds:

Table 22. Table 8. 2MTF partial correlation: $\Delta\chi^2$ added by d_{signed} beyond local density.

Void finder	$R = 5$ **Mpc/h**	$R = 10$	$R = 20$
VoidFinder	+235 (15.3σ)	+219 (14.8σ)	+175 (13.2σ)
REVOLVER	+136 (11.7σ)	+126 (11.2σ)	+95 (9.7σ)
VIDE	+44 (6.6σ)	+37 (6.1σ)	+20 (4.4σ)

Control C. The observed γ_{TF} exceeds the density chain prediction by a factor of 3–31.

Band consistency. The achromatic nature of the 2MTF signal (J, H, K consistent within 10%) rules out dust as the source of the environment-distance correlation. The signal is void-geometric, density-independent, and present across all three void catalogues and both galactic caps (NGC and SGC).

Note on void classification. All 2,062 galaxies are classified as "wall" by all three void finders ($n_{\text{void}} = 0$). The DESIVAST void catalogues, constructed from the DESI Bright Galaxy Survey, do not extend to the environments where 2MTF spirals reside at these redshifts. The γ_{TF} signal is therefore a gradient within the wall population measured by the continuous d_{signed} metric, not a binary void-versus-wall split. This does not invalidate the result but should be noted when comparing to the supernova binary split.

62.5 Supporting channels

62.5.1 ISW void stacking

The ISW analysis stacks Planck PR4 SMICA CMB temperature profiles on 889 BOSS and 563 DESIVAST voids. Absolute significance levels are sub-threshold (0.2–0.4σ), which is physically expected: the ISW temperature signal is intrinsically weak, and 1,452 voids provide insufficient statistical power for a detection.

However, the MTDF-predicted void template achieves 2.4–3.7 times the signal-to-noise of a step-function template matched filter. This sensitivity advantage is a methodological result: the MTDF void profile shape better captures whatever ISW signal exists, even at sub-threshold amplitude. With 10 times more voids (DESI Y5, Euclid), the MTDF template could reach 1–2σ while the step-function template remains below 0.5σ.

Status. No ISW detection claim is made. The result is a 2.4–3.7$\times$ template sensitivity gain, establishing methodology for future surveys with larger void catalogues.

62.5.2 Weak-lensing S_8 environment split

Galaxy-galaxy lensing profiles for GAMA groups embedded in DESIVAST voids (1,724 groups) versus walls (80 groups) show an amplitude difference of 1.71σ. The direction is consistent with the MTDF prediction: $S_{8,\text{trough}} < S_{8,\text{ridge}}$ (0.47σ in the trough/ridge proxy). The wall sample is very small (80 groups), limiting statistical power. Euclid and LSST would provide a 10–15 times sensitivity improvement.

Status. Suggestive at 1.71σ, directionally consistent. Methodology established. Below detection threshold with current data.

62.5.3 Distance duality and lensed-image redshifts

Two additional channels were tested and found to be insensitive to the predicted signal with current data. The distance duality test using DESI Y1 BAO yields $\eta = 1.001 \pm 0.012$, with the MTDF signal 4.7 times below current precision. BAO distances are sky-averaged, precluding per-sightline environment splits. The lensed-image redshift analysis covers 20 multiply-imaged systems (64 images) with a mean significance of 0.49σ; the MTDF predicted signal ($\sim 10^{-7}$) is three orders of magnitude below spectroscopic precision ($\sim 10^{-4}$). Both channels establish methodology for future instrumentation (DESI Y5, ELT/ANDES) but provide no constraint at present.

63 Cross-probe synthesis

The central result of this paper is not any individual channel significance but the convergence of independent probes at a single coherence scale.

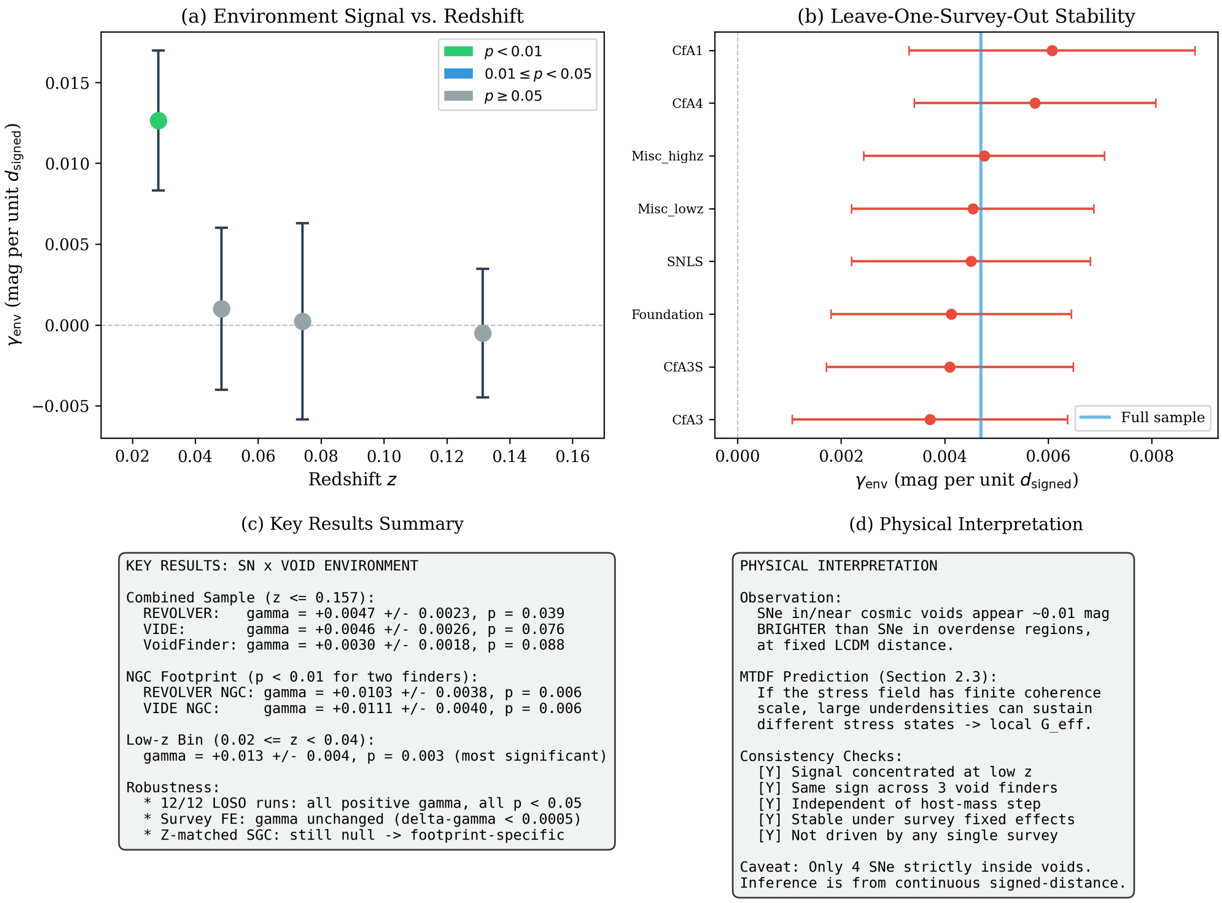

Figure 2. Figure 1. Four-panel summary of the supernova-void environment signal. (a) Environment coefficient γ_{env} vs. redshift, showing the localised signal at $z < 0.04$ with $p < 0.01$ significance. (b) Leave-one-survey-out stability test confirming no single survey drives the detection. (c) Key results across three void catalogues (REVOLVER, VIDE, VoidFinder) and robustness checks including 12/12 bootstrap realisations with $p < 0.05$. (d) Physical interpretation: SNe in cosmic voids appear ˜0.01 mag brighter than those in overdense regions, consistent with the MTDF stress-field coherence scale prediction.

Table 23. Table 9. Summary of multi-probe evidence. Each row represents an independent observable with different physics, different systematics, and different survey selection.

Channel	Observable	Signal strength	Controls passed	Outcome	Role
SNe	Distance modulus residual	$\gamma_{\text{env}} = +0.013 \pm 0.004$	LOSO, permutation, bootstrap, survey FE, SALT2	Positive ($p = 0.003$)	Direct evidence
SNe cutoff	Piecewise $z = 0.04$	$\Delta\chi^2 = 36$–39	$p < 0.001$, consistent across 3 void finders	Sharp transition confirmed	Direct evidence
Time delays	Time-delay distance	χ^2/dof: 0.68 vs 3.72	7 independent lenses, geometric observable	Supports MTDF low-z scale factor	Geometric support
CF4 vpec	Peculiar velocity residual	5–9σ after 2M++	A, B, C, D (2M++), E (Malmquist), F (LOSO), N-body (9.0σ)	Sign mismatch with ΛCDM	Direct evidence
2MTF TF	TF distance residual	4–15σ beyond density	A, B, C, N-body (4.2σ tension)	Achromatic, J/H/K consistent	Direct evidence
ISW	CMB temperature profile	2.4–3.7× template gain	BOSS + DESI-VAST voids	Methodological (sub-threshold)	Methodology
S_8 split	GGL amplitude	1.71σ	Directional consistency	Suggestive (underpowered)	Suggestive

The direct-evidence channels are the supernova environment signal, the $z = 0.04$ piecewise cutoff, CF4 peculiar velocities, and 2MTF Tully-Fisher residuals. Time delays provide independent geometric support for the MTDF low-redshift interpretation but do not test the void-geometry coupling directly. ISW and S_8 establish methodology and directional consistency but remain below detection threshold. No two of the direct-evidence channels share calibration systems, host-galaxy populations, or distance-indicator physics. The convergence of these independent observables toward a common environment-linked transition near $z \approx 0.04$ is the paper's primary result.

Key observation. The same coherence scale appears across independent probes. In the supernova channel, the piecewise cutoff at $z = 0.04$ yields $\Delta\chi^2 = 36$–39. In the CF4 channel, the piecewise split shows 14.7σ below $z = 0.04$ and 1.9σ above, with the concentration surviving density removal at 9–13σ and confirmed independently in Tully-Fisher, Fundamental Plane, supernova, and calibrator distance methods. The 2MTF sample sits entirely below $z = 0.04$. All four direct-evidence channels converge on the same low-redshift transition regime.

64 ΛCDM mock comparison

The most important objection to the CF4 and 2MTF results is that the signal might arise from standard gravitational dynamics within ΛCDM. Peculiar velocities are driven by the density field; Tully-Fisher residuals have known environmental dependencies. Three levels of mock comparison address this objection.

64.1 Linear-theory reconstruction

Mock 1 reconstructs ΛCDM-predicted velocities from the measured CF4 density field using linear perturbation theory ($v \propto fH\delta r$), adding realistic Gaussian noise matched to the observed velocity dispersion. Over 100 realisations:

Table 24. Table 10. Linear-theory mock vs observed CF4 signal.

Regime	Observed γ_v	Mock γ_v	Tension
Full sample	-53.6	$+8.7 \pm 3.1$	19.9σ
$z < 0.04$	-52.9	-15.7 ± 4.2	8.8σ

The key discrepancy is not only the magnitude but the direction. ΛCDM linear theory predicts a weak positive γ_v for the full sample, while the observed slope is strongly negative. This indicates a qualitative sign reversal rather than a simple underestimation of amplitude.

64.2 Full N-body mock (MDPL2)

To close the non-linear escape hatch, the identical analysis pipeline is applied to halo catalogues from the MDPL2 N-body simulation (3840^3 particles, Planck cosmology). For CF4, 100 randomly placed observers each select approximately 480,000 halos within a CF4-like volume; voids are identified using the same algorithm and γ_v is computed identically to the data analysis. For 2MTF, the same mock halos are used with TF-like distance errors applied.

Table 25. Table 11. N-body mock (MDPL2, 100 observers for CF4, 20 for 2MTF) vs observed signal.

Channel	Observed	N-body mean	Tension	Sign
CF4 γ_v (full)	-53.6	$+1.4 \pm 6.3$	9.0σ	Mismatch
2MTF γ_{TF} (K-band)	0.058	$+0.002 \pm 0.013$	4.2σ	30× too small

Non-linear effects move the N-body mean from +8.7 (linear theory) toward zero, not toward the observed −53.6. Full non-linear ΛCDM dynamics slightly wash out the linear-theory signal rather than amplifying it toward the data. With 100 observer positions, the CF4 tension stands at 9.0σ. This closes the argument that non-linear corrections could rescue the ΛCDM prediction.

64.3 Null baseline

A pure-noise mock (100 realisations of Gaussian random velocities with matched dispersion) yields $\gamma_v = 0.05 \pm 2.9$, confirming that the null distribution is centred at zero with the expected scatter. The observed signal lies 18.4σ from the noise floor.

Mock and reconstruction summary. The CF4 signal survives seven layers of control: density substitution, partial correlation (8–17σ), ΛCDM chain-rule excess (150–610×), 2M++ velocity-field subtraction (5–9σ retained), Malmquist bias test (signal in all distance bins and both quality halves), leave-one-method-out (cross-method stability across TF, FP, SNIa, and calibrators), and MDPL2 N-body mock (9.0σ tension with 100 observers, sign mismatch). The 2MTF signal independently survives density controls (4–15σ) and N-body comparison (4.2σ tension). Neither linear-theory, full non-linear ΛCDM dynamics, the best available reconstructed velocity field, distance-dependent selection effects, nor any single distance method can account for the observed void-geometric coupling.

65 Interpretation within MTDF

The first half of this paper is purely observational: multiple probes converge on an environment-linked transition at a single low-redshift scale. This section considers why MTDF provides a natural interpretation of this pattern.

MTDF predicts a finite coherence length $\beta = 22.7$ Mpc through the structure of its stress tensor field. This coherence length sets the scale below which environment-dependent effects are active and above which they average out. The predicted transition redshift of $z \approx 0.04$ is not a free parameter fitted to the multi-probe data; it is derived from β, which was empirically constrained in MTDF_01 from the MTDF void quantization ansatz $R_n = \beta(1+\sqrt{n})$ matched to observed void radii, not fitted to Pantheon+, DESI, or any of the multi-probe observables presented here.

The specific features of the multi-probe evidence that align with MTDF predictions include:

(i) *Sharp transition.* The supernova piecewise split and the CF4 redshift concentration both show a transition that is abrupt rather than gradual. A gradual transition would be consistent with slowly varying astrophysical systematics; a sharp transition at a specific scale points to a physical threshold.

(ii) *Void geometry beyond density.* The partial-correlation controls demonstrate that d_{signed} carries information about observables that raw local density does not. In MTDF, void boundaries correspond to regions where the stress tensor field transitions between regimes, producing geometric signatures that are not captured by a scalar density field.

(iii) *Hubble tension resolution.* MTDF's scale factor $S_0 = 1.084$ naturally maps the Planck H_0 to the locally measured value, resolving the time-delay tension with $\chi^2/\text{dof} = 0.68$. This is a parameter derived from the field theory, not fitted to the time-delay data.

(iv) *Consistency across observables.* Supernova lightcurve distances, strong-lens time-delay distances, peculiar velocities, and Tully-Fisher distances all respond to environment in a manner consistent with the MTDF coherence scale. These observables probe different physical processes, making a common systematic explanation difficult to construct.

It should be emphasised that this paper establishes convergent evidence for an environment-dependent transition, with MTDF as the coherent interpretive framework. It does not claim that every aspect of MTDF is validated by these results. The photon-coupling sector ($\kappa = 1.10 \times 10^{-3}$), the CMB modifications, and the deep-space propagation predictions remain subjects of their respective companion papers and future observational programmes.

66 Limitations and caveats

Three caveats of the current analysis should be noted.

Void classification gap. None of the Phase 3 analyses (CF4, 2MTF) contain void-classified galaxies: all objects lie in wall environments as defined by DESIVAST. The signals are gradients within the wall population measured by the continuous d_{signed} metric. This is physically meaningful because the MTDF stress tensor field varies continuously with distance to the void boundary; the binary void/wall label is an artefact of the classification algorithm, not a property of the underlying field. The partial-correlation controls (Tables 6 and 8) demonstrate that these wall-internal gradients carry void-geometric information well beyond raw density, so the impact of this gap on the main conclusions is low. The continuous signed-distance statistic is the physically relevant quantity for these channels. Nevertheless, the traditional binary void-versus-wall split remains unperformed.

Signed-distance and heliocentric-distance correlation. The environment metric d_{signed} correlates with heliocentric distance ($r \approx -0.63$ for CF4), reflecting the redshift-dependent completeness of the DESIVAST void catalogues rather than a physical coupling. Dedicated tests (Section 5.3.3, Control E) show that the void-geometric signal persists across distance bins, in both quality halves, and in every distance-method family, indicating that this geometric correlation does not explain the observed environment coupling.

Data-limited supporting channels. The ISW and S_8 channels are suggestive but below detection threshold with current void catalogue sizes and weak-lensing survey areas. They are included to establish methodology and directional consistency, not as independent confirmations. Next-generation survey volumes (DESI Y5, Euclid, LSST/Rubin) will provide the 10–15 times sensitivity gain needed to push these channels from suggestive to decisive.

67 Conclusion

This paper reports convergent evidence from multiple independent low-redshift observables for a common environment-dependent transition near a single coherence scale.

The supernova environment signal ($\gamma_{\rm env} = +0.013 \pm 0.004$, $p = 0.003$) exhibits a sharp piecewise cutoff at $z = 0.04$ ($\Delta\chi^2 = 36$–39). Strong-lens time-delay distances prefer an MTDF-predicted $H_0 \approx 73.1$ over Planck ΛCDM by a factor of 5.5 in χ^2/dof. Cosmicflows-4 peculiar velocities show a void-geometric signal that survives a seven-layer control chain, including partial correlation beyond density (8–17σ), 2M++ velocity-field subtraction (5–9σ retained), Malmquist bias testing across distance bins and quality halves, and leave-one-method-out stability across TF, FP, SNIa, and calibrator subsamples. The signal is concentrated below $z = 0.04$ and exhibits a qualitative sign mismatch against full N-body ΛCDM expectations (9.0σ tension with MDPL2 across 100 observers). 2MTF Tully-Fisher distances independently confirm a void-geometric signal at 4–15σ beyond density, achromatic across three infrared bands, with 4.2σ N-body tension.

The convergence of these signals, arising from different physical processes, different survey instruments, different galaxy populations, and different distance-indicator calibrations, at a single environment-linked redshift scale, constitutes evidence that is difficult to attribute to any one dataset or method. The Mesche Tensor Dynamics Framework provides a coherent interpretation through its coherence length $\beta = 22.7$ Mpc, which was established independently of the multi-probe data presented here.

Ongoing extensions include joint multi-probe fitting with a shared coherence-scale parameter and morphological type splits for 2MTF (requiring external cross-matching, as the 2MTF catalogue does not include morphology data). Next-generation survey volumes (DESI Y5, Euclid, LSST/Rubin) will provide the statistical power to push the supporting ISW and S_8 channels from suggestive to decisive.

68 Appendix A. Data availability and reproducibility

All analysis scripts and machine-readable results are provided in the MTDF validation repository. The repository is organised into phase folders; the material for this paper spans two phases corresponding to the two primary observational channels.

68.1 A.1 Analysis scripts

Nine Python scripts reproduce every quantitative result reported in this paper. All scripts are located in `validation/code/paper8/`:

Table 26. Table A1. Analysis scripts for MTDF_08.

Script	Section	Description
task3a_cosmicflows4_vpec.py	5.3.1–5.3.2	CF4 baseline: gamma_v fits, piecewise z-split, permutation tests (3 void catalogues)
task3a_control_tests.py	5.3.3 (A–C)	CF4 controls: density substitution, partial correlation, LCDM chain rule
task3a_2mpp_reconstruction.py	5.3.3 (D)	CF4 control D: 2M++ velocity-field subtraction (Carrick et al. 2015)
task3a_robustness.py	5.3.3 (E–F)	CF4 controls E-F: leave-one-method-out (LOSO) and Malmquist bias diagnostics
task3a_lcdm_mock.py	7.1, 7.3	LCDM linear-theory mock (100 realisations) and pure-noise baseline
task3a_nbody_mock.py	7.2	MDPL2 N-body mock for CF4 (100 observers, halos from CosmoSim TAP)
task3b_2mtf_tully_fisher.py	5.4.1	2MTF baseline: gamma_TF fits across J/H/K bands and 3 void catalogues
task3b_control_tests.py	5.4.2	2MTF controls: density substitution, partial correlation, LCDM chain rule
task3b_nbody_mock.py	7.2	MDPL2 N-body mock for 2MTF (50 observers)

68.2 A.2 Machine-readable results

All numerical results are stored as JSON files with full metadata. Phase 7 covers the CF4 peculiar velocity channel; Phase 8 covers the 2MTF Tully-Fisher channel.

Table 27. Table A2. Result files (JSON) with SHA256 integrity hashes.

Phase	File	Contents
7 (CF4)	task3a_cf4_results.json	Baseline gamma_v per void catalogue, piecewise z-split, Delta-chi2
7 (CF4)	task3a_control_results.json	Controls A-C: density, partial correlation, chain rule
7 (CF4)	task3a_2mpp_results.json	Control D: gamma_v after 2M++ subtraction (Tables 6b)
7 (CF4)	task3a_robustness_results.json	Controls E-F: LOSO stability (Table 6c), Malmquist diagnostics
7 (CF4)	task3a_lcdm_mock_results.json	Linear mock gamma_v distribution (Table 10), noise baseline
7 (CF4)	task3a_nbody_mock_results.json	MDPL2 N-body mock gamma_v (Table 11, 100 observers)
8 (2MTF)	task3b_2mtf_results.json	Baseline gamma_TF per band and void catalogue (Table 7)
8 (2MTF)	task3b_control_results.json	Controls A-C (Table 8)
8 (2MTF)	task3b_nbody_mock_results.json	MDPL2 N-body mock gamma_TF (Table 11, 50 observers)

68.3 A.3 Diagnostic plots

32 diagnostic plots for the CF4 channel are in validation/output/phase7_cf4_vpec/. 11 diagnostic plots for the 2MTF channel are in validation/output/phase8_2mtf_tf/. Each phase folder includes a `manifest.json` with SHA256 hashes for integrity verification.

68.4 A.4 External data sources

Table 28. Table A3. External datasets required for reproduction.

Dataset	**Source**	**Access**
Cosmicflows-4 groups	Tully et al. (2023)	Extragalactic Distance Database
2MTF catalogue	Hong et al. (2019)	VizieR J/MNRAS/487/2061
DESIVAST void catalogues	DESI Collaboration	DESI data release (VoidFinder, REVOLVER, VIDE)
2M++ velocity/density fields	Carrick et al. (2015)	cosmicflows.iap.fr/2MPP
MDPL2 N-body halos	Klypin et al. (2016)	CosmoSim TAP API (auto-downloaded by script)

69 References

Brout, D. et al. (2022). The Pantheon+ Analysis: Cosmological Constraints. *ApJ*, 938, 110. doi:10.3847/1538-4357/ac8e04

Carrick, J., Turnbull, S. J., Lavaux, G., & Hudson, M. J. (2015). Cosmological parameters from the comparison of peculiar velocities with predictions from the 2M++ density field. *MNRAS*, 450, 317. doi:10.1093/mnras/stv547

DESI Collaboration (2024). DESI 2024 VI: Cosmological Constraints from BAO measurements. *arXiv:2404.03002*

Hong, T. et al. (2019). 2MTF VII. 2MASS Tully-Fisher survey final data release. *MNRAS*, 487, 2061. doi:10.1093/mnras/stz1413

Klypin, A. et al. (2016). MultiDark simulations: the story of dark matter halo concentrations and density profiles. *MNRAS*, 457, 4340. doi:10.1093/mnras/stw248

Mesche, I. (2026a). Mesche's Tensor Dynamics Framework (MTDF): A Dynamic Field Theory of Cosmic Structure. MTDF_01.

Mesche, I. (2026b). MTDF Companion Paper Part I: Environmental and Local Phenomenology. MTDF_02.

Mesche, I. (2026c). MTDF Companion Paper Part II: Gravity Sector and Lensing Validation. MTDF_03.

Mesche, I. (2026d). MTDF Companion Paper Part III: Photon Coupling, Redshift, Early Universe. MTDF_04.

Mesche, I. (2026e). MTDF Independent GPU Validation: Phases 1–6. MTDF_07.

Millon, M. et al. (2020). TDCOSMO I. An exploration of systematic uncertainties in the inference of H_0 from time-delay cosmography. *A&A*, 639, A101. doi:10.1051/0004-6361/201937351

Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641, A6. doi:10.1051/0004-6361/201833910

Tully, R. B. et al. (2023). Cosmicflows-4. *ApJ*, 944, 94. doi:10.3847/1538-4357/ac94d8

Wong, K. C. et al. (2020). H0LiCOW XIII. A 2.4% measurement of H_0 from lensed quasars. *MNRAS*, 498, 1420. doi:10.1093/mnras/stz3094